

Discrete Initial Groups and Omnific Normalization

Sign-tree surgery, a sharp image classification,
an Ehrlich–Kaplan question, and convex factors

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Abstract

We give an explicit construction proposed as an affirmative solution of Ehrlich and Kaplan’s question whether every discretely ordered initial subgroup of the surreal numbers is order-isomorphic to an initial subgroup of the omnific integers (Question 2 of the arXiv preprint, Question 9.1 of the journal version). The construction is not ordinary multiplication by the reciprocal of the least positive element. Instead, a sign-sequence operation moves the minimum exponent to zero, while preserving all right-ancestor relations away from that minimum; a separate rescaling changes only the bottom coefficient group to the ordinary integers. The Ehrlich–Kaplan characterization of initial Hahn subgroups then establishes initiality of the image.

The argument yields more than existence. For a prescribed least positive element $2^{-n}\omega^{-\alpha}$, we characterize the images exactly by containment of an explicit dyadic-spine subgroup $K_\alpha \subseteq \mathbf{Oz}$. We identify the smallest and largest initial groups with that least positive element, give sharp cardinality consequences, construct an initial realization of the quotient by the least positive cyclic subgroup, and extend the additive normalization to rectangular surcomplex modules. All normal-form transports preserve leading truncations and set-indexed strong summability. Counterexamples explain why scalar normalization, exponent translation, and unrestricted inversion do not suffice.

A second manuscript realizes every convex subquotient of an initially realizable group by sign-tree compression, and shows that a Presburger group is initially realizable exactly when its residues in $\widehat{\mathbb{Z}}$ are ordinary integers. A third manuscript shows that a bounded dyadic formula defines the standard natural numbers in discretely ordered rings with division by 2 and 3 and a normalized additive map to $\mathbb{Z}$, that the additive group of a nonstandard model of bounded arithmetic has no nonzero homomorphism to $\mathbb{Z}$, and that bounded truth in a computable such ring is as hard as true arithmetic. With the second manuscript’s criterion, the additive group of every nonstandard model of $\mathsf{I}\Delta_0$ is not initially realizable, whereas open induction does not suffice. A fourth manuscript identifies the profinite residue image of a nonstandard model of Peano arithmetic: it consists exactly of the residue sequences computable from one member of the standard system, which it recovers as its idempotents. For Borel models, the real and p -adic points computable from the standard system form analytic subfields that leave a perfect algebraic-independence gap, which survives countably many extra constants and coefficient-preserving Hahn extension; it yields continuum many algebraically independent omnific integers that vanish in every finite quotient.

Status and scope. This is an unrefereed research manuscript with full proposed proofs, not an independently certified resolution or a claim of established priority. The principal external mathematical input is explicitly identified. Finite regression checks accompany the text; they do not certify the transfinite argument. No new Lean formalization is claimed. A targeted literature and repository review found the cited question and its motivating example, but did not establish exhaustive novelty. At placement in the collection every step was checked by hand and no gap was found (section 27.5); that check is not a refereeing, and the literature search was targeted, not exhaustive. The second manuscript (section 1.5), the third (section 1.6) and the fourth (section 1.7) have the same status.

Keywords: surreal numbers; omnific integers; initial subgroups; sign sequences; Hahn series; ordered abelian groups.

Mathematical focus: additive and simplicity-theoretic structure, not multiplicative or exponential equivalence.

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1 The problem and the main results

1.1 A specific published question

An additive subgroup of **No** is *initial* when it contains every proper sign-sequence prefix of each of its members. This is a much stronger requirement than being an ordered subgroup. An initial subgroup is *discrete* when it is nonzero and has a least positive element. Throughout, “isomorphism” of ordered groups preserves addition and order; it need not preserve the ambient simplicity relation.

Ehrlich and Kaplan ask:

“Is every discrete initial subgroup of **No** isomorphic to an initial subgroup of **Oz**?”

This is Question 2 in Section 9 of the explicitly versioned preprint [1], printed page 18. The journal article appeared in 2018 [2]. We use the preprint’s numbering throughout to avoid mixing versions. The question also appears in Kaplan’s thesis [3].

In the journal version the same question is Question 9.1, with the same wording and the same motivating example. The results of [1] used below are numbered differently there; the correspondence, checked when this report was placed in the collection against the author-hosted text of the journal version [2], is:

arXiv v1 [1]	journal version [2]	use here
Lemma 3	Lemma 5.1	initial subgroups of $\mathbb{R}$, imported fact 2.3
Theorem 1	Theorem 5.1	the initiality criterion, imported fact 2.2
Proposition 9	Proposition 5.2	shape of the least positive element
Question 2	Question 9.1	the question addressed here
Question 3	Question 9.2	the initial-monoid problem, not addressed
Theorem 6	Theorem 8.1	optimality with class models (source 02)
Question 1	Question 8.1	set-model optimality, not answered

Below, the arXiv numbers are used, except where the wording of the journal version matters (after imported fact 2.2).

The distinction between a group and a ring matters. A discrete initial subgroup can contain $1/2$ or a nonzero infinitesimal, whereas an omnific integer has neither a nonintegral constant coefficient nor a negative normal-form exponent. The task is therefore to change the initial *realization* of the ordered group, not to prove that its original realization already lies in **Oz**.

The paper addresses precisely this existence question and several structural refinements. It does not claim a classification of all ordered abelian groups admitting initial embeddings, nor a solution of the separate initial-monoid problem (Question 3 of [1], Question 9.2 of the journal version).

1.2 Normalization and its exact range

Write $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{D} = \mathbb{Z}[1/2]$. If α is an ordinal, let

$$p_\alpha = -\alpha = \langle - \rangle^\alpha, \quad q_\beta = \langle + \rangle \frown \langle - \rangle^\beta \quad (\beta \in \mathbf{On}).$$

Here $\frown$ means concatenation of sign sequences, not surreal addition. For example, $q_0 = 1$, $q_1 = 1/2$, and $q_m = 2^{-m}$ for finite m . Define the finite-support additive subgroup

$$K_\alpha = \mathbb{Z} \oplus \bigoplus_{\beta < \alpha} \mathbb{D} \omega^{q_\beta} \subseteq \mathbf{Oz}. \quad (1)$$

The direct sum means that each element has only finitely many nonzero summands. In particular, $K_0 = \mathbb{Z}$.

Theorem A (Omnific normalization). Let $A \subseteq \mathbf{No}$ be a nonzero discrete initial additive subgroup. There are unique $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$ such that its least positive element is

$$\varepsilon = 2^{-n}\omega^{-\alpha}.$$

An explicitly defined map $N_{\alpha,n}$ restricts to an order-preserving group isomorphism

$$N_{\alpha,n} : A \longrightarrow H, \quad H \subseteq \mathbf{Oz} \text{ initial}, \quad N_{\alpha,n}(\varepsilon) = 1.$$

The map extends to a real-linear order-isomorphism between the full normal-form spaces supported in $[-\alpha, \infty)$ and $[0, \infty)$. This extension preserves support order types, all leading truncations, and set-indexed strong summability in both directions.

Theorem B (Sharp image classification). Fix $\alpha \in \mathbf{On}$ and $n \in \mathbb{N}$. Under the same explicit map $N_{\alpha,n}$, the initial groups with least positive element $2^{-n}\omega^{-\alpha}$ correspond bijectively, with inclusions preserved and reflected, to the initial groups $H \subseteq \mathbf{Oz}$ satisfying

$$K_\alpha \subseteq H.$$

The inverse is explicit. Thus the dyadic-spine condition is both necessary and sufficient, not merely an artifact of a sufficient construction.

Theorem C (Removing the bottom cyclic layer). For every group A in [Theorem A](#), the subgroup $\mathbb{Z}\varepsilon$ is convex, and the ordered quotient $A/\mathbb{Z}\varepsilon$ has an explicit initial realization in $\mathbf{No}$. The corresponding short exact sequence splits as an additive group, with a canonical splitting relative to the given Conway normal forms. Equivalently,

$$A \cong B \overrightarrow{\times} \mathbb{Z}$$

for an explicitly constructed initial subgroup $B \subseteq \mathbf{No}$, where the first coordinate is dominant in the lexicographic order.

The main construction is in [sections 4 and 5](#); the proofs of the three theorems are completed in [sections 6 to 8](#), respectively. [Theorem B](#) contains more information than [Theorem A](#): it identifies exactly which normalized realizations can be moved to any prescribed least positive scale by this construction.

1.3 Why a change of exponents is necessary

Multiplication by ε^{-1} certainly gives an ordered-group isomorphism. Initiality, however, refers to sign prefixes rather than numerical intervals, and multiplication does not generally preserve it. Even $A = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\omega$ is an initial group for which $2A = \mathbb{Z} + 2\mathbb{Z}\omega$ is not initial: 2ω belongs to the image but its simpler predecessor ω does not.

Our operation treats two different issues separately. First, it relocates the minimum of the *exponent tree* to zero. Second, it rescales only the real coefficient at that minimum. No claim of multiplicativity is made, and none is needed for the published ordered-group question.

1.4 Conventions shared with the collection

This report is built from four manuscripts: source 01, the normalization manuscript ([sections 1 to 10 and Appendix A](#)), source 02, the convex-factor manuscript ([section 1.5, sections 11 to 18 and Appendix C](#)), source 03, the standard-cut manuscript ([section 1.6 and sections 19 to 22](#)), and source 04, the transcendence-gap manuscript ([section 1.7 and sections 23 to 26](#)). Their provenance, the placement reviews and the relation to other reports are in [section 27.5](#). Statements added in the merge of the first two sources, and proved here, are marked [merge]; statements added when source 03 was written in, and proved here, are marked [write]; the one statement added when source 04 was written in is marked [write 04]. The following conventions are fixed here once; the notation of source 02 is fixed in [section 11.1](#), that of source 03 in [section 19.2](#), and that of source 04 in [section 23.2](#).

1. *Initial* always means closed under proper sign-sequence prefixes, as in the definition of the definable-surreals report (dsn:sec:foundations). It never means an initial segment of the numerical order, and it has nothing to do with the *initial forms* (leading forms of polynomial solutions) of the Diophantine report (odg:prop:initial).
2. $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$, with Π the class of purely infinite normal forms, is the omnific ring of the collection (odg:def:rings, osq:eq:Ozdef, found:eq:omnific); (4) below is the same description. For $x \in \mathbf{Oz}$, $\text{ct}(x)$ is its coefficient at exponent 0, as there. Only the additive and order structure of $\mathbf{Oz}$ is used; its ring structure plays no role.
3. The symbols Θ_α , Ψ_α , q_β and $N_{\alpha,n}$ are local to this report. They are unrelated to the theta series and to the nome or dilation parameter q of other reports.

1.5 Convex factors and profinite obstructions

Theorem C removes the least positive cyclic subgroup. The second source of this report, *Convex Factors and Profinite Obstructions for Initial Surreal Groups*, removes an arbitrary convex subgroup. Call an ordered abelian group *initially realizable* if it is order-isomorphic to an additive subgroup of $\mathbf{No}$ closed under all proper sign-sequence prefixes, and write $G \in \text{Init}$ for this property; the zero group is included. A *convex factor* of G is C'/C , where $C \subseteq C'$ are convex subgroups of G and the quotient has its usual ordered group structure.

Theorem D (Convex-factor closure). If $G \in \text{Init}$ and $C \subseteq C'$ are convex subgroups of G , then $C'/C \in \text{Init}$. For an actual initial realization $G \subseteq \mathbf{No}$, the construction is explicit: restrict normal forms to the corresponding convex exponent interval and compress its sign tree.

Moreover, every convex subgroup sequence

$$0 \longrightarrow C \longrightarrow G \longrightarrow G/C \longrightarrow 0$$

splits as a sequence of abelian groups. In the chosen normal-form realization the section is order-preserving and the decomposition is lexicographic. The coordinate projections and compressed factor realizations satisfy precise composition laws under nested restrictions.

The theorem concerns *initial realizability*, not literal initiality of the original subgroup C or of a selected complement. It also does not assert a splitting invariant under every abstract automorphism. The explicit splitting is attached to a chosen initial realization. With $C' = G$ it answers [question 28.1](#): every convex subgroup of an initial group has an explicit initial realization of the quotient by a normal-form transport, and for $C = \mathbb{Z}e$ that transport is the construction of [section 8](#) ([proposition 13.5](#)). It also answers [question 28.2](#): no extra hypothesis is needed at a limit stage ([corollary 13.6](#)).

A $\mathbb{Z}$ -group is a discretely ordered abelian group G with least positive element e such that $G/\mathbb{Z}e$ is divisible. These are exactly the ordered-group models of additive Presburger arithmetic [10]. Their coherent standard residues define

$$\text{res}_G : G \longrightarrow \widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z}, \quad \text{res}_G(e) = 1.$$

Here $\mathbb{Z}$ on the right is the ordinary, diagonally embedded copy of the integers, not a nonstandard integer substructure.

Theorem E (Presburger initiality criterion). For a set-sized $\mathbb{Z}$ -group G , the following are equivalent:

$$\begin{aligned} G \in \text{Init}; \quad & G \text{ has an initial realization in } \mathbf{Oz}; \quad & \mathbb{Z}e \text{ is a direct summand of } G; \\ \text{res}_G(G) = \mathbb{Z}; \quad & G \cong G^{\text{div}} \overrightarrow{\times} \mathbb{Z}, \quad & G^{\text{div}} = \bigcap_{n \geq 1} nG. \end{aligned}$$

The group G^{div} is divisible. In the split case its complement to $\mathbb{Z}e$ is unique.

This is a global compatibility condition, not a condition on any one finite modulus. Every $\mathbb{Z}$ -group has the same individual quotients $G/nG \cong \mathbb{Z}/n\mathbb{Z}$. Initiality distinguishes whether the residues of a *single element*, across all standard moduli, come from one ordinary integer.

Theorem F (Explicit countermodels). There are $2^{\aleph_0}$ pairwise nonisomorphic countable rigid $\mathbb{Z}$ -groups which are not initially realizable, although they are all elementarily equivalent to $\mathbb{Z}$. There are also countable densely ordered groups, elementarily equivalent to an explicit initial subgroup of $\mathbf{No}$, which are not initially realizable.

No recursively saturated $\mathbb{Z}$ -group, and no nonprincipal ultrapower over $\mathbb{N}$ of a $\mathbb{Z}$ -group, is initially realizable. The ordered additive group completion of any nonstandard model of Peano arithmetic is likewise not initially realizable.

The proofs are in [section 13](#) (Theorem D), [section 14](#) (Theorem E) and [section 16](#) (Theorem F). In source 02 these three theorems are Theorems A, B and C; they are lettered D, E and F here because source 01's Theorems A–C keep their letters.

What is new in source 02, and what is not. The primary external structural input is again the Ehrlich–Kaplan characterization of initial Hahn subgroups ([imported fact 2.2](#)). Their treatment of convex subgroups asks which subgroups in the *given* realization are themselves initial. The construction of source 02 changes the realization of a factor and preserves its ordered additive structure. Source 01 already gives the normalization of [Theorem A](#), the integer-bottom suspension and the relative splitting criterion for discrete groups ([theorem 8.5](#) and [corollary 8.6](#)); its [section 8](#) explicitly leaves arbitrary convex quotients outside its conclusions. Source 02 does not claim those results again as new. It replaces the special positive-root deletion by a compression operation valid for arbitrary convex exponent intervals; in particular it handles cuts with no endpoint and convex factors not accessible by a finite iteration of least-positive quotients.

The profinite residue map and its divisible kernel belong to the classical structure theory of $\mathbb{Z}$ -groups; see especially [10, Sections 2 and 4]. Likewise, the extension parameter $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ is classical [11, Example 8.27]. Source 02's arithmetic contribution is their explicit connection with initial surreal realizability, together with the displayed constructions and obstructions. The existence of rigid Presburger models is not claimed as a new phenomenon. The results are not announced as a resolution of the set-model version of Ehrlich–Kaplan's optimality question (Question 1 of [1], Question 8.1 of the journal version); a precise comparison is in [section 18.2](#). A targeted review found no matching formulation of the main convex-factor theorem, but this is not an exhaustive priority search.

1.6 Standard integers and bounded arithmetic

Source 02 shows that the ordered additive group of a nonstandard model of Peano arithmetic is not initially realizable ([theorem 16.6](#)), and asks for the weakest arithmetic fragment with that property ([question 28.7](#)). The third source of this report, *Standard Integers in a Single Infinite Interval*, is weak arithmetic of discretely ordered rings R with Euclidean division by ordinary integers and an additive map $\chi : R \rightarrow \mathbb{Z}$ with $\chi(1) = 1$, not necessarily order-preserving; $\mathbf{Oz}$ with its constant term is one. Its main results are:

- a fixed bounded formula Std^{dy} , built from the divisor-theoretic predicate “no odd divisor at least 3”, defines exactly the standard natural numbers whenever division by 2 and 3 and such a χ are available, and is inductive under division by 2 alone ([theorems 19.4](#) and [19.7](#) and [lemma 19.6](#));
- with division by every ordinary integer, the integer characters are 0 or $\mathbb{Z}\chi$ with χ a ring map, and the difference ring of a nonstandard model of $\text{I}\Delta_0$ has none at all; so no such model has a unit-preserving additive map into $\mathbf{Oz}$ ([theorems 20.1](#) and [20.4](#) and [corollary 21.4](#));

- every arithmetic sentence compiles to a bounded formula with one infinite bound and quadratic atoms, and bounded truth in the computable ring $\mathbb{Z} + X\mathbb{Q}[X]$ is many-one equivalent to true arithmetic (theorems 22.2 and 22.4);
- a counterexample to a uniqueness lemma of a recent preprint, with a repair under division by 2 (propositions 21.1 and 21.2).

Combined with Theorem E, the second item lowers the hypothesis of theorem 16.6 from Peano arithmetic to $\text{I}\Delta_0$, and indeed to a recursively axiomatized theory T^{dy} whose only induction is one bounded instance, while open induction does not suffice (corollary 20.11 and proposition 20.12, [write]). Bounded definability of the standard cut in such rings, including **Oz**, was already known [14, Theorem 2.9], and so was the failure of induction beyond open induction in **Oz** [18]; what source 03 adds is stated in section 19.1.

1.7 Standard systems and perfect transcendence gaps

Source 03 shows that the residue image $\text{im res}_{R_M} \subseteq \widehat{\mathbb{Z}}$ of a nonstandard model M of $\text{I}\Delta_0$ is not contained in $\mathbb{Z}$ (corollary 20.9) and asks which subrings of $\widehat{\mathbb{Z}}$ occur as such images (question 28.11). The fourth source of this report, *Perfect Transcendence Gaps in Borel Arithmetic: Standard Systems, p -adic Shadows, Hahn Fields, and Omnific Integers*, answers this for nonstandard models of full Peano arithmetic: the image is exactly the ring $\widehat{\mathbb{Z}}(\mathcal{S})$ of profinite integers whose factorial residue sequences are computable from one member of the standard system $\mathcal{S} = \text{SSy}(M)$, and the idempotents of that ring recover $\mathcal{S}$ (theorem 25.3 and corollary 25.4). Its other main results are:

- if $k \subseteq \mathbb{R}$ or $k \subseteq \mathbb{Q}_p$ is an analytic subfield, relatively algebraically closed, then adjoining any countable set of constants and taking the relative algebraic closure leaves a proper, null and meager field, over which every nonempty open set contains an algebraically independent Cantor set (theorem 23.1); the mechanism is that an analytic subfield cannot have positive countable transcendence degree (lemma 24.1);
- for a nonstandard Borel model of PA, the real and p -adic points computable from members of $\text{SSy}(M)$ form such fields (theorem 23.12), by Glazer’s theorem that Borel models lack full standard system, imported (imported fact 23.6);
- the p -adic residue image of R_M is exactly the valuation ring of the p -adic field so obtained (theorem 25.1);
- a coefficient-preserving Hahn transfer gives continuum many purely infinite omnific integers $r\omega$, algebraically independent over a restricted Hahn field and zero in every finite quotient, although the family is discrete in the surreal order topology (theorems 26.1 and 26.3).

The identification of the image covers Peano arithmetic only; for $\text{I}\Delta_0$ the question stays open, and it is re-scoped in a dated note after its statement. With Scott’s classical theorem the images of the countable nonstandard models of PA are exactly the rings $\widehat{\mathbb{Z}}(\mathcal{S})$ of countable Scott sets (corollary 25.5, [write 04]). Source 04 did not know this report: one of its corollaries is weaker than theorem 20.4, its Hahn transfer follows from a lemma of another report, and part of its comparison of completions is a set-sized version of facts already proved in Lean for **Oz** (section 23.1).

2 Foundations and the imported initiality criterion

2.1 Sets, classes, signs, and two orders

We work in NGB with global choice, in the conventional class-sized formulation of surreal numbers. Every individual surreal is a set-length sign sequence. Every individual normal form has set support.

All strongly summable families considered below are indexed by sets. Statements about families of class-sized groups are understood as correspondence statements about class parameters, not as assertions that a class of all classes exists.

For sign sequences u, v , write $u \prec v$ for a proper prefix and $u \preceq v$ for a prefix allowing equality. Their ordinary numerical order is determined at the first disagreement, with

$$- < \text{undefined} < +.$$

The empty sequence is the number 0. Put

$$\text{Pred}(x) = \{y : y \prec x\}, \quad \text{R}_s(x) = \{y \prec x : x < y\}, \quad \text{L}_s(x) = \{y \prec x : y < x\}.$$

In sign language, a prefix obtained by stopping just before a minus sign is a right ancestor; stopping just before a plus sign gives a left ancestor. Each of these ancestor collections is a set.

A subclass $E \subseteq \mathbf{No}$ is initial precisely when $\text{Pred}(x) \subseteq E$ for every $x \in E$. This does not mean that E is downward closed in the numerical order. For instance, $\mathbf{No}_{\geq 0}$ is initial as a sign tree but is a numerical final segment.

If u and v have lengths λ and μ , then $u \frown v$ has length $\lambda +_{\text{ord}} \mu$. All additions of *lengths* in this paper are ordinal additions. Concatenation is associative and preserves comparison between tails sharing the same prefix. At a limit ordinal there is no last sign to remove: “remove the first sign” or “remove a specified concatenated block” always means restricting to the appropriate interval and taking its order type. These conventions agree with the sign-sequence framework of [5].

2.2 Hahn spaces without an exponent group

For an ordered subclass $E \subseteq \mathbf{No}$, denote by $\mathcal{H}_{\mathbb{R}}(E)$ the additive space of formal expressions

$$x = \sum_{\gamma \in S} r_{\gamma} \omega^{\gamma}, \quad r_{\gamma} \in \mathbb{R} \setminus \{0\}, \quad (2)$$

where $S \subseteq E$ is a set reverse-well-ordered by $<$. Conway normal forms identify these expressions with the corresponding surreal numbers. The ordering is the sign of the coefficient at the greatest exponent of a nonzero difference. For $x \neq 0$, write

$$\deg_H(x) = \max \text{supp}(x), \quad \text{lc}(x) = r_{\deg_H(x)}.$$

We do not assert that E is an additive group. Accordingly, $\mathcal{H}_{\mathbb{R}}(E)$ is used here as an additive, real-linear Hahn space, not necessarily as a ring. This distinction permits nonadditive exponent reindexings.

Suppose $S = \{\gamma_{\xi} : \xi < \lambda\}$ is enumerated in strictly decreasing order. A leading truncation of x is

$$x_{<\eta} = \sum_{\xi < \eta} r_{\gamma_{\xi}} \omega^{\gamma_{\xi}} \quad (\eta \leq \lambda).$$

“Truncation closed” will always refer to these leading truncations, including those at limit indices, not to closure under arbitrary subseries. The normal-form and monomial facts used here are part of the standard theory summarized in [1]; we never choose a noncanonical Hahn embedding of an arbitrary ordered group.

Definition 2.1 (Strong summability). A set-indexed family $(x_i)_{i \in I}$ in a Hahn space is strongly summable if $\bigcup_i \text{supp}(x_i)$ is reverse-well-ordered and each exponent occurs in only finitely many supports. Its sum is the normal form obtained by the finite coefficient sum at each exponent, with zero coefficients discarded.

For a subgroup $G \subseteq \mathcal{H}_{\mathbb{R}}(E)$, set

$$G_{\gamma} = \{r \in \mathbb{R} : r\omega^{\gamma} \in G\}.$$

It is *cross-sectional* on E when $\omega^{\gamma} \in G$ for every $\gamma \in E$.

Imported fact 2.2 (Ehrlich–Kaplan, concrete form). *An initial subgroup of $\mathbf{No}$ is a truncation-closed, cross-sectional Hahn subgroup on its initial exponent class $\Gamma(G) = \{\gamma : \omega^{\gamma} \in G\}$. Its nonzero coefficient groups are initial subgroups of $\mathbb{R}$, and*

$$\gamma, \eta \in \Gamma(G), \quad \eta \in R_s(\gamma) \implies \mathbb{D} \subseteq G_{\eta}. \quad (3)$$

Conversely, these conditions imply that the canonical normal-form image is initial in $\mathbf{No}$.

This is the form of [1, Theorem 1 and its proof] needed here. The last sentence is important: the sufficiency proof establishes initiality of the *specified canonical image*, not just existence of some other initial embedding.

The journal version states this concrete form outright. Its Theorem 5.1, the counterpart of [1, Theorem 1], says that a subgroup of $\mathbf{No}$ is initial if and only if it is isomorphic “via the canonical isomorphism” to a truncation-closed, cross-sectional Hahn subgroup whose exponent class is an initial ordered subclass of $\mathbf{No}$ and whose coefficient groups satisfy the two conditions above [2]. The canonical isomorphism is the normal-form identification of $\mathbf{No}$ with the Hahn group, so the journal statement is [imported fact 2.2](#) as used here. The criterion is imported as a published statement. Neither the source manuscript nor the placement review re-proves it ([section 27.5](#)).

Imported fact 2.3 (Real coefficient groups). *A nonzero initial additive subgroup of $\mathbb{R}$ either contains $\mathbb{D}$ or is $2^{-m}\mathbb{Z}$ for an integer $m \geq 0$ [1, Lemma 3] (Lemma 5.1 of the journal version). In particular, a discrete one has least positive element 2^{-m} .*

No proof in the present paper replaces [imported fact 2.2](#) by the weaker assertion that initiality of the exponent set alone is sufficient. The coefficient groups and the right-ancestor condition are essential.

2.3 Omnific integers and coefficient isolation

We use the normal-form description

$$\mathbf{Oz} = \left\{ \sum_{\gamma > 0} r_{\gamma} \omega^{\gamma} + m : m \in \mathbb{Z} \right\}, \quad (4)$$

with set reverse-well-ordered positive support and arbitrary real coefficients. This is Conway’s canonical integer part, as recalled in [3, Section 6.2], and the collection’s $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ ([section 1.4](#)). It is an initial additive subgroup: apply [imported fact 2.2](#) to $E = \mathbf{No}_{\geq 0}$, coefficient group $\mathbb{Z}$ at zero and $\mathbb{R}$ at every positive exponent. Zero is never a right ancestor of a nonnegative number, so the exceptional coefficient group causes no violation of (3). Every positive element of $\mathbf{Oz}$ is at least 1; this discreteness is also `odg:prop:units` of the Diophantine report.

Lemma 2.4 (Isolating a normal-form term). *Let G be an additive subgroup closed under leading truncations. If $x \in G$ has a nonzero coefficient r_{γ} at exponent γ , then $r_{\gamma} \omega^{\gamma} \in G$.*

Proof. If $\gamma = \gamma_{\eta}$ in the decreasing enumeration of the support, subtract the truncation at η from the truncation at $\eta + 1$. Both belong to G . This also works when η is a limit ordinal. $\square$

This lemma permits removing a specified individual term. It does not license an arbitrary infinite subseries of an element of G .

3 Locating the least positive scale

Lemma 3.1 (A minimum in an initial tree). *If a nonempty initial subclass $E \subseteq \mathbf{No}$ has a numerical minimum p , then $p = -\alpha$ for a unique ordinal α . Moreover, the cone $C_\alpha = \{x \in \mathbf{No} : x \geq -\alpha\}$ is initial.*

Proof. A plus sign in p would give a proper prefix smaller than p . Initiality would place that prefix in E , contradicting minimality. Thus every sign in p is minus.

For the second assertion, let $p = -\alpha$, $x \geq p$, and $v \prec x$. If $v < p$, then at the first disagreement with the all-minus sequence p one would have to find either a sign smaller than minus, which is impossible, or a minus sign after p has ended. The latter would force x itself to be below p . Hence $v \geq p$. $\square$

Proposition 3.2 (Bottom coefficient structure). *Let A be a nonzero discrete initial subgroup with least positive member ε . There are unique $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$ such that, writing $p = -\alpha$ and $c = 2^{-n}$,*

$$\varepsilon = c\omega^p, \quad \min \Gamma(A) = p, \quad A_p = c\mathbb{Z}. \quad (5)$$

Every support of every member of A lies in C_α .

Proof. By imported fact 2.2, A is truncation closed. Write the leading term of ε as $r\omega^p$ with $r > 0$. This term belongs to A . If the remaining tail t were nonzero, then $|t| \in A$ and $0 < |t| < \varepsilon$, since the leading exponent of t is strictly smaller than p . This contradicts the choice of ε . Consequently $\varepsilon = r\omega^p$.

The exponent p lies in $\Gamma(A)$ by the cross-sectional normal-form description. If $\gamma < p$ belonged to $\Gamma(A)$, then $0 < \omega^\gamma < r\omega^p = \varepsilon$, another contradiction. Thus p is the minimum of $\Gamma(A)$. By lemma 3.1, $p = -\alpha$ for a unique ordinal α .

The real coefficient group A_p has least positive element r . By imported fact 2.3, it is $2^{-n}\mathbb{Z}$ for a unique $n \in \mathbb{N}$, and $r = 2^{-n}$. All supports lie in $\Gamma(A)$, proving the last assertion. $\square$

The shape of the least positive element is consistent with [1, Proposition 9 and its proof] (Proposition 5.2 of the journal version); the argument above also records the support and coefficient data needed later.

Corollary 3.3 (Forced ancestors). *Under (5), for every $\beta < \alpha$ one has*

$$-\beta \in \Gamma(A), \quad \mathbb{D}\omega^{-\beta} \subseteq A.$$

Proof. The number $-\beta$ is a right ancestor of $-\alpha$. Initiality gives the exponent, and (3) gives all dyadic coefficients there. $\square$

The bottom coefficient group can be discrete, but all these ancestors must carry dyadically divisible coefficient groups. This is the information that survives normalization as the subgroup K_α .

4 Moving a minimum to zero in the sign tree

4.1 Definition and inverse

Fix $\alpha \in \mathbf{On}$, put $p = -\alpha$, and keep $C_\alpha = [p, \infty)$. Every $x \in C_\alpha$ belongs to exactly one of the following cases: $x = p$; $x = p \smallfrown \langle + \rangle \smallfrown u$ for a unique tail u ; or p is not a prefix of x . Indeed, an extension through $p \smallfrown \langle - \rangle$ would lie below the cone.

Definition 4.1 (Root relocation). Define $\Theta_\alpha : C_\alpha \rightarrow \mathbf{No}_{\geq 0}$ by

$$\Theta_\alpha(x) = \begin{cases} 0, & x = p, \\ \langle + \rangle \frown p \frown u, & x = p \frown \langle + \rangle \frown u, \\ \langle + \rangle \frown x, & p \not\preceq x. \end{cases} \quad (6)$$

The inverse candidate $\Psi_\alpha : \mathbf{No}_{\geq 0} \rightarrow C_\alpha$ is

$$\Psi_\alpha(0) = p, \quad \Psi_\alpha(\langle + \rangle \frown v) = \begin{cases} p \frown \langle + \rangle \frown u, & v = p \frown u, \\ v, & p \not\preceq v. \end{cases} \quad (7)$$

The symbols in these formulas are sign blocks. For $\alpha = 0$, the block p is empty and both maps are the identity. For $\alpha > 0$, one may picture the operation as deleting the old minimum node, moving its right subtree into the vacated position, and adding a new common root. The formulas, not this picture, govern limit-length cases.

Lemma 4.2 (The inverse is well-defined). *The maps (6) and (7) are inverse bijections between C_α and $\mathbf{No}_{\geq 0}$.*

Proof. Every nonempty image under Θ_α begins with plus and is therefore positive. Conversely, let $y = \langle + \rangle \frown v > 0$. If $p \preceq v$, the specified preimage extends p through plus and is greater than p . If $p \not\preceq v$, then $v > p$: a sequence below an all-minus word must extend that entire word through minus, contrary to the assumption. Thus Ψ_α takes values in the cone.

If $x = p$, the composite $\Psi_\alpha \Theta_\alpha$ returns p . If $x = p \frown \langle + \rangle \frown u$, removing the first sign of $\langle + \rangle \frown p \frown u$ leaves $p \frown u$, and the first inverse case returns x . In the remaining case, removing the first sign leaves x , to which the second inverse case applies. The same two cases for v prove $\Theta_\alpha \Psi_\alpha(y) = y$ for positive y , and zero is immediate. Uniqueness of tails follows from the order types of the corresponding ordinal intervals, so none of these steps requires a successor length. $\square$

Lemma 4.3 (Order preservation). *The bijection Θ_α is strictly increasing.*

Proof. The minimum p maps to the minimum 0. On the block

$$U = \{p \frown \langle + \rangle \frown u : u \in \mathbf{No}\},$$

comparison is comparison of the tails u , before and after the map. On the remaining class $V = C_\alpha \setminus (U \cup \{p\})$, adding one common initial plus sign also preserves comparison.

It remains to compare the two blocks. Every member of U is smaller than every member v of V : before the end of p , such a v either stops or first has plus where p has minus. In both cases v is greater than every extension of p . After applying Θ_α , cancel the common initial plus sign. One then compares an extension $p \frown u$ of p with v , and the same first disagreement proves the same inequality. When $\alpha = 0$, the block V is empty. This covers all cases. $\square$

4.2 The exact predecessor identity

The following identity is the central combinatorial statement. It makes the exception at the old minimum completely explicit.

Theorem 4.4 (Predecessor and right-ancestor transport). *For every $x \in C_\alpha$ with $x > p$,*

$$\text{Pred}(\Theta_\alpha(x)) = \Theta_\alpha[\text{Pred}(x) \cup \{p\}]. \quad (8)$$

Consequently,

$$R_s(\Theta_\alpha(x)) = \Theta_\alpha[R_s(x)], \quad (9)$$

$$L_s(\Theta_\alpha(x)) = \Theta_\alpha[L_s(x) \cup \{p\}]. \quad (10)$$

At $x = p$ the image has no proper ancestors, while

$$R_s(p) = \text{Pred}(p) = \{-\beta : \beta < \alpha\}, \quad \Theta_\alpha[R_s(p)] = \{q_\beta : \beta < \alpha\}. \quad (11)$$

For $x, y > p$, simplicity is preserved and reflected:

$$x \prec y \iff \Theta_\alpha(x) \prec \Theta_\alpha(y).$$

Proof. First suppose $p \not\preceq x$. No prefix $v \preceq x$ can equal or extend p ; otherwise $p \preceq x$. Thus $\Theta_\alpha(v) = \langle + \rangle \frown v$ for all proper prefixes v of x . The proper prefixes of $\langle + \rangle \frown x$ are exactly the empty word, together with these $\langle + \rangle \frown v$. The empty word is $\Theta_\alpha(p)$, proving (8) in this case.

Now suppose $x = p \frown \langle + \rangle \frown u$. The proper prefixes of x are

$$\{-\beta : \beta \leq \alpha\} \cup \{p \frown \langle + \rangle \frown v : v \prec u\}.$$

Under Θ_α these become, respectively,

$$\{q_\beta : \beta < \alpha\} \cup \{0\}, \quad \{\langle + \rangle \frown p \frown v : v \prec u\}.$$

These are precisely the proper prefixes of $\langle + \rangle \frown p \frown u$. When u is empty the second collection is empty; when u has limit length it includes exactly its proper ordinal restrictions. This proves (8) without a missing limit case.

Filter (8) according to numerical comparison with $\Theta_\alpha(x)$. By lemma 4.3, comparison of images is comparison of preimages. Since $p < x$, the extra element $\Theta_\alpha(p) = 0$ is a left ancestor, never a right ancestor. This gives (9) and (10). Formula (11) follows directly from the all-minus word p . Finally, if $x, y > p$, membership of $\Theta_\alpha(x)$ in the predecessor set of $\Theta_\alpha(y)$ is, by injectivity and (8), exactly membership of x in $\text{Pred}(y)$. $\square$

Remark 4.5 (A stronger identity that would be false). For $\alpha > 0$ one must not assert (9) at $x = p$. Already $p = -1$ has right ancestor 0, whose image is 1, while $\Theta_\alpha(p) = 0$ has no ancestors at all. The missing relations at precisely this one point account for the extra condition in Theorem B.

Corollary 4.6 (Initiality of the image). *If Γ is initial and has minimum $p = -\alpha$, then $\Theta_\alpha[\Gamma]$ is initial and has minimum 0.*

Proof. The minimum assertion follows from the order-isomorphism. The image of p has no predecessors. For $x > p$ in Γ , the right side of (8) lies in $\Theta_\alpha[\Gamma]$, because both $\text{Pred}(x)$ and p lie in Γ . $\square$

Proposition 4.7 (The exact inverse obstruction). *Let $\Delta \subseteq \mathbf{No}_{\geq 0}$ be nonempty and initial. Put*

$$\Lambda_\alpha = \{0\} \cup \{q_\beta : \beta < \alpha\}.$$

Then $\Psi_\alpha[\Delta]$ is initial if and only if $\Lambda_\alpha \subseteq \Delta$.

Proof. The empty word 0 belongs to Δ , so p belongs to the inverse image. If the inverse image is initial, it contains every $-\beta$ with $\beta < \alpha$; applying Θ_α gives $q_\beta \in \Delta$.

Conversely, suppose $\Lambda_\alpha \subseteq \Delta$. All predecessors of p are then in the inverse image. For any other inverse-image element $x = \Psi_\alpha(y)$, where $y > 0$, (8) gives

$$\text{Pred}(x) \cup \{p\} = \Psi_\alpha[\text{Pred}(y)] \subseteq \Psi_\alpha[\Delta].$$

Hence every inverse-image element has all its predecessors in that image. $\square$

4.3 Exponent lengths and a finite picture

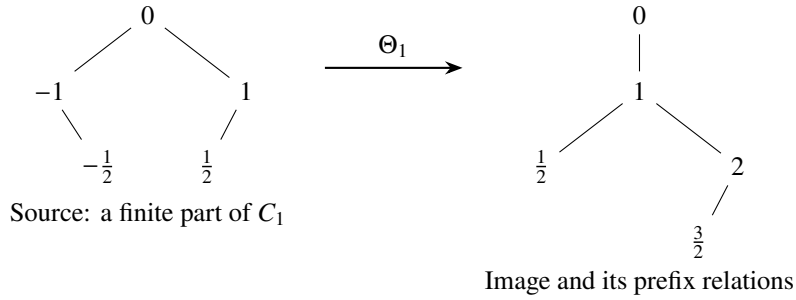
Proposition 4.8 (An exponent-birthday bound). *For every $x \in C_\alpha$,*

$$\ell(\Theta_\alpha(x)) \leq 1 +_{\text{ord}} \ell(x).$$

In particular, an infinite exponent length never increases.

Proof. At $x = p$ the image length is zero. In the outside case the length is exactly $1 + \ell(x)$. In the inside case, if $\ell(u) = \mu$, the source length is $\alpha + 1 + \mu$, and the image length is $1 + \alpha + \mu$. Monotonicity of ordinal addition gives $1 + \alpha + \mu \leq 1 + \alpha + 1 + \mu$. Finally, $1 + \lambda = \lambda$ for any infinite ordinal λ : write λ as a nonzero limit ordinal followed by a finite tail. $\square$

This is a bound for *exponents*. It is not a theorem bounding the birthday of the entire transformed surreal normal form.



Here $-1 \mapsto 0$, $0 \mapsto 1$, $1 \mapsto 2$, $-\frac{1}{2} \mapsto \frac{1}{2}$, and $\frac{1}{2} \mapsto \frac{3}{2}$. The numerical labels are exponents, not the monomials with those exponents.

5 Strong transport of normal forms

Definition 5.1 (Bottom-normalized transport). For $\alpha \in \mathbf{On}$, $n \in \mathbb{N}$, $p = -\alpha$, and $c = 2^{-n}$, define

$$\mathcal{N}_{\alpha,n} \left(\sum_{\gamma \geq p} r_\gamma \omega^\gamma \right) = c^{-1} r_p + \sum_{\gamma > p} r_\gamma \omega^{\Theta_\alpha(\gamma)}. \quad (12)$$

A coefficient not in the support is interpreted as zero. The inverse is

$$\mathcal{N}_{\alpha,n}^{-1} \left(s_0 + \sum_{\delta > 0} s_\delta \omega^\delta \right) = c s_0 \omega^p + \sum_{\delta > 0} s_\delta \omega^{\Psi_\alpha(\delta)}. \quad (13)$$

Theorem 5.2 (Ambient transport). *The formulas above define inverse real-linear order-isomorphisms*

$$\mathcal{N}_{\alpha,n} : \mathcal{H}_{\mathbb{R}}(C_\alpha) \longrightarrow \mathcal{H}_{\mathbb{R}}(\mathbf{No}_{\geq 0}).$$

They preserve support cardinality and decreasing support order type, commute with every leading truncation, and preserve and reflect strong summability of set-indexed families. For every such strongly summable family,

$$\mathcal{N}_{\alpha,n} \left(\sum_{i \in I} x_i \right) = \sum_{i \in I} \mathcal{N}_{\alpha,n}(x_i). \quad (14)$$

For a nonzero x ,

$$\deg_H(\mathcal{N}_{\alpha,n}(x)) = \Theta_\alpha(\deg_H(x)), \quad \text{lc}(\mathcal{N}_{\alpha,n}(x)) = \begin{cases} c^{-1} \text{lc}(x), & \deg_H(x) = p, \\ \text{lc}(x), & \deg_H(x) > p. \end{cases} \quad (15)$$

Proof. An order-isomorphism takes a reverse-well-ordered support to a reverse-well-ordered support and preserves its order type. The support is still a set: it is the image of a set under a definable class function. Every coefficient is multiplied by a nonzero real number, so no support point is lost and no new point is created. These observations show that both displayed formulas are legitimate normal forms. Their inverse identities follow coefficient by coefficient from lemma 4.2.

Both addition of normal forms and the proposed map are coefficientwise, with the same fixed multiplier c^{-1} at the bottom point. Hence the map is real-linear. The greatest nonzero coefficient retains its sign, because all multipliers are positive. Thus the map and its inverse preserve order. This proves (15) as well.

A leading truncation is specified by an initial segment of the decreasing support enumeration. The same enumeration indices specify the corresponding truncation of the transported support. Applying the map therefore commutes with truncation, including at a limit index.

For a set-family (x_i) , one has the exact equality

$$\bigcup_i \text{supp}(\mathcal{N}_{\alpha,n}(x_i)) = \Theta_\alpha \left[\bigcup_i \text{supp}(x_i) \right].$$

Reverse well-ordering of these unions is equivalent in both directions. For each γ , the index set of family members contributing at γ is exactly the index set contributing at $\Theta_\alpha(\gamma)$ after transport. The point-finiteness requirement is therefore also equivalent. When it holds, coefficient extraction reduces (14) to a finite real sum, through which the fixed coefficient multiplier commutes. This proves the entire assertion without invoking convergence in an order topology. $\square$

Remark 5.3 (No implicit completion of a subgroup). The ambient Hahn spaces in theorem 5.2 are full spaces. A particular initial subgroup A need not be closed under all strong sums or all infinite subseries. We define its image as the actual set or class $\mathcal{N}_{\alpha,n}[A]$, not as the space of all series with the same coefficient groups. If a strongly summable family in A has sum outside A , (14) still holds in the ambient spaces, but is not an assertion of an internal summation operation on A .

Remark 5.4 (What is and is not preserved). The map preserves the ordered chain of Archimedean classes, with their exponents reindexed by Θ_α . It is not a homomorphism of additive *exponent* groups. Consequently (15) should not be described as preservation of a multiplicative valuation with the same value-group structure.

6 Proof of omnific normalization

We now prove Theorem A. Let A be as there, and obtain α, n, p, c from proposition 3.2. Write

$$\Gamma = \Gamma(A), \quad \Delta = \Theta_\alpha[\Gamma], \quad H = \mathcal{N}_{\alpha,n}[A].$$

The ambient transport restricts to an ordered-group isomorphism onto H . It remains to prove that this actual image is an initial subgroup of $\mathbf{Oz}$.

Lemma 6.1 (Transported coefficients). *The group H is truncation closed and cross-sectional on Δ , and its coefficient groups are*

$$H_0 = \mathbb{Z}, \quad H_{\Theta_\alpha(\gamma)} = A_\gamma \quad (\gamma > p). \quad (16)$$

Its exponent class is exactly Δ .

Proof. Truncation closure follows from theorem 5.2. For $\gamma > p$, one has $\mathcal{N}_{\alpha,n}(\omega^\gamma) = \omega^{\Theta_\alpha(\gamma)}$. At the bottom, $\mathcal{N}_{\alpha,n}(c\omega^p) = 1$. Thus all the leaders on Δ belong to H . No image element can have a support point outside Δ .

The formula at a positive image exponent leaves its coefficient unchanged, and the inverse formula has the same property. This proves both inclusions in the second identity of (16). At zero the coefficient group is $c^{-1}A_p = \mathbb{Z}$. $\square$

Proof of Theorem A. By corollary 4.6, Δ is initial with minimum zero. By lemma 6.1, the image is truncation closed and cross-sectional, and all its coefficient groups are initial real subgroups.

Consider an obligation in the dyadic right-ancestor condition for H : let $\delta, \eta \in \Delta$ and $\eta \in R_s(\delta)$. The point δ cannot be zero. Write $\delta = \Theta_\alpha(\gamma)$ with $\gamma > p$. By (9), there exists $\xi \in R_s(\gamma)$ with $\eta = \Theta_\alpha(\xi)$. Since $\xi > \gamma > p$, its coefficient group was not rescaled. The initiality of A gives

$$\mathbb{D} \subseteq A_\xi = H_\eta.$$

Every target obligation is therefore satisfied. The concrete sufficiency in imported fact 2.2 proves that H is initial in **No**.

All exponents of H are nonnegative and its zero coefficient is an integer. Hence $H \subseteq \mathbf{Oz}$ by (4). Its minimum positive element is 1, either by order-isomorphism or directly from its normal forms. Finally, $N_{\alpha,n}(\varepsilon) = 1$, and all support and summability claims are the restrictions of theorem 5.2. Uniqueness of α, n was established in proposition 3.2. $\square$

Corollary 6.2 (The cited existence question). *Every discrete initial subgroup of **No** is isomorphic, as an ordered abelian group, to an initial subgroup of **Oz**.*

Remark 6.3 (An initial image is not a simplicity isomorphism). The conclusion is exactly an ordered-group isomorphism to an initial image. For example, on $A = \frac{1}{2}\mathbb{Z}$ the map $N_{0,1}$ sends 1 to 2 and $1/2$ to 1. Although $1 \prec 1/2$, one does not have $2 \prec 1$. Thus the normalization is not generally an isomorphism of the full simplicity-enriched structures. Any initial cyclic subgroup of **Oz** is $\mathbb{Z}$, and the order-isomorphism from $\frac{1}{2}\mathbb{Z}$ to $\mathbb{Z}$ is forced. Thus this example rules out a generally simplicity-preserving version. The published question does not demand that stronger conclusion.

7 Exact images, extremal groups, and admissible scales

7.1 The compulsory dyadic spine

Lemma 7.1 (The spine group is initial). *For every ordinal α , the group K_α of (1) is initial in **No**, has least positive element 1, and lies in **Oz**. Its exponent class is Λ_α .*

Proof. The proper prefixes of q_β are 0 and the q_η with $\eta < \beta$. Thus Λ_α is initial. The finite-support group on this class is cross-sectional and truncation closed. Its coefficient group is $\mathbb{Z}$ at zero and $\mathbb{D}$ at every other point. Any right ancestor lies among those positive spine points, where the required dyadic containment holds. Apply imported fact 2.2. The remaining claims follow from its normal forms. $\square$

Proposition 7.2 (Necessary image data). *If A has least positive element $2^{-n}\omega^{-\alpha}$ and is initial, then $K_\alpha \subseteq N_{\alpha,n}[A]$.*

Proof. The image contains 1. For each $\beta < \alpha$, corollary 3.3 gives $\mathbb{D}\omega^{-\beta} \subseteq A$. These terms are above the bottom exponent, so their coefficients are unchanged, and $\Theta_\alpha(-\beta) = q_\beta$. Thus the image contains every generating summand of K_α and hence all their finite sums. $\square$

Proof of Theorem B. Necessity has just been proved. Conversely, let $H \subseteq \mathbf{Oz}$ be initial with $K_\alpha \subseteq H$, and put $A = N_{\alpha,n}^{-1}[H]$. Since H contains 1, it has least positive element 1 and $H_0 = \mathbb{Z}$. Let $\Delta = \Gamma(H)$. The inclusion of K_α implies $\Lambda_\alpha \subseteq \Delta$, so $\Gamma = \Psi_\alpha[\Delta]$ is initial by proposition 4.7. Its minimum is p .

By ambient transport, A is an additive, truncation-closed Hahn subgroup. Its bottom coefficient group is $c\mathbb{Z}$, which contains 1 since $c = 2^{-n}$; all other coefficient groups are inherited from H . It is therefore cross-sectional and has initial real coefficient groups.

We check the source right-ancestor condition. If $x \in \Gamma$ satisfies $x > p$ and $y \in R_s(x)$, then $\Theta_\alpha(y) \in R_s(\Theta_\alpha(x))$. The coefficient group at y is unchanged from that at $\Theta_\alpha(y)$, so it contains $\mathbb{D}$ by the initiality of H . If instead $x = p$, its right ancestors are exactly the $-\beta$ for $\beta < \alpha$. The hypothesis $K_\alpha \subseteq H$ says that all coefficient groups at their images q_β contain $\mathbb{D}$; again these coefficients are unchanged. This checks the only obligations not already covered by (9).

The criterion [imported fact 2.2](#) now proves A initial. Its least positive element is $N_{\alpha,n}^{-1}(1) = c\omega^p$, since the inverse is increasing. The inverse identities show uniqueness, and a fixed bijection preserves and reflects inclusion of its images. This proves the correspondence. $\square$

7.2 Smallest and largest initial realizations at a fixed scale

Define

$$L_{\alpha,n} = c\mathbb{Z}\omega^p \oplus \bigoplus_{\beta < \alpha} \mathbb{D}\omega^{-\beta}, \quad (17)$$

$$M_{\alpha,n} = \{x \in \mathcal{H}_{\mathbb{R}}(C_\alpha) : [\omega^p]x \in c\mathbb{Z}\}. \quad (18)$$

Here $[\omega^p]x$ denotes coefficient extraction. The direct sum in (17) again permits only finite supports; the class in (18) permits every set reverse-well-ordered support in the cone.

Theorem 7.3 (Extremal groups). *The groups $L_{\alpha,n}$ and $M_{\alpha,n}$ are initial with least positive element $c\omega^p$. Every initial group A with that least positive element satisfies*

$$L_{\alpha,n} \subseteq A \subseteq M_{\alpha,n}.$$

Moreover,

$$N_{\alpha,n}[L_{\alpha,n}] = K_\alpha, \quad N_{\alpha,n}[M_{\alpha,n}] = \mathbf{Oz}, \quad M_{\alpha,n} = c\omega^p \mathbf{Oz}. \quad (19)$$

Proof. The first two identities in (19) follow directly from the formulas: every positive image coefficient is unrestricted in M , whereas precisely the finite dyadic spine terms occur for L . By [Theorem B](#), the inverse images of K_α and $\mathbf{Oz}$ are initial and have the required minimum. The lower bound on A follows from [corollary 3.3](#) and $A_p = c\mathbb{Z}$; the upper bound follows from [proposition 3.2](#).

For the last identity, ordinary multiplication by $c\omega^p$ sends a normal form on $[0, \infty)$ to one on $[p, \infty)$ by the arithmetic translation $\delta \mapsto p + \delta$. Its bottom coefficient belongs to $c\mathbb{Z}$, and the other coefficients range over all of $\mathbb{R}$ because multiplication by $c \neq 0$ is a bijection of $\mathbb{R}$. Conversely, translate any support in the cone back by $-p$ and divide its coefficients by c . The resulting series belongs to $\mathbf{Oz}$. $\square$

The equality $M_{\alpha,n} = \varepsilon \mathbf{Oz}$ does not make the theorem trivial. It concerns the *largest* group, whose higher coefficients are all real and whose support class is the full cone. Arithmetic translation can destroy initiality for smaller initial subgroups, even though it behaves correctly for this maximal one.

Corollary 7.4 (Sharp size obstruction). *Every set-sized initial group A with least positive element $2^{-n}\omega^{-\alpha}$ satisfies*

$$|A| \geq \max(\aleph_0, |\alpha|).$$

For every ordinal α and every $n \in \mathbb{N}$ this bound is attained by $L_{\alpha,n}$. In particular, a countable initial group can have such a least positive element only when α is countable.

Proof. The group contains the countably infinite cyclic subgroup generated by its least positive member and contains distinct terms $\omega^{-\beta}$ for $\beta < \alpha$. This gives the lower bound. The finite-support sum defining $L_{\alpha,n}$ has countable coefficient groups and an index set of size $|\alpha| + 1$, giving exactly the displayed cardinality. Its initiality and minimum follow from [theorem 7.3](#). $\square$

This is a restriction on a specified initial realization. The ordinal α is not an invariant of the abstract ordered group under arbitrary isomorphisms: normalization itself changes the least positive element to 1.

7.3 Dyadic depth and the spectrum of inverse scales

Definition 7.5 (Dyadic-spine depth). Let $H \subseteq \mathbf{Oz}$ be a nonzero initial subgroup. If there exists an ordinal β such that $\mathbb{D}\omega^{q\beta} \not\subseteq H$, let $d(H)$ be the least such β . Otherwise write $d(H) = \mathbf{On}$ as a formal unbounded-depth marker, not as a surreal or a set ordinal.

Corollary 7.6 (Admissible scales). *If $d(H)$ is an ordinal, then for every $\alpha \in \mathbf{On}$ and every $n \in \mathbb{N}$,*

$$\mathcal{N}_{\alpha,n}^{-1}[H] \text{ is initial} \iff \alpha \leq d(H).$$

If $d(H) = \mathbf{On}$, every ordinal α is allowed. Whenever the inverse is initial, its least positive member is $2^{-n}\omega^{-\alpha}$. The parameter n imposes no further obstruction.

Proof. By [Theorem B](#), the condition is exactly that all $\mathbb{D}\omega^{q\beta}$ for $\beta < \alpha$ lie in H . This holds precisely through the first missing index, inclusively, and fails at all larger indices. In the unbounded case there is no missing index. $\square$

For example, $d(K_\alpha) = \alpha$, since the exponent q_α is absent while all the earlier required dyadic terms are present. Thus the endpoint in [corollary 7.6](#) can be a limit ordinal and must not be replaced with a strict inequality. For a set-sized H , a first missing index exists: the distinct monomials $\omega^{q\beta}$ cannot all belong to a set. The subgroup $\bigcup_{\alpha \in \mathbf{On}} K_\alpha$, interpreted as the class of all finite dyadic-spine sums with an integer constant, is an example with unbounded depth. It is initial by [imported fact 2.2](#), but is much smaller than $\mathbf{Oz}$, for example it does not contain ω^2 .

8 The quotient by the least positive cyclic subgroup

8.1 Convexity and an actual splitting

Fix A , p , c , and $\varepsilon = c\omega^p$ as before. Define

$$\pi_p : A \longrightarrow \mathbb{Z}, \quad \pi_p(x) = c^{-1}[\omega^p]x, \quad A^{>p} = \ker \pi_p.$$

Although its notation emphasizes the higher exponents, $A^{>p}$ also contains zero.

Lemma 8.1 (Bottom-layer decomposition). *The map π_p is a surjective additive homomorphism, $\pi_p(m\varepsilon) = m$, and*

$$A = A^{>p} \oplus \mathbb{Z}\varepsilon$$

as additive groups. The subgroup $\mathbb{Z}\varepsilon$ is convex. Every nonzero positive element of $A^{>p}$ dominates all integer multiples of ε .

Proof. Coefficient extraction is additive, and $A_p = c\mathbb{Z}$. This proves all assertions about π_p . Given $x \in A$, the term $\pi_p(x)\varepsilon$ belongs to A and subtraction produces an element of $A^{>p}$. The intersection of the two displayed summands is zero.

A nonzero element of $A^{>p}$ has greatest exponent strictly above p . Its absolute value therefore exceeds $m\varepsilon$ for every positive integer m . If $0 \leq x \leq m\varepsilon$ in A for some integer $m \geq 0$, its component in $A^{>p}$ must consequently be zero; thus $x \in \mathbb{Z}\varepsilon$. Translation and sign reversal prove convexity. $\square$

The summand $A^{>p}$ need not itself be initial as a concrete subclass of **No**. For example, with $A = \mathbb{Z} + \mathbb{Z}\omega$, it is $\mathbb{Z}\omega$, which omits the proper prefix 1 of ω . A second exponent operation is needed for an *initial* quotient realization.

8.2 Deleting a positive root

Every positive surreal exponent has a unique form $\langle + \rangle \frown v$. Define

$$\rho : \mathbf{No}_{>0} \longrightarrow \mathbf{No}, \quad \rho(\langle + \rangle \frown v) = v.$$

Unlike Θ_α , this map has no exceptional point inside its domain. It preserves comparison and preserves and reflects prefix relations between positive source points.

Lemma 8.2 (Positive-root deletion). *If $\Delta \subseteq \mathbf{No}_{\geq 0}$ is initial, then $E = \rho[\Delta \setminus \{0\}]$ is initial, possibly empty. For $\delta > 0$,*

$$R_s(\rho(\delta)) = \rho[R_s(\delta)].$$

Proof. If $v \prec \rho(\delta)$, then $\langle + \rangle \frown v \prec \delta$ is a positive prefix and lies in Δ . Its image is v , establishing initiality. Every proper prefix of $\delta = \langle + \rangle \frown u$ is either 0, which is smaller than δ , or $\langle + \rangle \frown v$ with $v \prec u$. Comparisons of the latter with δ are comparisons of v with u . This proves the right-ancestor identity. $\square$

Let $H = N_{\alpha,n}[A] \subseteq \mathbf{Oz}$ and

$$H^+ = \{h \in H : \text{ct}(h) = 0\}.$$

The superscript $+$ here means “strictly positive exponents,” not “positive elements.” To avoid any order-cone interpretation, define its transported group explicitly by

$$B = \left\{ \sum_{\delta > 0} s_\delta \omega^{\rho(\delta)} : \sum_{\delta > 0} s_\delta \omega^\delta \in H^+ \right\}. \quad (20)$$

Proposition 8.3 (An initial model of the quotient). *The class B in (20) is an initial additive subgroup of **No**. The normal-form map in that formula induces ordered-group isomorphisms*

$$A/\mathbb{Z}\varepsilon \cong H/\mathbb{Z} \cong H^+ \cong B.$$

Proof. The first isomorphism is induced by $N_{\alpha,n}$. Since constants of H are integers and all integers lie in H , every coset in $H/\mathbb{Z}$ has a unique representative with zero constant coefficient. The quotient ordering agrees with the ordering of those representatives, because a nonzero positive-exponent term dominates all integers.

The group H^+ is truncation closed and cross-sectional on $\Delta \setminus \{0\}$. Its coefficient groups at those points are the same as those of H . Reindexing by the order-isomorphism ρ produces a truncation-closed, cross-sectional subgroup with the same coefficient groups. Its exponent class is initial by lemma 8.2. The right-ancestor identity in that lemma transfers every dyadic obligation from H to B . Hence B is initial by imported fact 2.2. If there are no positive exponents, the group is $\{0\}$ and the same conclusion is immediate. $\square$

Proof of Theorem C. Combine lemma 8.1 and proposition 8.3. The explicit isomorphism sends $x \in A$ to

$$(\rho_* (N_{\alpha,n}(x) - \pi_p(x)), \pi_p(x)),$$

where ρ_* denotes the normal-form reindexing of (20). The higher component determines the sign whenever it is nonzero; otherwise the sign is the sign of the integer component. This is precisely the indicated lexicographic order. $\square$

Corollary 8.4 (Finite iteration). *Starting with an initial group, repeatedly quotient by its least positive cyclic subgroup whenever the current quotient is nonzero and discrete. At every finite stage reached in this way, the quotient has an explicit initial realization in **No**.*

Proof. Apply Theorem C at the first stage and then apply it to the initial realization just constructed at each subsequent stage. This is ordinary finite induction and makes no assertion at a limit stage. $\square$

Limit stages are treated in corollary 13.6, which rests on source 02: at every stage, limit stages included, the quotient has an explicit initial realization, it is the one above at finite stages, and the realizations at different stages are related by compression.

8.3 The converse construction: adjoining an integer bottom layer

There is also an explicit converse to the lexicographic decomposition. For an initial subgroup $B \subseteq \mathbf{No}$, including $B = \{0\}$, put

$$\sigma(\gamma) = \langle + \rangle \frown \gamma, \quad \sigma_* \left(\sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\sigma(\gamma)}, \quad J(B) = \mathbb{Z} \oplus \sigma_*[B].$$

The map σ_* changes exponents but not coefficients. Its image has no constant term, and each nonzero image element dominates every ordinary integer in absolute value.

Theorem 8.5 (Integer-bottom suspension). *The construction $B \mapsto J(B)$ gives an inclusion-preserving and inclusion-reflecting correspondence between initial subgroups of **No** and nonzero initial subgroups of **Oz**. Its inverse sends H to the initial group obtained by deleting the leading plus sign from every positive exponent of $\ker(\text{ct}|_H)$. Moreover, $J(B) \cong B \overrightarrow{\times} \mathbb{Z}$ as ordered abelian groups.*

Proof. For nonzero B , let $E = \Gamma(B)$. The exponent class of $J(B)$ is $\{0\} \cup \sigma[E]$. It is initial: the prefixes of $\langle + \rangle \frown \gamma$ are zero and $\langle + \rangle \frown v$ for $v \prec \gamma$. The group is truncation closed and cross-sectional. Its coefficient group at zero is $\mathbb{Z}$, while the coefficient group at $\sigma(\gamma)$ is B_{γ} . Every right ancestor of $\sigma(\gamma)$ has the form $\sigma(v)$ with $v \in R_s(\gamma)$, so its coefficient group contains $\mathbb{D}$. The criterion imported fact 2.2 proves initiality. For $B = \{0\}$, the construction is simply $J(B) = \mathbb{Z}$.

Conversely, if $H \subseteq \mathbf{Oz}$ is initial and nonzero, it contains $\mathbb{Z}$ and each constant coefficient of H is integral. Thus $H = \mathbb{Z} \oplus \ker(\text{ct}|_H)$. By proposition 8.3, deleting the initial plus sign on the positive exponents of that kernel gives an initial group B . Reattaching the sign recovers the kernel exactly, including its actual infinite-support membership conditions. Hence $J(B) = H$, and the two constructions are inverse. Inclusions are preserved and reflected because both operations are fixed reindexings of the nonconstant normal forms. Finally, $(b, m) \mapsto \sigma_*(b) + m$ is additive and has the stated lexicographic order. $\square$

Corollary 8.6 (A relative criterion for discrete ordered groups). *Let G be a nonzero discrete ordered abelian group with least positive member e . Then G admits an initial realization in **No** if and only if $G/\mathbb{Z}e$ admits an initial realization and the exact sequence*

$$0 \longrightarrow \mathbb{Z}e \longrightarrow G \longrightarrow G/\mathbb{Z}e \longrightarrow 0$$

splits as a sequence of abelian groups.

Proof. If G admits an initial realization, the least positive element is carried to the least positive element there. Apply Theorem C and transport its splitting and quotient realization back. For any discrete ordered group, $\mathbb{Z}e$ is convex: finite induction shows that $[0, me]$ consists of $0, e, \dots, me$ for $m \in \mathbb{N}$. Conversely, a group-theoretic splitting identifies G with $(G/\mathbb{Z}e) \overrightarrow{\times} \mathbb{Z}$ as an ordered group: a lift of a nonzero positive quotient element lies above every element of $\mathbb{Z}e$, by convexity. Choose an initial realization B of the quotient and apply theorem 8.5. $\square$

This criterion is relative: it reduces initial realizability to the quotient together with a splitting condition, rather than deciding initial realizability of every possible quotient.

No result of this section states that every convex quotient of every initial group has an initial realization; that was the extent of source 01. The cyclic least-positive case is special here because it has a normal-form coefficient projection and the exact root-deletion operation above. Source 02 removes the restriction: Theorem D realizes every convex factor, with the coefficient projection replaced by the difference of two leading truncations and root deletion replaced by sign-tree compression, and proposition 13.5 shows that for $\mathbb{Z}\varepsilon$ its construction is exactly the one of this section.

9 Surcomplex and linear-algebraic consequences

9.1 Complex normal forms and rectangular modules

Let $\mathbf{No}[i] = \mathbf{No} \oplus i\mathbf{No}$ with $i^2 = -1$. Combining the real and imaginary supports gives a normal form with complex coefficients and surreal exponents; the union of the two reverse-well-ordered supports is again reverse-well-ordered. Conjugation acts coefficientwise. Define the Gaussian omnific module

$$\mathbf{Oz}[i] = \{a + ib : a, b \in \mathbf{Oz}\}.$$

For an additive group $A \subseteq \mathbf{No}$, put $A[i] = A \oplus iA$. It is a $\mathbb{Z}[i]$ -module because multiplication by i sends (a, b) to $(-b, a)$.

Definition 9.1 (Rectangular initiality). A rectangular subclass $A + iA$ is coordinatewise initial if A is initial in $\mathbf{No}$. This is a deliberately coordinate-dependent notion; no canonical single sign-tree order on $\mathbf{No}[i]$ is being postulated.

Theorem 9.2 (Surcomplex additive normalization). *Let A be a nonzero discrete initial subgroup and $H = \mathcal{N}_{\alpha,n}[A]$. The map*

$$\mathcal{N}_{\alpha,n}^{\mathbb{C}}(a + ib) = \mathcal{N}_{\alpha,n}(a) + i\mathcal{N}_{\alpha,n}(b)$$

is a $\mathbb{Z}[i]$ -linear isomorphism

$$A[i] \longrightarrow H[i] \subseteq \mathbf{Oz}[i].$$

It commutes with conjugation and has a complex-linear extension to the full complex-coefficient Hahn spaces on C_α and $\mathbf{No}_{\geq 0}$. That extension preserves and reflects set-indexed strong summability and commutes with its sums. The image is coordinatewise initial. For fixed α, n , its exact possible rectangular images are the $H[i]$ with $H \subseteq \mathbf{Oz}$ initial and $K_\alpha \subseteq H$.

Proof. Additivity and multiplication by i commute with the coordinatewise map, so it is $\mathbb{Z}[i]$ -linear. The inverse is coordinatewise $\mathcal{N}_{\alpha,n}^{-1}$. Conjugation negates the imaginary coordinate, and additivity gives the asserted compatibility.

In complex normal forms, the map is exactly

$$\sum_{\gamma \geq p} z_\gamma \omega^\gamma \longmapsto c^{-1} z_p + \sum_{\gamma > p} z_\gamma \omega^{\Theta_\alpha(\gamma)}.$$

The coefficient multiplier c^{-1} is real. It commutes with every complex scalar and with conjugation. The support argument in theorem 5.2 is unchanged; there is no additional infinite sum of constants because strong summability remains point-finite. Coordinatewise initiality follows from the initiality of H , and the range assertion is exactly Theorem B applied to the real factor. $\square$

9.2 What linear systems can be transported

Corollary 9.3 (Exact transport of integral linear equations). *Let $M \in \mathbb{Z}^{r \times s}$, let $b \in A^r$, and apply $N_{\alpha,n}$ coordinatewise. Then*

$$Mx = b \iff MN_{\alpha,n}(x) = N_{\alpha,n}(b).$$

The corresponding statement holds for $M \in \mathbb{Z}[i]^{r \times s}$ and $b \in A[i]^r$ under $N_{\alpha,n}^{\mathbb{C}}$. In the real case, coordinatewise inequalities are preserved as well.

Proof. Every row is a finite integer or Gaussian-integer linear combination. The relevant module isomorphism commutes with each such combination, and its injectivity gives the reverse implication. The real order assertion follows from the ordered-group isomorphism. $\square$

The right-hand side is transported too. In general $N_{\alpha,n}(1) \neq 1$, so this corollary must not be rewritten as preservation of equations with every ordinary constant fixed. Nor does it apply to nonlinear polynomials, arbitrary surreal scalar matrices, norms, or inner products.

10 Examples and failure modes

10.1 The motivating two-level example

Ehrlich and Kaplan's example in the paragraph preceding their question is

$$A = \mathbb{D} + \mathbb{Z}\omega^{-1}, \quad \varepsilon = \omega^{-1}.$$

Our construction has $\alpha = 1$, $n = 0$, $\Theta_1(-1) = 0$, and $\Theta_1(0) = 1$. It gives

$$d + m\omega^{-1} \mapsto d\omega + m, \quad H = \mathbb{D}\omega + \mathbb{Z} = K_1.$$

Thus the proposed general construction specializes to their known map, rather than treating that map as a new result [1, Section 9]. The example and the map stand in Section 9 of the journal version as well, immediately before its Question 9.1.

10.2 Rescale the bottom coefficient, not the whole series

Consider

$$A = \tfrac{1}{2}\mathbb{Z} + \mathbb{Z}\omega.$$

The exponent class $\{0, 1\}$ is initial, there are no right-ancestor dyadic obligations, and its coefficient groups $\tfrac{1}{2}\mathbb{Z}$ and $\mathbb{Z}$ are initial. Hence A is initial by imported fact 2.2. Its least positive element is $1/2$.

Scalar multiplication by 2 yields $\mathbb{Z} + 2\mathbb{Z}\omega$, which omits ω despite containing 2ω . Equivalently, its coefficient group at exponent 1 is not initial in $\mathbb{R}$. By contrast,

$$N_{0,1}(a/2 + b\omega) = a + b\omega$$

has initial image $\mathbb{Z} + \mathbb{Z}\omega$. Only the bottom coefficient changes.

10.3 An exponent translation fails at the first infinite branch

Let

$$\Gamma = \{-1\} \cup \mathbb{N} \cup \{\omega\}$$

be an exponent class, and let A be the finite-support group with coefficient $\mathbb{Z}$ at -1 , coefficient $\mathbb{D}$ at 0 , and coefficient $\mathbb{Z}$ at every positive member of Γ . This exponent class is initial: -1 contributes the prefix 0 , and the ordinal ω contributes all finite ordinals. Its only right-ancestor obligation is at -1 , forcing the chosen dyadic group at zero. Thus A is initial, with least positive element ω^{-1} .

Multiplication by ω replaces the support class by

$$\Gamma + 1 = \{0, 1, 2, \dots\} \cup \{\omega + 1\}.$$

This class is not initial because it contains $\omega + 1$ but omits its prefix ω . The resulting group is not initial. In contrast, root relocation satisfies

$$\Theta_1(-1) = 0, \quad \Theta_1(k) = k + 1 \ (k < \omega), \quad \Theta_1(\omega) = \omega.$$

The last identity uses $\langle + \rangle \frown \langle + \rangle^\omega = \langle + \rangle^\omega$; it is precisely where ordinal concatenation differs from arithmetic translation. The correct image has exponent class $\mathbb{N} \cup \{\omega\}$, coefficient $\mathbb{Z}$ at zero, $\mathbb{D}$ at one, and $\mathbb{Z}$ at the other exponents. It is initial.

10.4 A genuine limit-support example

Take $\alpha = \omega$, $n = 0$, and consider the group of all series

$$x = \sum_{k < \omega} a_k \omega^{-k} + m \omega^{-\omega}, \quad a_k \in \mathbb{D}, \quad m \in \mathbb{Z}. \quad (21)$$

Zeros in this expression are ignored. The allowed support is a subset of the reverse-well-ordered sequence $0, -1, -2, \dots, -\omega$. The exponent class is initial, its ancestor coefficients are dyadic, and its bottom coefficient is integral, so the full group (21) is initial.

Since $\Theta_\omega(-k) = q_k = 2^{-k}$ for finite k and $\Theta_\omega(-\omega) = 0$, normalization gives the exact strong-sum identity

$$\mathcal{N}_{\omega,0}(x) = \sum_{k < \omega} a_k \omega^{2^{-k}} + m. \quad (22)$$

For example, when $a_k = 1$ for every k and $m = 1$, the source and image have support order type $\omega + 1$. The decreasing exponents $1, 1/2, 1/4, \dots, 0$ are a legitimate reverse-well-ordered support; no analytic limiting process is involved.

The finite-support minimal group $L_{\omega,0}$ is a proper subgroup of the group in (21). Its image K_ω consists only of finite sums of the positive terms appearing in (22), together with an integer constant. The full limit-support example therefore illustrates why the image cannot be reconstructed from its coefficient groups alone.

10.5 Inverse initiality needs the whole dyadic spine

Let $H = \mathbb{Z}$ and $\alpha = 1$, $n = 0$. The inverse formula produces

$$\mathcal{N}_{1,0}^{-1}[\mathbb{Z}] = \mathbb{Z} \omega^{-1},$$

which is not initial: 1 is simpler than ω^{-1} but is missing. Here $K_1 \not\subseteq H$, so Theorem B correctly rules out initiality.

There is also a purely coefficient-level obstruction. Let $H = \mathbb{Z} + \mathbb{Z}\omega$ and keep $\alpha = 1, n = 0$. Its exponent class contains $\Lambda_1 = \{0, 1\}$, so the inverse exponent class is initial. Nevertheless, the inverse group is

$$\mathbb{Z} + \mathbb{Z}\omega^{-1},$$

which is not initial: its monomial ω^{-1} forces every dyadic constant through (3), but its constant group is only $\mathbb{Z}$. Equivalently, $1/2$ is a missing prefix of ω^{-1} . Thus the exponent condition in proposition 4.7 alone does not suffice for the group-level inverse.

10.6 Multiplication and ordinary constants are not preserved

For $A = \mathbb{D} + \mathbb{Z}\omega^{-1}$, one has $1 \in A$ and $\mathcal{N}_{1,0}(1) = \omega$. The product $1 \cdot 1$ also belongs to A , but

$$\mathcal{N}_{1,0}(1 \cdot 1) = \omega \neq \omega^2 = \mathcal{N}_{1,0}(1)\mathcal{N}_{1,0}(1).$$

Even on products that happen to stay in the source, multiplicativity fails. The surcomplex extension inherits this limitation. In particular, no preservation of the complex modulus or of multiplication by general surcomplex scalars follows from its Gaussian-linearity.

11 Convex subgroups as exponent cuts

This section and the next seven are source 02's (section 1.5). They extend section 8 from the least positive cyclic subgroup to every convex subgroup.

11.1 Conventions of the second source

All statements about arbitrary set-sized ordered groups can be read in ZFC. For the uniform class constructions we work, as in section 2, in NBG with global choice, allowing class parameters. A quotient of a proper-class group is not treated as a set of proper-class cosets. When such notation is used, the quotient is represented by the explicitly constructed normal-form complement. The class operations restrict to set operations on every individual support. In particular, none of the proofs sums a proper class of terms.

Signs, prefixes, Pred, R_s , normal forms, leading truncations, strong summability, G_γ and cross-sectionality are as in section 2. For two sign sequences x, y , let $x \wedge y$ be their longest common prefix; if one is a prefix of the other, their meet is the shorter one. Source 02 uses the Ehrlich–Kaplan criterion in exactly the concrete form of imported fact 2.2, including its proof that the canonical normal-form image is initial, and the description of initial real coefficient groups in imported fact 2.3; on an active exponent such a group is nonzero and contains 1. It imports one more result.

Imported fact 11.1 (Ehrlich's divisible embedding). *Every divisible ordered abelian group has an initial realization in No [7].*

This is Ehrlich's earlier embedding theorem, also recalled in [1]. It is used below only for set-sized groups. Neither source re-proves imported fact 2.2 or imported fact 11.1.

Remark 11.2 (Actual membership is part of the data). A truncation-closed subgroup need not contain every series allowed by its individual coefficient groups. In particular it need not be closed under arbitrary infinite subseries or all ambient strong sums. All transported groups below are actual images, not silently enlarged Hahn completions. This is the caution of remark 5.3, stated by both sources.

Several symbols of source 02 are renamed here, because the letters are already in use in sections 1 to 10. The renaming changes no statement.

here	in source 02	reason for the change
res_G	ρ_G	ρ is root deletion (lemma 8.2)
G^{div}	$D(G)$	$\mathbb{D}$ is $\mathbb{Z}[1/2]$
$\zeta, \zeta_* \in \widehat{\mathbb{Z}}; G_\zeta$	$\alpha, \alpha_*; G_\alpha$	α is the ordinal of the bottom exponent $-\alpha$
τ_*	p_*	$p = -\alpha$ is the bottom exponent
$W = \mathbb{Q}\omega^2 + \mathbb{Z}\omega + \mathbb{Q}$	K	K_α is the spine group
$\widetilde{G}_\zeta = G_\zeta \overrightarrow{\times} \mathbb{Q}$	H_α	H is the image in Theorem A
$C \subseteq C'$	$C_1 \subseteq C_2$	C_α is the cone $[-\alpha, \infty)$
$\Gamma_C, \Gamma^C = \Gamma \setminus \Gamma_C$	Λ_C, Δ	Λ_α is the spine exponent class
$C^\natural = G[\Gamma^C]$	H_C	as for H
b_E, ι_x	c_E, e_x	$c = 2^{-n}$; e is a least positive element
π_1	π	π_p is the bottom coefficient projection
Theorems D, E, F	Theorems A, B, C	source 01's letters are kept

Kept unchanged: e for the least positive element of an abstract discrete group (as in corollary 8.6); Π for the purely infinite normal forms, zero included (as in section 1.4); $J(B)$ for the suspension of theorem 8.5; “coordinatewise initial” of definition 9.1, which source 02 calls “rectangularly initial”. In sections 14 to 17 the letter n is an ordinary modulus, not the dyadic exponent of proposition 3.2. The profinite residue res_G is the residue res_I of the trigonometry report (`trigonometry:per:eq:residue`), and G^{div} its I^{div} (`trigonometry:per:prop:divisible`).

11.2 Exponent cuts and the canonical splitting

Throughout sections 11 to 13 let $G \subseteq \mathbf{No}$ be initial, with active exponent class $\Gamma = \Gamma(G)$. For an exponent subclass E define the ambient coefficient restriction

$$P_E(x) = \sum_{\gamma \in \text{supp}(x) \cap E} r_\gamma \omega^\gamma.$$

It is always an ambient normal form. Its membership in G needs proof.

Lemma 11.3 (Archimedean detection). *Let C be a convex subgroup of G and set*

$$\Gamma_C = \{\gamma \in \Gamma : \omega^\gamma \in C\}.$$

Then Γ_C is a numerical lower segment of Γ . For nonzero $x \in G$, with greatest exponent γ , the following are equivalent:

$$x \in C; \quad \gamma \in \Gamma_C; \quad \text{supp}(x) \subseteq \Gamma_C.$$

Proof. The numbers $|x|$ and ω^γ are Archimedean equivalent: each is bounded by a positive ordinary integer multiple of the other. Indeed their ratio has positive, nonzero real leading coefficient and an infinitesimal remainder. Convexity and subgroup closure therefore show that $x \in C$ exactly when $\omega^\gamma \in C$. If $\delta < \gamma$ lies in Γ , then $0 < \omega^\delta < \omega^\gamma$, so membership of the latter in C forces membership of the former. This proves the lower-segment assertion and the support equivalence. Zero belongs to every subgroup separately. $\square$

Lemma 11.4 (Restriction to a convex interval). *If E is a convex subclass of the ordered class Γ , then $P_E(G) \subseteq G$. More precisely, P_E is the difference of two leading-truncation projections. Its image equals*

$$G[E] := \{x \in G : \text{supp}(x) \subseteq E\}.$$

This subgroup is truncation closed and cross-sectional on E .

Proof. There is nothing to prove when E is empty. Otherwise set

$$U = \{\gamma \in \Gamma : (\exists \delta \in E) \delta \leq \gamma\}, \quad V = \{\gamma \in \Gamma : (\forall \delta \in E) \delta < \gamma\}.$$

Both are upper segments, $V \subseteq U$, and convexity gives $E = U \setminus V$. For every reverse-well-ordered support, restriction to an upper segment retains a leading segment. Truncation closure, with zero and the entire form admitted, gives $P_U(x), P_V(x) \in G$. Hence $P_E(x) = P_U(x) - P_V(x) \in G$. The image description follows from $P_E^2 = P_E$. Leading truncations of an element supported in E remain supported there, and every ω^γ with $\gamma \in E$ lies in the image. $\square$

Theorem 11.5 (Canonical splitting in a chosen realization). *For a convex subgroup C of G , put $\Gamma^C = \Gamma \setminus \Gamma_C$. Then*

$$C = G[\Gamma_C], \quad C^\natural := G[\Gamma^C], \quad G = C^\natural \oplus C.$$

The map $x \mapsto P_{\Gamma^C}(x)$ realizes G/C as the ordered subgroup $C^\natural$, and addition induces an ordered-group isomorphism

$$C^\natural \overrightarrow{\times} C \longrightarrow G, \quad (h, c) \longmapsto h + c.$$

Proof. The first equality is lemma 11.3. Since Γ^C is an upper segment, $P_{\Gamma^C}(x)$ is a leading truncation in G . The difference $x - P_{\Gamma^C}(x)$ is supported in Γ_C and belongs to C . The supports of the two summands are disjoint, proving uniqueness. If $h \neq 0$ and $c \in C$, the leading exponent of h exceeds every exponent of c . Consequently $h + c$ has the sign of h ; when $h = 0$ it has the sign of c . This is the lexicographic ordering. The kernel of P_{Γ^C} is C , so it gives the stated quotient model and the inclusion of $C^\natural$ is an order-preserving section. $\square$

For a discrete initial A and $C = \mathbb{Z}\varepsilon$ one has $\Gamma_C = \{p\}$ and $C^\natural = A^{>p}$, and theorem 11.5 is lemma 8.1 (see proposition 13.5).

Remark 11.6 (Which projection preserves order?). The upper projection P_{Γ^C} is nondecreasing. The lower projection P_{Γ_C} usually is not: for example, a positive element $\omega - 1$ has negative constant projection. Splitting the abelian-group sequence does not make both component projections positive maps.

Proposition 11.7 (Convex factors and commuting projections). *If $C \subseteq C'$ are convex in G , then $\Gamma_C \subseteq \Gamma_{C'}$ and $E = \Gamma_{C'} \setminus \Gamma_C$ is convex in Γ . Restriction induces an order-isomorphism*

$$C'/C \cong G[E].$$

For arbitrary exponent subclasses E, F the ambient identities are

$$P_E P_F = P_{E \cap F} = P_F P_E. \tag{23}$$

If E, F are convex intervals, all these maps preserve G .

Proof. The lower-segment descriptions give the interval assertion. Every $x \in C'$ splits uniquely into its E -part and its Γ_C -part. The first is a representative of the quotient, and its leading coefficient determines the sign of a nonzero coset. The identities (23) are coefficientwise. Intersections of convex intervals are convex, so lemma 11.4 handles membership. $\square$

These arguments already yield an intrinsic obstruction: an ordered abelian group with a nonsplit convex subgroup sequence cannot be initially realizable. The next step shows that the explicit complements and interval subgroups can themselves be given initial realizations.

12 Compression of a meet-closed sign tree

The central construction is elementary at the level of signs, but its retained positions must be indexed by their *order type*, not by ordinary integer subtraction.

Definition 12.1 (Retained-ancestor compression). A nonempty subclass $E \subseteq \mathbf{No}$ is *meet closed* if $x \wedge y \in E$ for all $x, y \in E$. For $x \in E$ let

$$I_E(x) = \{\beta < \ell(x) : x \upharpoonright \beta \in E\}, \quad \kappa_E(x) = \text{otp}(I_E(x)).$$

Let $\iota_x : \kappa_E(x) \rightarrow I_E(x)$ be the unique increasing enumeration. Define $b_E(x)$ by

$$\ell(b_E(x)) = \kappa_E(x), \quad b_E(x)(\xi) = x(\iota_x(\xi)). \quad (24)$$

Thus the sign at position β is retained exactly when its preceding prefix belongs to E .

The set $I_E(x)$ is a subset of the ordinal $\ell(x)$, even when E is a proper class. The definition is therefore elementwise legitimate.

Lemma 12.2 (Convex intervals are meet closed). *If Γ is initial in $\mathbf{No}$ and $E \subseteq \Gamma$ is a nonempty numerically convex subclass, then E is meet closed.*

Proof. If one of $x, y \in E$ is a prefix of the other, their meet is an endpoint. Otherwise, their common prefix $m = x \wedge y$ belongs to Γ by initiality. At the first differing sign, one continuation is minus and the other plus. Thus m lies strictly between x and y numerically, and convexity puts m in E . $\square$

Lemma 12.3 (Retained prefix formula). *If $y = x \upharpoonright \beta \in E$ with $\beta < \ell(x)$, then*

$$I_E(y) = I_E(x) \cap \beta.$$

Writing $\xi = \text{otp}(I_E(x) \cap \beta)$, one has

$$b_E(y) = b_E(x) \upharpoonright \xi, \quad \xi < \kappa_E(x), \quad \iota_x(\xi) = \beta.$$

Proof. The prefixes of y are exactly the prefixes of x of length less than β . Since $\beta \in I_E(x)$, it is the next retained position after those earlier positions. The signs in the two compressed sequences agree at all earlier retained positions by definition. $\square$

Theorem 12.4 (Compression theorem). *For nonempty meet-closed $E \subseteq \mathbf{No}$, the map $b_E : E \rightarrow \mathbf{No}$ is an order-isomorphism onto an initial subclass $E^b := b_E[E]$. For $x, y \in E$,*

$$y < x \iff b_E(y) < b_E(x), \quad (25)$$

$$b_E(x \wedge y) = b_E(x) \wedge b_E(y), \quad (26)$$

$$R_s(b_E(x)) = b_E[R_s(x) \cap E]. \quad (27)$$

It also preserves and reflects left-ancestor relations.

Proof. First suppose $y < x$. The retained prefix formula shows that $b_E(y)$ is a proper prefix of $b_E(x)$. At its first new position the compressed continuation has sign $x(\ell(y))$. Consequently the numerical comparison between y and x is unchanged.

Now suppose x, y are incomparable by prefix and put $m = x \wedge y$. Meet closure gives $m \in E$. The retained positions preceding $\ell(m)$ are identical for x and y , and the position $\ell(m)$ itself is retained in both. Their compressed sequences therefore have common prefix $b_E(m)$ and then opposite signs. The first differing sign has not changed. This proves order preservation, injectivity, and (26) in the

incomparable case. The comparable case of the meet identity follows from the retained prefix formula. These alternatives also prove the reverse implication in (25): incomparable points remain incomparable.

Let z be an arbitrary proper prefix of $b_E(x)$, of length $\xi < \kappa_E(x)$. Put $\beta = \iota_x(\xi)$ and $y = x \upharpoonright \beta$. By definition $y \in E$, and the retained prefix formula gives $b_E(y) = z$. Thus E^b is initial, at successor and limit prefix lengths alike. Combining this description of every prefix with order preservation proves (27); the left-ancestor assertion is identical with the inequality reversed. $\square$

Remark 12.5 (No unjustified limit step). A limit ordinal $\xi < \kappa_E(x)$ still has the position $\iota_x(\xi)$. The proof uses that position directly. It does not assume E is closed under limits of chains or that an infinite sequence of deletions converges to a surreal. This is why the construction applies to cuts without endpoints.

Proposition 12.6 (Uniqueness and nested coherence). *The map b_E is the unique isomorphism of the numerical and prefix orders from E onto an initial subclass of $\mathbf{No}$. If $F \subseteq E$ is nonempty and meet closed, then*

$$b_{b_E[F]}(b_E(x)) = b_F(x) \quad (x \in F). \quad (28)$$

Proof. In an initial image, the proper prefixes of the image of x correspond exactly to the proper ancestors of x in E , in their simplicity order. Hence the image length must be $\kappa_E(x)$. The sign at the position corresponding to $y \prec x$ is plus exactly when $y < x$. Both the positions and signs are forced; they agree with (24).

By (26), the subclass $b_E[F]$ is meet closed. In a second compression, a sign of $b_E(x)$ is retained exactly when its preceding prefix is $b_E(y)$ for some $y \in F$. By the retained prefix formula and prefix reflection, these are exactly the original positions β with $x \upharpoonright \beta \in F$. Thus the two-stage operation retains the same well-ordered positions, in the same order, as direct compression by F . This proves (28). $\square$

Example 12.7 (Compression is not exponent translation). Take the full finite sign tree of lengths at most two, in numerical order:

$$-- < - < -+ < 0 < +- < + < ++.$$

On the convex interval $E = \{-+, 0, +- \}$, compression gives

$$-+ \mapsto -, \quad 0 \mapsto 0, \quad +- \mapsto +.$$

The original exponents are $-1/2, 0, 1/2$, and their images are $-1, 0, 1$. On $E = \{+, ++ \}$ the images are $0, +$. The definition is a tree operation, not a uniform rule of subtracting a real or surreal endpoint.

Example 12.8 (Meet closure cannot be omitted). For $E = \{-, + \}$ neither point has a proper prefix in E . Both therefore compress to 0, so injectivity fails. This set omits the meet 0. Convexity in an initial exponent class prevents precisely this defect.

The compression theorem is compatible with the general viewpoint of surreal substructures and convex partitions in [5]. It does not assert that a set-sized convex interval is a copy of the whole proper-class structure $\mathbf{No}$; its target is an arbitrary initial subclass, often small.

13 Initial realizations of all convex factors

For the empty interval we set $E^b = \emptyset$, $\mathcal{F}_E(G) = \{0\}$, and let Φ_E be the unique map between the zero Hahn spaces. This convention also covers equal convex subgroups in the factor statements.

Theorem 13.1 (Hahn transport on an interval). *Let $G \subseteq \mathbf{No}$ be initial with exponent class Γ , and let $E \subseteq \Gamma$ be a nonempty convex interval. Define*

$$\Phi_E \left(\sum_{\gamma \in E} r_\gamma \omega^\gamma \right) = \sum_{\gamma \in E} r_\gamma \omega^{b_E(\gamma)}. \quad (29)$$

Then Φ_E is an ordered real-linear isomorphism $\mathcal{H}_{\mathbb{R}}(E) \rightarrow \mathcal{H}_{\mathbb{R}}(E^b)$. It restricts to an ordered-group isomorphism of $G[E]$ onto an initial subgroup $\mathcal{F}_E(G)$ of $\mathbf{No}$. It preserves support order types, leading coefficients, leading truncations, and strong summability in both directions.

Proof. By lemma 12.2 and theorem 12.4, the exponent map is an order-isomorphism with initial image. It therefore sends each set-sized reverse-well-ordered support to another such support, with the same order type. Its inverse has the same property. Coefficients are unchanged, so addition and real scalar multiplication commute with the map, and the leading coefficient retains its sign.

Set $B = \Phi_E[G[E]]$. By lemma 11.4, the source subgroup is cross-sectional and truncation closed. These properties pass to B , and its coefficient group at $b_E(\gamma)$ is exactly G_γ . The latter is initial in $\mathbb{R}$. Every right ancestor of $b_E(\gamma)$ has the form $b_E(\delta)$ with $\delta \in R_s(\gamma) \cap E$, by (27). The original dyadic obligation (3) supplies the required coefficient group. The sufficiency direction of imported fact 2.2 now shows that the actual image B is initial.

Leading truncations retain the same initial sets of support indices before and after reindexing. For a set-family (x_i) , the union of its transported supports is the image under b_E of the original union. Reverse well-ordering is equivalent in both directions. At any exponent, the set of indices contributing a nonzero coefficient is unchanged. Point-finiteness is therefore also equivalent. In a strongly summable family, every resulting coefficient is a finite real sum, and the transport commutes with that sum. $\square$

This is the analogue, for an interval and without a rescaled bottom coefficient, of theorem 5.2; the proof has the same structure.

Proof of Theorem D. Transport G to an actual initial realization. Apply proposition 11.7 to obtain $C'/C \cong G[\Gamma_{C'} \setminus \Gamma_C]$. If the exponent interval is empty, the factor is zero. Otherwise theorem 13.1 gives the required initial image. Splitting is theorem 11.5; the composition laws are (23) and (28). $\square$

Corollary 13.2 (Subgroups, quotients, and transfinite cuts). *The class $\mathbf{Init}$ is closed under convex subgroups, convex ordered quotients, and their subquotients. For a chain of convex subgroups $(C_\xi)_{\xi < \lambda}$ of an initially realizable group, any union or intersection which is a convex subgroup has the same conclusion for itself and for the corresponding quotient. No successor-stage restriction is necessary.*

Proof. The first assertions are special cases of Theorem D. A union of an increasing chain and an intersection of convex subgroups are convex. The theorem applies directly to their exponent cuts, rather than asserting that images previously chosen at earlier stages have a particular union or intersection. $\square$

Corollary 13.3 (Coherent compressed factors). *For nonempty convex intervals $F \subseteq E \subseteq \Gamma$, the direct map Φ_F agrees with first transporting by Φ_E and then compressing the interval $b_E[F]$. Thus the two constructions give the same normal forms, not just abstractly isomorphic groups.*

Proof. The coefficient at every retained exponent is unchanged in both constructions, and (28) identifies the final exponents. Actual membership follows from the restriction and image definitions. $\square$

Remark 13.4 (Strong sums and factor projection). The reindexing Φ_E preserves and reflects strong summability on its ambient space. The restriction P_E preserves strong summability but need not reflect it, because terms outside E are discarded. Neither assertion enlarges G to a group closed under all ambient strong sums.

13.1 Examples with an infinite exponent cut

Let $G = \mathbb{Q}[\omega]$ additively, meaning finite sums $\sum_{j=0}^m q_j \omega^j$. The exponent class $\mathbb{N}$ consists of all finite all-plus sequences and is initial. There are no right-ancestor obligations within this class, so the criterion shows that G is initial. The convex subgroup of degree less than k has quotient represented by degrees at least k ; compression sends the ordinal exponent $k + j$ to j . This recovers an initial quotient isomorphic to G .

For a genuinely transfinite example, take finite-support rational forms on the initial exponent class of all ordinals below $\omega + \omega$. The lower segment of finite exponents defines a convex subgroup. On the upper interval $[\omega, \omega + \omega)$, compression deletes the first ω ancestor positions and sends $\omega + n$ to n . The quotient again has an initial rational-polynomial realization. This calculation uses sign sequences, not an assumption that general surreal subtraction preserves initiality.

13.2 The cyclic case is root relocation followed by root deletion

The following statement, added in the merge, identifies source 02's construction for $C = \mathbb{Z}\varepsilon$ with the construction of section 8. Neither source states it.

Proposition 13.5 (Compression of the bottom point; [merge]). *Let A be a nonzero discrete initial subgroup, with $\alpha, n, p = -\alpha, c, \varepsilon = c\omega^p, \Theta_\alpha, N_{\alpha,n}, \rho, \pi_p, A^{>p}, H$ and B as in sections 3 to 6 and 8, and let $\Gamma = \Gamma(A)$ and $C = \mathbb{Z}\varepsilon$. Then $\Gamma_C = \{p\}$, $E := \Gamma^C = \Gamma \setminus \{p\}$, $C^\natural = A[E] = A^{>p}$, and*

$$b_E(\gamma) = \rho(\Theta_\alpha(\gamma)) \quad (\gamma \in E), \quad \text{that is,} \quad b_E(p \smallfrown \langle + \rangle \smallfrown u) = p \smallfrown u, \quad b_E(x) = x \quad (p \not\leq x). \quad (30)$$

Consequently $\Phi_E(P_E(x)) = \rho_(N_{\alpha,n}(x) - \pi_p(x))$ for every $x \in A$, the realization $\mathcal{F}_E(A)$ of $A/\mathbb{Z}\varepsilon$ is the group B of (20), and the splitting of theorem 11.5 for $C = \mathbb{Z}\varepsilon$ is that of lemma 8.1 and Theorem C.*

Proof. Since $\omega^p = 2^n \varepsilon \in \mathbb{Z}\varepsilon$ and no other monomial $\omega^\gamma, \gamma \in \Gamma$, lies in the convex subgroup $\mathbb{Z}\varepsilon$ (its positive members are the multiples of $c\omega^p$), $\Gamma_C = \{p\}$. By lemma 11.3, an element of A lies in $A[E]$ exactly when its support avoids p , that is, when $\pi_p(x) = 0$; so $A[E] = A^{>p}$.

First route: the formula. By proposition 3.2 every $x \in E$ lies in the cone C_α and $x > p$, so, as in section 4, either $p \not\leq x$ or $x = p \smallfrown \langle + \rangle \smallfrown u$. Every proper prefix of x lies in the initial class Γ , hence in E unless it equals p . If $p \not\leq x$, no prefix equals p , every position is retained and $b_E(x) = x = \rho(\langle + \rangle \smallfrown x) = \rho(\Theta_\alpha(x))$. If $x = p \smallfrown \langle + \rangle \smallfrown u$, the only prefix equal to p is $x \upharpoonright \alpha$, so exactly the plus sign at position α is deleted and $b_E(x) = p \smallfrown u = \rho(\langle + \rangle \smallfrown p \smallfrown u) = \rho(\Theta_\alpha(x))$. Positions are read as ordinal intervals, so limit lengths of u need no separate case.

Second route: uniqueness. On E , the map $\rho \circ \Theta_\alpha$ is an order-isomorphism (lemma 4.3, and ρ preserves comparison), it preserves and reflects prefixes (theorem 4.4 for points above p , and lemma 8.2), and its image $\rho[\Theta_\alpha[\Gamma] \setminus \{0\}]$ is initial (corollary 4.6 and lemma 8.2). By the uniqueness in proposition 12.6 it equals b_E .

For $x = \sum_\gamma r_\gamma \omega^\gamma \in A$, (12) gives $N_{\alpha,n}(x) - \pi_p(x) = \sum_{\gamma > p} r_\gamma \omega^{\Theta_\alpha(\gamma)}$, since the constant term of $N_{\alpha,n}(x)$ is $c^{-1}r_p = \pi_p(x)$. Deleting the root of each exponent gives $\sum_{\gamma \in E} r_\gamma \omega^{\rho \circ \Theta_\alpha(\gamma)}$, which is $\Phi_E(P_E(x))$ by (30). An element of H has zero constant coefficient exactly when it is the image of an element of $A^{>p}$, so $H^+ = N_{\alpha,n}[A^{>p}]$ and $B = \rho_*[H^+] = \Phi_E[A[E]] = \mathcal{F}_E(A)$. The two splittings have the same complement $A^{>p} = C^\natural$ and the same cyclic summand. $\square$

Thus the construction of source 02, applied to $C = \mathbb{Z}\varepsilon$, compresses away the bottom exponent and gives exactly source 01's $\rho \circ \Theta_\alpha$ on exponents, with the same coefficients. The two constructions also agree by the uniqueness of compression, and a finite computation at placement confirmed (30) on words of small length (section 27.5).

13.3 Limit stages of iterated bottom-layer removal

Corollary 8.4 (source 01) covers finite stages only. The following corollary, formulated in the merge from source 02's corollaries 13.2 and 13.3 and proposition 13.5, covers every stage.

Corollary 13.6 (Transfinite iteration; [merge]). *Let $A \subseteq \mathbf{No}$ be initial with exponent class Γ . Define subgroups C_ξ of A by recursion: $C_0 = \{0\}$; if A/C_ξ is nonzero and discrete, $C_{\xi+1}$ is the preimage in A of the least positive cyclic subgroup of A/C_ξ , and otherwise the recursion stops at ξ ; at a limit λ , $C_\lambda = \bigcup_{\xi < \lambda} C_\xi$. Put $E_\xi = \Gamma \setminus \Gamma_{C_\xi}$. Then, at every stage ξ reached, limit stages included:*

- (i) C_ξ is convex, and $x + C_\xi \mapsto \Phi_{E_\xi}(P_{E_\xi}(x))$ is an ordered-group isomorphism of A/C_ξ onto the initial group $\mathcal{F}_{E_\xi}(A)$;
- (ii) for finite ξ this is the realization produced by ξ applications of Theorem C, as in corollary 8.4;
- (iii) for $\xi < \eta$ with $E_\eta \neq \emptyset$, the realization at stage η is obtained from that at stage ξ by compressing the interval $\mathfrak{b}_{E_\xi}[E_\eta]$: $\Phi_{E_\eta} = \Phi_{\mathfrak{b}_{E_\xi}[E_\eta]} \circ \Phi_{E_\xi}$ on normal forms supported in E_η .

No hypothesis beyond the initiality of A is needed at a limit stage.

Proof. (i) The preimage of a convex subgroup under the order-preserving quotient map by a convex subgroup is convex, and the union of a chain of convex subgroups is a convex subgroup; so every C_ξ is convex, and proposition 11.7 and theorems 11.5 and 13.1 give the realization, with $\mathcal{F}_\emptyset(A) = \{0\}$ when E_ξ is empty. (iii) is corollary 13.3 for $E_\eta \subseteq E_\xi$.

(ii) By induction on finite k . For $k = 0$, $E_0 = \Gamma$ is initial, so every position is retained and Φ_Γ is the identity. Suppose the stage- k realization is $B_k = \mathcal{F}_{E_k}(A)$, an initial group with exponent class $\mathfrak{b}_{E_k}[E_k]$. The convex subgroup $C_{k+1}/C_k \cong \mathbb{Z}$ corresponds by proposition 11.7 to the interval $\Gamma_{C_{k+1}} \setminus \Gamma_{C_k}$, which is nonempty and cannot contain two points (their monomials would be Archimedean inequivalent in a cyclic group); so it is the single point $\min E_k$, and $E_{k+1} = E_k \setminus \{\min E_k\}$. The least positive cyclic subgroup of B_k is the image of C_{k+1}/C_k , with bottom exponent $\mathfrak{b}_{E_k}(\min E_k)$. By proposition 13.5, applied to B_k , Theorem C realizes $B_k/(\text{that subgroup})$ by compressing $\mathfrak{b}_{E_k}[E_k]$ minus its bottom point, which is $\mathfrak{b}_{E_k}[E_{k+1}]$. By (iii) the result is $\mathcal{F}_{E_{k+1}}(A)$. $\square$

This answers question 28.2: the limit stage needs no hypothesis, its quotient is realized directly from the cut Γ_{C_λ} rather than as a limit of earlier realizations, and it is coherent with every earlier stage. The recursion itself uses no choice: each least positive element is unique.

14 The residue obstruction for Presburger groups

14.1 The bottom cyclic group and coherent residues

Let G be a $\mathbb{Z}$ -group with least positive element e . The subgroup $\mathbb{Z}e$ is convex: finite induction shows that the interval $[0, me]$ contains exactly $0, e, \dots, me$ for every ordinary $m \in \mathbb{N}$ (as in the proof of corollary 8.6). Its quotient is torsion free, as every ordered abelian group is. In particular $\mathbb{Z}e$ is pure in G :

$$ny \in \mathbb{Z}e \implies y \in \mathbb{Z}e.$$

Lemma 14.1 (Residues). *For each ordinary $n \geq 1$ and $x \in G$ there is a unique $r_n(x) \in \{0, \dots, n-1\}$ such that $x - r_n(x)e \in nG$. The resulting homomorphisms $G \rightarrow \mathbb{Z}/n\mathbb{Z}$ are compatible and define res_G . Moreover*

$$\ker \text{res}_G = \bigcap_{n \geq 1} nG.$$

Proof. Divisibility of $G/\mathbb{Z}e$ gives $x = ny + ke$ for some $y \in G$ and $k \in \mathbb{Z}$. Reduce k modulo n . If two remainders worked, then $(r - r')e = ny$. Purity puts y in $\mathbb{Z}e$, so n divides $r - r'$ and the chosen range forces equality. Addition and compatibility under divisibility of moduli follow from the same uniqueness. The kernel identity is immediate from the zero remainder condition. $\square$

Lemma 14.2 (The divisible kernel, classical). *The subgroup $G^{\text{div}} := \ker \text{res}_G$ is divisible and is the largest divisible subgroup of G .*

Proof. Let $a \in G^{\text{div}}$ and fix $k \geq 1$. Choose $b \in G$ with $kb = a$. For every $n \geq 1$, the membership $a \in knG$ gives $c \in G$ such that $a = knc$. Torsion freeness yields $b = nc$, so $b \in nG$ for all n . Thus $b \in G^{\text{div}}$. Any divisible subgroup is contained in every nG , and therefore in G^{div} . $\square$

This is the divisible-kernel part of [10, Proposition 4.1 and Remark 4.2]; the elementary proof is included to make the subsequent equivalences explicit.

Proposition 14.3 (Split bottom versus ordinary residues). *For a $\mathbb{Z}$ -group G , the following are equivalent:*

- (i) $\mathbb{Z}e$ is a direct summand of G ;
- (ii) $\text{res}_G(G) = \mathbb{Z} \subseteq \widehat{\mathbb{Z}}$;
- (iii) $G = G^{\text{div}} \oplus \mathbb{Z}e$.

In case (iii), the order is $G^{\text{div}} \xrightarrow{\sim} \mathbb{Z}$. Every group complement to $\mathbb{Z}e$ equals G^{div} .

Proof. Assume $G = V \oplus \mathbb{Z}e$. The quotient map identifies V with $G/\mathbb{Z}e$, so V is divisible and $V \subseteq G^{\text{div}}$. For $v + me$ its coherent residue is m . Hence $\text{res}_G(G) = \mathbb{Z}$, and its kernel is exactly V .

Conversely, if every coherent residue is an ordinary integer, then for each x there is $m \in \mathbb{Z}$ such that $\text{res}_G(x) = m$. It follows that $x - me \in G^{\text{div}}$. The intersection $G^{\text{div}} \cap \mathbb{Z}e$ is zero, since no nonzero ordinary integer is divisible by every positive ordinary integer. Thus (iii) holds, and implies (i). Convexity of $\mathbb{Z}e$ gives the lexicographic ordering for any complement: a nonzero positive quotient lift is larger than every member of $\mathbb{Z}e$. The first argument also proves uniqueness of that complement. $\square$

14.2 Initial omnific realization in the split case

Recall $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ from (4) and section 1.4. The symbol Π denotes purely *infinite* forms, with zero included, not positive elements and not infinitesimals. Source 02 uses the integer-bottom suspension $J(B) = \mathbb{Z} \oplus \sigma_*[B]$ of theorem 8.5: for initial $B \subseteq \mathbf{No}$, $J(B)$ is an initial subgroup of $\mathbf{Oz}$ order-isomorphic to $B \xrightarrow{\sim} \mathbb{Z}$. It re-proves this from imported fact 2.2 by the argument of the first paragraph of the proof of theorem 8.5, so that Theorem E does not depend on source 01's separate normalization proof. The statement and proof are printed once, there.

Proof of Theorem E. If $G \in \text{Init}$, convex splitting (theorem 11.5) applied to $\mathbb{Z}e$ in an initial realization makes the bottom cyclic group a direct summand. Proposition 14.3 proves the algebraic and residue equivalences. Conversely, in the split case G^{div} is divisible and ordered. Choose an initial realization B of it by imported fact 11.1. The suspension $J(B)$ is an initial omnific realization of G . An initial subgroup of $\mathbf{Oz}$ is, by the meaning of initial here, an initial subgroup of $\mathbf{No}$. $\square$

Source 01's relative criterion gives the same answer by a second route. For a $\mathbb{Z}$ -group the quotient $G/\mathbb{Z}e$ is divisible, hence initially realizable by imported fact 11.1, so corollary 8.6 reduces initial realizability of G to the splitting of $\mathbb{Z}e$: the first and third conditions of Theorem E.

Corollary 14.4 (Rigidity on the initial side). *The only initially realizable rigid $\mathbb{Z}$ -group is $\mathbb{Z}$ itself.*

Proof. In $G^{\text{div}} \overrightarrow{\times} \mathbb{Z}$, the map $(d, m) \mapsto (2d, m)$ is an ordered-group automorphism, because G^{div} is divisible and torsion free. It is nontrivial unless $G^{\text{div}} = 0$. The standard group $\mathbb{Z}$ has no nonidentity order-preserving group automorphism. $\square$

15 An explicit factorial family of nonsplit models

15.1 Presentation inside an ordinary rational plane

Factorial moduli are cofinal for divisibility, so a profinite integer $\zeta \in \widehat{\mathbb{Z}}$ is specified by its compatible residues modulo $n!$. Let $S_n = S_n(\zeta)$ be the unique integer with

$$0 \leq S_n < n!, \quad S_n \equiv \zeta \pmod{n!}, \quad n \geq 1.$$

Then $S_1 = 0$ and

$$S_{n+1} = S_n + a_n n!, \quad a_n \in \{0, 1, \dots, n\}. \quad (31)$$

In the lexicographically ordered rational plane $\mathbb{Q} \overrightarrow{\times} \mathbb{Q}$, put

$$e = (0, 1), \quad x_n = \left(\frac{1}{n!}, \frac{S_n}{n!} \right), \quad G_\zeta = \bigcup_{n \geq 1} (\mathbb{Z}x_n \oplus \mathbb{Z}e). \quad (32)$$

The groups in the union are nested because

$$(n+1)x_{n+1} = x_n + a_n e. \quad (33)$$

Proposition 15.1 (Structure of the factorial family). *The group G_ζ is countable, has least positive element e , and first-coordinate projection π_1 gives an exact sequence of ordered groups*

$$0 \longrightarrow \mathbb{Z}e \longrightarrow G_\zeta \xrightarrow{\pi_1} \mathbb{Q} \longrightarrow 0. \quad (34)$$

Consequently it is a $\mathbb{Z}$ -group. Its first generator has residue $\text{res}_{G_\zeta}(x_1) = -\zeta$.

Proof. Each stage is a free rank-two group, and the union is countable. Its first-coordinate image is $\bigcup_n (n!)^{-1}\mathbb{Z} = \mathbb{Q}$. If $kx_n + me$ has first coordinate zero, then $k = 0$; hence the kernel is exactly $\mathbb{Z}e$. A positive element of nonzero first coordinate exceeds every integer multiple of e , and the positive elements of the kernel are $e, 2e, \dots$. This proves discreteness and convexity. The quotient is $\mathbb{Q}$, hence divisible. Finally, $n!x_n = x_1 + S_n e$, so $x_1 \equiv -S_n e \pmod{n!G_\zeta}$. Compatibility over factorial moduli gives the residue assertion. $\square$

Theorem 15.2 (Exact splitting test). *For $\zeta \in \widehat{\mathbb{Z}}$,*

$$G_\zeta \in \text{Init} \iff (34) \text{ splits} \iff \zeta \in \mathbb{Z}.$$

Proof. A retraction $r : G_\zeta \rightarrow \mathbb{Z}$ with $r(e) = 1$ is determined by the integers $b_n = r(x_n)$. The relations imply

$$(n+1)b_{n+1} = b_n + a_n, \quad n!b_n = b_1 + S_n. \quad (35)$$

Thus $b_1 + \zeta$ is zero modulo every $n!$, so $\zeta = -b_1 \in \mathbb{Z}$. Conversely, when $\zeta = m \in \mathbb{Z}$, put $b_n = (S_n - m)/n!$. These are integers satisfying all transition relations. On each stage define $r(kx_n + le) = kb_n + l$; compatibility makes this a well-defined retraction on the union. The equivalence with initial realizability follows from Theorem E. $\square$

15.2 A completely explicit noninteger parameter

Define

$$\zeta_* = \sum_{k=1}^{\infty} k! \quad \text{in } \widehat{\mathbb{Z}}, \quad S_n = \sum_{k=1}^{n-1} k!. \quad (36)$$

For each fixed modulus, all sufficiently late factorials vanish, so this is a well-defined profinite sum. It is not an ordinary convergent real series and is not a Hahn sum of omnific integers.

Lemma 15.3. *One has $0 \leq S_n < n!$, $S_n \rightarrow +\infty$, and $S_n/n! \rightarrow 0$ in the ordinary real sense. In particular $\zeta_* \notin \mathbb{Z}$.*

Proof. The first inequality is direct for $n = 1, 2$. If $S_n < n!$, then $S_{n+1} = S_n + n! < 2n! \leq (n+1)!$ for $n \geq 1$, with the small equality case harmless because the first inequality is strict. For $n \geq 3$, induction gives $S_n \leq 2(n-1)!$: it holds at $n = 3$, and

$$S_{n+1} \leq 2(n-1)! + n! \leq 2n! \quad (n \geq 2).$$

Hence $S_n/n! \leq 2/n$. Unboundedness follows from the summand $(n-1)!$. If ζ_* were a nonnegative ordinary integer m , its least residues modulo all sufficiently large $n!$ would be the fixed value m , contrary to unboundedness. If it were $-m$ with $m > 0$, those residues would be $n! - m$, whose ratio to $n!$ tends to one, contrary to the proved ratio limit. $\square$

For this parameter all digits a_n equal 1, and $(n+1)x_{n+1} = x_n + e$. The first generators are shown below. Thus the abstract obstruction has a presentation entirely in terms of pairs of ordinary rational numbers.

n	$n!$	S_n	x_n
1	1	0	(1, 0)
2	2	1	(1/2, 1/2)
3	6	3	(1/6, 1/2)
4	24	9	(1/24, 3/8)
5	120	33	(1/120, 11/40)
6	720	153	(1/720, 17/80)
7	5040	873	(1/5040, 97/560)

Remark 15.4 (Every finite retraction problem is solvable). Fix N . Taking $b_1 = -S_N$ makes $b_n = (S_n - S_N)/n!$ integral for every $n \leq N$ and satisfies every transition visible through stage N . There is nevertheless no single integer b_1 satisfying all stages. The obstruction is precisely the failure of this infinite compatibility problem to have a solution in $\mathbb{Z}$, although it has the profinite solution $-\zeta_*$.

15.3 Extension classes and rigidity

Proposition 15.5 (Classical extension parameter, explicit form). *There is an isomorphism $G_\zeta \rightarrow G_\eta$ inducing the identity on both endpoints of (34) if and only if $\zeta - \eta \in \mathbb{Z}$.*

Proof. Such a map must send e to e and x_n^ζ to $x_n^\eta + b_n e$ with $b_n \in \mathbb{Z}$. Comparing $n!$ times these elements gives

$$n!b_n = b_1 + S_n(\zeta) - S_n(\eta).$$

Thus $b_1 + \zeta - \eta = 0$ in $\widehat{\mathbb{Z}}$. Conversely, if this equality holds for an ordinary integer b_1 , the displayed formula defines integers b_n satisfying the transition relations. They define compatible isomorphisms on the stages and hence an isomorphism on the union; replacing b_n by $-b_n$ gives the inverse. $\square$

This is the concrete presentation behind the familiar parameter $\widehat{\mathbb{Z}}/\mathbb{Z}$. No new computation of the functor Ext is being claimed. For completeness, arbitrary extensions of $\mathbb{Q}$ by $\mathbb{Z}$ admit the same kind of presentation: choose lifts of $1/n!$, adjust their differences by integers, and collect the resulting compatible residues. Only the explicit family and its equivalence relation are needed here.

Theorem 15.6 (Continuum many countable rigid countermodels). *There are $2^{\aleph_0}$ pairwise nonisomorphic countable rigid $\mathbb{Z}$ -groups not in Init , among the G_ζ .*

Proof. The profinite integers have cardinality $2^{\aleph_0}$, and each coset modulo $\mathbb{Z}$ is countable. Proposition 15.5 therefore gives continuum many endpoint-marked extension classes. An ordered-group isomorphism must preserve the least positive element and induces an order-preserving automorphism of $\mathbb{Q}$. Every such automorphism is multiplication by an element of $\mathbb{Q}_{>0}$, a countable set of choices. Thus one ordered-group isomorphism class can account for at most countably many of the marked extension classes. There remain continuum many ordered-group types, with only the zero extension class split.

Let $\zeta \notin \mathbb{Z}$. If G_ζ^{div} were nonzero, its image in $\mathbb{Q}$ would be a nonzero divisible subgroup, hence all of $\mathbb{Q}$. The intersection $G_\zeta^{\text{div}} \cap \mathbb{Z}e$ is zero. Consequently the restriction of π_1 would be an isomorphism $G_\zeta^{\text{div}} \rightarrow \mathbb{Q}$, providing a section and contradicting theorem 15.2. Hence $G_\zeta^{\text{div}} = 0$ and res_{G_ζ} is injective. Every ordered-group automorphism preserves e and all its residues, so injectivity of the residue map forces that automorphism to be the identity. $\square$

The final argument is the residue-injectivity rigidity mechanism studied in [10]. Here it also supplies a particularly explicit family on the noninitial side of Theorem E.

16 Elementary equivalence, dense models, and omitted types

16.1 Initial realizability is not elementary

Corollary 16.1. *Initial realizability is not preserved by elementary equivalence, even among countable $\mathbb{Z}$ -groups. It is not first-order axiomatizable as a property of ordered abelian groups.*

Proof. The standard group $\mathbb{Z}$ is initial. Every G_ζ is a $\mathbb{Z}$ -group, hence elementarily equivalent to $\mathbb{Z}$ by the characterization of Presburger models [10]. For $\zeta \notin \mathbb{Z}$ it is not initially realizable. A first-order axiomatizable class is closed under elementary equivalence. $\square$

The same contrast already occurs with the explicit initial nonstandard $\mathbb{Z}$ -group $\mathbb{Q}\omega + \mathbb{Z}$, order-isomorphic to $\mathbb{Q} \overrightarrow{\times} \mathbb{Z}$. It has the same Presburger theory as every nonsplit G_ζ .

Theorem 16.2 (Dense elementary-equivalence counterexample). *For $\zeta \notin \mathbb{Z}$, let $\widetilde{G}_\zeta = G_\zeta \overrightarrow{\times} \mathbb{Q}$. Then $\widetilde{G}_\zeta$ is countable and densely ordered, is not initially realizable, and is elementarily equivalent as an ordered group to the initial subgroup*

$$W = \mathbb{Q}\omega^2 + \mathbb{Z}\omega + \mathbb{Q} \subseteq \mathbf{No}.$$

Proof. The final $\mathbb{Q}$ layer is convex and densely ordered. Between two elements with distinct upper coordinates one can increase the lower coordinate of the smaller element; with equal upper coordinates one uses density of $\mathbb{Q}$. Thus the whole group is densely ordered. The quotient by the bottom copy of $\mathbb{Q}$ is G_ζ . If $\widetilde{G}_\zeta$ were initially realizable, convex-factor closure would make G_ζ initially realizable, a contradiction.

Since $G_\zeta \equiv \mathbb{Q} \overrightarrow{\times} \mathbb{Z}$, adjoining a disjoint second sort interpreted as the fixed ordered group $\mathbb{Q}$ preserves elementary equivalence. This can be verified by induction on formulas: formulas in disjoint sorts reduce to Boolean combinations of formulas in the individual sorts. The Cartesian product with

coordinatewise addition and lexicographic order is uniformly interpretable in that two-sorted structure. It follows that

$$\tilde{G}_\zeta \equiv (\mathbb{Q} \vec{\times} \mathbb{Z}) \vec{\times} \mathbb{Q} \cong W.$$

Finally, the exponent class $\{0, 1, 2\}$ is initial and has no right ancestors internally, and the coefficient groups $\mathbb{Q}, \mathbb{Z}, \mathbb{Q}$ are initial real groups. Finite support gives truncation closure and cross-sectionality, so the criterion proves that W is initial. $\square$

16.2 A single recursive type witnessing the obstruction

With the computable integers S_n of (36), consider the partial type in one variable

$$\tau_*(X) = \{X \equiv S_n e \pmod{n!G} : n = 1, 2, \dots\}. \quad (37)$$

Each congruence abbreviates the first-order formula $\exists Y (X = n!Y + S_n e)$ in the ordered-group language with e named. The least positive element is itself definable, so naming it does not change the substantive distinction.

Theorem 16.3 (An omitted computable type). *The type τ_* is finitely satisfiable in every $\mathbb{Z}$ -group and omitted by every initially realizable $\mathbb{Z}$ -group. It is a recursive type under a fixed standard coding of these canonical formulas. Consequently no recursively saturated $\mathbb{Z}$ -group is initially realizable.*

Proof. A finite set of congruences is satisfied by $S_N e$ for any sufficiently large N , because $S_N \equiv S_n \pmod{n!}$ for $n \leq N$. A realization of all congruences has coherent residue ζ_* . That residue is not in $\mathbb{Z}$, by lemma 15.3, whereas Theorem E requires the entire residue image to be $\mathbb{Z}$. Thus the type is omitted in every initially realizable $\mathbb{Z}$ -group.

The sequences $n!$ and S_n are computable. Using canonical integer numerals, the set of displayed formulas is decidable: read the modulus, decide whether it is a factorial, and check the corresponding residue, handling the duplicate modulus 1 at the start by our convention $n \geq 1$. Standard definitions of recursive saturation therefore apply. A recursively saturated model realizes every finitely satisfiable recursive type over finitely many parameters, contradicting the omission. $\square$

Corollary 16.4 (Countable saturation). *If a $\mathbb{Z}$ -group is countably saturated, its residue map is surjective onto $\widehat{\mathbb{Z}}$. In particular it is not initially realizable.*

Proof. For an arbitrary $\zeta \in \widehat{\mathbb{Z}}$, use its compatible residues $S_n(\zeta)$ instead of S_n in (37). The resulting countable type is finitely satisfiable by ordinary multiples of e . Countable saturation supplies a realization. Taking ζ_* yields the final assertion without any assumption on the continuum hypothesis. $\square$

Theorem 16.5 (Every nonprincipal sequential ultrapower fails). *Let G be any $\mathbb{Z}$ -group and U any nonprincipal ultrafilter on $\mathbb{N}$. Then $G^{\mathbb{N}}/U$ is not initially realizable.*

Proof. By Łoś's theorem it is a $\mathbb{Z}$ -group. In the ultrapower take $a = [(S_N e)_{N \geq 1}]_U$. For any fixed n , the congruence $S_N e \equiv S_n e \pmod{n!G}$ holds for all $N \geq n$, hence on a set in U . Therefore a realizes τ_* . The previous theorem rules out an initial realization. The argument uses only cofinite sets and does not invoke an ultrapower saturation theorem. $\square$

16.3 An obstruction for nonstandard Peano models

Theorem 16.6 (Nonstandard arithmetic cannot have an initial additive realization). *Let M be a nonstandard model of Peano arithmetic, and let $G(M)$ be the ordered abelian group completion of its additive monoid. Then $G(M)$ is not initially realizable in **No**.*

Proof. The ordinary division algorithm, as proved in Peano arithmetic, gives unique division with remainder by every positive standard integer. Consequently $G(M)$ is a $\mathbb{Z}$ -group with least positive element 1.

Choose a nonstandard $h \in M$. Peano arithmetic defines the factorial function and the primitive-recursive partial sums

$$F(h) = \sum_{k=1}^h k!.$$

The sum here is the internally coded finite sum in M , not an external infinite group sum. For each fixed ordinary n , arithmetic proves that $n!$ divides $k!$ whenever $k \geq n$, and therefore proves

$$h \geq n \implies F(h) \equiv S_n \pmod{n!}.$$

This follows by induction on the finite-sum length inside the model; all the displayed factorial and summation recursions are provably total in Peano arithmetic. Since h exceeds every standard n , $F(h)$ realizes the external type τ_* in $G(M)$. Theorem 16.3 excludes initial realizability. $\square$

The arithmetic infrastructure used here is standard; factorials and division can in fact be handled in much weaker arithmetic, as noted in [12]. The fragment is not optimized here. The conclusion is stronger than merely observing that the usual omnific multiplication does not make the positive cone of $\mathbf{Oz}$ a nonstandard Peano model: it excludes *any* such model having an initially realizable ordered additive group. [Added at the write of source 03: by corollary 20.11 the same conclusion holds for every nonstandard model of $\mathsf{I}\Delta_0$, by a different residue argument, and by proposition 20.12 it fails for open induction; see the note after question 28.7.]

Proof of Theorem F. The discrete family and rigidity are theorem 15.6 and Presburger elementary equivalence. The dense examples are theorem 16.2. The remaining assertions are theorems 16.3, 16.5 and 16.6. $\square$

17 Omnific and surcomplex consequences

17.1 The precise residue image of omnific integers

Proposition 17.1. *The additive group of $\mathbf{Oz}$ has least positive element 1, its purely infinite subgroup Π is divisible, and for every ordinary $n \geq 1$*

$$\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}, \quad \text{res}_{\mathbf{Oz}}(z) = \text{ct}(z) \in \mathbb{Z} \subseteq \widehat{\mathbb{Z}}.$$

The type τ_ is finitely satisfiable in $\mathbf{Oz}$ and has no realization there.*

Proof. The normal form (4) gives $z = y + m$ uniquely with $y \in \Pi$ and $m \in \mathbb{Z}$. Every nonzero purely infinite form dominates all real constants in absolute value, proving the least-positive claim. For a positive ordinary n , dividing the real coefficients of y by n leaves its support strictly positive. Thus Π is divisible and $z \equiv m \pmod{n\mathbf{Oz}}$. The kernel of the residue map to $\mathbb{Z}/n\mathbb{Z}$ is exactly $n\mathbf{Oz}$, proving both formulas. The final assertion follows by the same explicit residue argument as theorem 16.3. $\square$

This is a class-function statement; it does not require a set of all omnific integers or a set quotient by a proper-class subgroup. The finite quotient is represented directly by its ordinary remainder.

Corollary 17.2 (Unit-preserving omnific embeddings). *For a set-sized $\mathbb{Z}$ -group G with least positive element e , the following are equivalent: $G \in \mathsf{Init}$; there is an order-preserving additive embedding $f : G \rightarrow \mathbf{Oz}$ with $f(e) = 1$; there is any additive homomorphism $f : G \rightarrow \mathbf{Oz}$ with $f(e) = 1$.*

Proof. An initial omnific realization supplies the stated embedding and takes the least positive element to 1. Conversely, for any such homomorphism, $\text{ct} \circ f : G \rightarrow \mathbb{Z}$ is a retraction on $\mathbb{Z}e$. Thus the bottom cyclic group splits, and Theorem E applies. $\square$

In particular, the ordered additive completion of a nonstandard Peano model cannot even admit a unit-preserving additive embedding into $\mathbf{Oz}$, whether or not the image is initial. (Source 03 proves this for nonstandard models of $\mathbf{l}\Delta_0$, corollary 21.4.) There is no contradiction with ordinary ordered-group embeddings that move the distinguished unit: the copy $(a, b) \mapsto a\omega^2 + b\omega$ embeds each G_ζ into $\mathbf{Oz}$ but sends e to ω , not to 1.

[Added 3 October 2026, batch 87: two members of the factorial family reappear in the report docs/foundations-and-computation/polish-models-of-omnific-arithmetic, in its Parts VII and VIII (AI-assisted, unrefereed, not formalized). Its source 13 presents G_{ε_2} , with $\varepsilon_2 \in \widehat{\mathbb{Z}}$ the profinite integer whose 2-adic coordinate is 1 and whose odd coordinates are 0, as $\mathbb{Z}_{(2)} \times \mathbb{Z}[1/2]$ with unit $(1, 1)$ (pma:nsp:prop:nonsplit-core), and says so. Its source 14 builds G_{ζ_*} for the parameter (36), with the two coordinates exchanged, and proves proposition 15.1 and theorem 15.2 for it again with the estimate of lemma 15.3 (pma:atr:lem:D), without citing this report. Placing a countable nonsplit $\mathbb{Z}$ -group, with the discrete topology, below $\mathbb{R}$ or $\mathbb{Q}^{\mathbb{N}}$ gives uncountable perfect Polish models of Presburger arithmetic with continuous addition, locally compact or homeomorphic to Baire space, that do not split (pma:nsp:thm:countable-core-transfer). By corollary 17.2 none of them has a unit-preserving additive map to $\mathbf{Oz}$. Source 13 recalls this with a direct proof and a two-element certificate for its core; source 14 proves it again for the cone of $\mathbb{R} \overrightarrow{\times} G_{\zeta_*}$ (pma:atr:prop:surreal), and its non-unital embedding of that group into $\mathbf{Oz}$ restricts on G_{ζ_*} , after the exchange of coordinates, to the copy $(a, b) \mapsto a\omega^2 + b\omega$ above.]

17.2 Period groups of surcomplex phases

The trigonometry report studies the normalized period group $I(\Psi)$ of a phase Ψ on a field F (trigonometry:per:def:periodgroup). It is an additive integer part of F with least positive element 1, it sits in the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow I \rightarrow \Pi_F \rightarrow 0$ given by the principal part (trigonometry:per:thm:dictionary), and for set-sized F it is a $\mathbb{Z}$ -group (after trigonometry:per:prop:division). Its residue map res_I is the map res_G of lemma 14.1 for $G = I$, $e = 1$ (trigonometry:per:eq:residue), and its profinite defect $\partial_\Psi : \Pi_F \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ (trigonometry:per:thm:defect) is that residue taken on a lift and read modulo $\mathbb{Z}$. Source 02's criterion therefore decides which period groups are initially realizable. The statement is the merge's.

Corollary 17.3 (Initially realizable period groups; [merge]). *Let F be $\mathbf{No}$ or one of the set-sized fields of trigonometry:per:sec:phases, and let Ψ be a phase on F with normalized period group $I = I(\Psi)$, ordered as a subgroup of F . The following are equivalent:*

- (i) I is initially realizable;
- (ii) the profinite defect ∂_Ψ vanishes;
- (iii) the period sequence $0 \rightarrow \mathbb{Z} \rightarrow I \rightarrow \Pi_F \rightarrow 0$ splits.

In that case $I \cong \Pi_F \overrightarrow{\times} \mathbb{Z}$ as ordered groups; for $F = \mathbf{No}$ and for the initial fields $\mathbf{No}(\lambda)$ this is an order-isomorphism onto the canonical period group $\mathbf{Oz} \cap F$, which is initial.

Proof. (ii) $\Leftrightarrow$ (iii) is trigonometry:per:thm:split, conditions (1) and (2).

(i) $\Rightarrow$ (iii). The element 1 is the least positive element of I and $I \cap O_F = \mathbb{Z}$. An element of I outside $\mathbb{Z}$ is infinite, so its sign is the sign of its principal part; hence the principal part induces an ordered isomorphism $I/\mathbb{Z} \cong \Pi_F$. If I is initially realizable, the convex subgroup $\mathbb{Z}$ is a direct summand by theorem 11.5 (or by corollary 8.6), and the period sequence splits.

(iii) $\Rightarrow$ (i). A splitting gives $\mathcal{I} = V \oplus \mathbb{Z}$ with V mapped isomorphically onto Π_F by the principal part. By convexity of $\mathbb{Z}$ the order is lexicographic, and since the principal part preserves signs on V , the map $v + m \mapsto \text{pr}(v) + m$ is an order-isomorphism $\mathcal{I} \rightarrow \Pi_F \oplus \mathbb{Z}$, the latter ordered as a subgroup of F , which is $\Pi_F \overrightarrow{\times} \mathbb{Z}$. For $F = \mathbf{No}$ the target is $\Pi \oplus \mathbb{Z} = \mathbf{Oz}$, and for $F = \mathbf{No}(\lambda)$ it is $\mathbf{Oz} \cap F$; both are initial ($\mathbf{Oz}$ by (4), and an intersection of initial classes is initial). For a set-sized F , Π_F is a divisible ordered group, so it has an initial realization B by imported fact 11.1, and $J(B)$ realizes $\Pi_F \overrightarrow{\times} \mathbb{Z}$ (theorem 8.5).

Second route for set-sized F : $\mathcal{I}$ is a set-sized $\mathbb{Z}$ -group with $\text{res}_{\mathcal{I}}(\mathcal{I}) = q^{-1}(\partial_{\Psi}(\Pi_F))$, q the quotient map $\widehat{\mathbb{Z}} \rightarrow \widehat{\mathbb{Z}}/\mathbb{Z}$ (`trigonometry:per:thm:defect`); this is $\mathbb{Z}$ exactly when $\partial_{\Psi} = 0$, and Theorem E gives (i) $\Leftrightarrow$ (ii). $\square$

So a period group is initially realizable exactly when its profinite defect is zero. The corollary is additive: it says nothing about multiplicative closure of the periods, which the trigonometry report shows to be independent of the defect (`trigonometry:per:sec:tests`), and it does not make a noncanonical split period group equal to $\mathbf{Oz} \cap F$ as a subclass. The nonsplit period groups are further instances of the obstruction of Theorem E.

17.3 Rectangular surcomplex factors

Let $\mathbf{No}[i]$, $A[i]$ and coordinatewise initiality be as in section 9 (definition 9.1). This is an explicitly coordinate-dependent notion; no single canonical sign-tree order on $\mathbf{No}[i]$ is postulated.

Theorem 17.4 (Rectangular convex-factor transport). *Let $G \subseteq \mathbf{No}$ be initial and $C \subseteq C'$ convex in G . For the interval $E = \Gamma_{C'} \setminus \Gamma_C$, the map*

$$a + ib \mapsto \Phi_E(P_E a) + i\Phi_E(P_E b), \quad a, b \in C',$$

induces a $\mathbb{Z}[i]$ -linear isomorphism

$$C'[i]/C[i] \cong \mathcal{F}_E(G)[i].$$

It commutes with conjugation. The target is coordinatewise initial, and the ambient complex-coefficient Hahn transport preserves and reflects set-indexed strong summability.

Proof. The real factor map has kernel C in each coordinate. The induced map is therefore bijective on the rectangular quotient. Additivity and compatibility with $(a, b) \mapsto (-b, a)$ give $\mathbb{Z}[i]$ -linearity, and negating the imaginary coordinate gives conjugation compatibility. Initiality of the real target is theorem 13.1. A complex normal form is obtained by the union of the two real supports, still reverse well ordered. Reindexing changes neither that order nor the finite set of contributors at any exponent, so the strong-sum proof is the same coefficientwise proof as in the real case. $\square$

Proposition 17.5 (Gaussian omnific residues). *For each ordinary $n \geq 1$,*

$$\mathbf{Oz}[i]/n\mathbf{Oz}[i] \cong \mathbb{Z}[i]/n\mathbb{Z}[i], \quad \text{res}_{\mathbf{Oz}[i]}(a + ib) = \text{ct}(a) + i \text{ct}(b) \in \mathbb{Z}[i] \subseteq \widehat{\mathbb{Z}}[i].$$

Proof. Apply proposition 17.1 to both coordinates. Multiplication by i interchanges them with a sign, so the coordinate residue map is $\mathbb{Z}[i]$ -linear. Its kernel modulo n is $n\mathbf{Oz}[i]$. $\square$

These are additive and module-theoretic conclusions. The exponent compression need not respect addition of exponents, so it does not in general preserve products of normal forms. No field, ring, differential, or exponential equivalence is inferred from theorem 17.4.

18 Boundaries of the convex-factor theory

18.1 Why these conclusions do not classify every ordered group

Convex splitting is necessary for initial realizability, but no sufficiency theorem for that condition alone is proved here. Even a truncation-closed cross-sectional Hahn presentation must meet the coefficient and right-ancestor conditions. The proof of Theorem E succeeds because the remaining quotient is divisible and hence has an initial realization by an established theorem. For a general discrete group that quotient need not be divisible.

Arbitrary subgroup closure is false. In fact the rational-coordinate map $(a, b) \mapsto a\omega + b$ embeds every G_ζ into the initial group $\mathbb{Q}\omega + \mathbb{Q}$. For $\zeta \notin \mathbb{Z}$ this gives a concrete noninitially-realizable subgroup of a two-level initial group. The structural theorem requires convex factors or separately verified coordinate restrictions. Likewise, a convex subgroup may fail literal initiality in its original position. In the initial group $\mathbb{D} + \mathbb{Z}\omega^{-1}$ of section 10, the subgroup $\mathbb{Z}\omega^{-1}$ is convex and does not contain the simpler number 1. It nevertheless has the initial realization $\mathbb{Z}$, as the theorem predicts.

18.2 The full optimality question is still separate

Ehrlich and Kaplan prove an optimality theorem allowing class models, for theories between ordered abelian groups and divisible ordered abelian groups, and ask for a set-model counterpart (Section 8 of [1]: Theorem 6 and Question 1; in the journal version, Theorem 8.1 and Question 8.1 [2]). The explicit dense groups of theorem 16.2 are set-sized noninitial examples, but are not a construction uniform over every proper intermediate theory.

In particular, Presburger arithmetic asserts discreteness and is not a subtheory of divisible ordered abelian groups. Thus Theorem E and theorem 15.6 alone cannot answer their full question by simply substituting Presburger arithmetic for an intermediate theory. The dense variant in theorem 16.2 removes discreteness of the whole group, but still does not establish the required uniformity over all intermediate theories. Source 02 makes no claim to have resolved that published question in its entirety.

Source 02 answers none of Ehrlich and Kaplan’s questions. Question 2 (journal 9.1) is the subject of source 01 (corollary 6.2), and source 02 does not use that answer: from source 01 it takes only the suspension, which it re-proves. Question 3 (journal 9.2), the initial-monoid problem, is not touched. Question 1 (journal 8.1) is the one discussed here.

Remark 18.1 (The two formulations of the optimality theorem; [merge]). The arXiv version quantifies over theories T with $T_{\text{OAG}} \subseteq T \subseteq T_{\text{DOAG}}$; since $\text{Th}(W)$ proves non-divisibility, it is not such a theory. The journal version, read at placement in the authors’ text hosted on the second author’s web page, states Theorem 8.1(i) for theories in the language $\{<, +, 0\}$ containing the theory T_D of nontrivial densely ordered abelian groups, and Question 8.1 asks whether its “only if” part can be established in NBG with set models alone. In that formulation $\text{Th}(W)$ is in scope, and theorem 16.2 gives, for every theory $T \supseteq T_D$ in that language that holds in W , a countable model of T that is not initially realizable. This settles the set-model “only if” part for those theories only, as Ehrlich and Kaplan’s own example of a p -divisible group does for others; it gives no construction for every theory lacking full divisibility, and the question stays open in both versions.

18.3 Product and completion pitfalls

The group G_ζ for $\zeta \notin \mathbb{Z}$ has both convex kernel $\mathbb{Z}$ and quotient $\mathbb{Q}$ initially realizable, but is not itself initially realizable. Thus closure under convex factors has no unrestricted converse for extensions. The missing datum is the extension class.

By contrast, theorem 11.5 gives an actual complement whenever the middle group is initial. This

complement depends on the normal-form realization; a group automorphism can change a splitting in general. Only the Presburger split case has the uniqueness proved in proposition 14.3.

The maps P_E for infinitely many disjoint intervals may form an ambient strongly summable family when applied to one fixed normal form, but this does not identify an arbitrary initial group with the full Hahn product of its factors. Such an identification would reintroduce the unsupported completion assumption excluded in remark 11.2.

19 The standard cut in a single infinite interval

19.1 The third source

Sections 19 to 22 are source 03, *Standard Integers in a Single Infinite Interval: Residue Characters, Bounded Induction, and Omnific Arithmetic* (20 pages, 3 October 2026). Its subject is weak arithmetic of discretely ordered rings carrying an additive map to the ordinary integers. It does not mention initial subgroups or this report; it was placed here because, combined with Theorem E, it answers part of question 28.7. That combination is stated and proved at this write in section 20.5 (corollary 20.11 and proposition 20.12). Everything else in these four sections is source 03's, with the symbols renamed as in section 19.2.

Motivation, as source 03 gives it. Elliot Glazer's *A Topological Tennenbaum Theorem* [13] studies how much regularity a presentation of arithmetic can support. For uncountable Polish models of $\mathbb{Q} + \mathbb{I}\Delta_0 + \text{Exp}$ with Borel operations, neither operation is continuous [13, Theorem 2], and [13, Corollary 1] already excludes continuous addition with Borel multiplication for $\mathbb{Q} + \mathbb{I}\text{Open}$. Source 03 investigates an algebraic obstruction that arises *before* any choice of topology. Omnific constant extraction is an additive map to $\mathbb{Z}$ taking 1 to 1. Can the additive structure of a nonstandard model of bounded arithmetic map into such a ring while preserving the unit? The answer is no (theorem 20.4 and corollary 21.4). A related question asks whether syntactically bounded formulas must have low computational complexity in a computable nonstandard ordered ring. The answer is again no: one interval can contain the full difficulty of standard arithmetic truth (theorems 22.2 and 22.4). The obstruction concerns the existence of maps, not their regularity, and is complementary to Glazer's theorem rather than a strengthening of it (remark 20.8). Source 03 answers neither of Glazer's printed questions and does not discuss his speculation, printed after his Question 2, that an affirmative answer to his Question 1 would lead to a negative answer to Question 2. For Question 2 an affirmative construction is proposed in the collection's report docs/foundations-and-computation/polish-models-of-omnific-arithmetic, written from manuscripts of the same delivery (section 27.5).

What was already known. Source 03 separates a new proof or application from an established phenomenon, and this report narrows its claims further (section 27, Novelty). Jeřábek noted in 2011 that $\mathbf{Oz}$ satisfies open induction but fails stronger induction, even bounded-existential induction [18]. Enayat, Łętyk and Visser [14, Sections 2.5, 4.2 and 5.2] develop bounded definitions of the standard cut in related ordered rings, including Shepherdson's model, and use bounded interpretations in their study of restricted-complexity completions. In particular, their Theorem 2.9 states that the natural numbers are Δ_0 -definable in the nonnegative part of every *ordered Dorroh ring*, by the parameter-free formula “every $y \leq x$ is a sum of four squares of elements $\leq y$ ”. $\mathbf{Oz}$ is an ordered Dorroh ring, with witnessing ideal Π , and so is every ring considered below that carries a normalized integer character (remark 20.5). Hence a bounded definition of $\mathbb{N}$ in $\mathbf{Oz}_{\geq 0}$, and with it a failed bounded induction instance, were known before source 03; the repository itself has an *existential* definition of $\mathbb{Z}$ in $\mathbf{Oz}$ (odg:thm:standarddef) and failed existential induction instances (odg:thm:induction, odg:rem:ringinduction). Classical

work of Phillips [15] places the additive structure of nonstandard arithmetic, and four-square obstructions to proposed ring constructions, in a much older line of research.

Consequently neither “the standard integers are definable” nor “omnific integers fail induction” is presented as a discovery. What source 03 contributes is: the explicit dyadic formula Std^{dy} and its finite odd-factor certificate (theorem 19.4); its inductiveness under division by 2 alone, without any character (lemma 19.6), with the single-instance boundary (theorem 19.7) and the sharpness examples (proposition 19.8); the classification and vanishing of integer characters and its consequences (theorems 20.1 and 20.4, proposition 20.2, and corollaries 20.6, 20.7, 20.9 and 21.4); the single-bound quadratic compiler and the exact truth complexity (theorems 22.2 and 22.4 and corollary 22.5); and a correction, with a repair, of a uniqueness lemma in [14] (propositions 21.1 and 21.2). Priority for each exact formulation was not established by its literature search, and all novelty claims are proposed refinements subject to comparison with the wider literature.

A map of the results. The proofs have four largely independent layers. First, a finite odd-factor certificate rules out every nonstandard solution of a divisor-theoretic power-of-two predicate. Second, an elementary successor argument shows that the resulting standard-cut formula is inductive without using the constant-term map. Third, residue calculations turn arbitrary additive characters into normalized ones and explain why every nonstandard $\text{I}\Delta_0$ model is excluded. Finally, relativizing arithmetic to the standard cut yields the single-interval translation and exact computability statements.

Result	Scope and limitation
Dyadic detector, theorem 19.4	Division by 2, 3 and a normalized additive integer character suffice; no normal-form theory is needed.
Single induction boundary, theorem 19.7	One explicit bounded induction instance forces standardness in that class.
Additive obstruction, theorem 20.4 and corollary 20.6	No nonzero integer character for a nonstandard $\text{I}\Delta_0$ difference ring; no unit-preserving additive map into a ring retracting to $\mathbb{Z}$.
Initial realizability, corollary 20.11 and proposition 20.12 [write]	The additive completion of a nonstandard $\text{I}\Delta_0$ model is not initially realizable; open induction does not suffice.
Source correction, propositions 21.1 and 21.2	Distinct ideals with quotient $\mathbb{Z}$ refute unrestricted uniqueness; division by 2 repairs it.
Single-bound compiler, theorem 22.2	Every arithmetic sentence is translated effectively; this is not one universal truth formula.
Exact complexity, theorem 22.4	A quadratic-atomic, one-bound fragment in $\mathcal{A}_0 = \mathbb{Z} + X\mathbb{Q}[X]$ is many-one equivalent to true arithmetic.

19.2 Conventions of the third source

Source 03 works in ordinary set mathematics unless it explicitly discusses the full surreal class (section 21.6). Several of its symbols are renamed here, because the letters are in use elsewhere in this report or in the collection. The renaming changes no statement.

here	in source 03	reason for the change
χ, χ_R (integer character)	ρ	ρ is root deletion (lemma 8.2)
res_R	κ_R	it is res_G of lemma 14.1 for $G = (R, +)$, $e = 1$; κ_E is an order type in section 12
$R^{\text{div}} = \ker \chi$	D	$\mathbb{D}$ is $\mathbb{Z}[1/2]$; G^{div} as in lemma 14.2
$\text{Pow}^{\text{dy}}, \text{Std}^{\text{dy}}$	P, S	the Diophantine report's Std is a different, existential formula (odg:thm:standarddef)
Ω (positive infinite bound)	H	H is the image group in Theorems A and B
$\mathcal{A}, \mathcal{A}_0, \mathcal{A}_{\text{Sh}}, \mathcal{A}(k, \Gamma)$	$A, A_0, A_{\text{Sh}}, A(k, \Gamma)$	A is the initial group of Theorem A
$\chi_{\mathcal{A}} : \mathcal{A} \rightarrow \mathbb{Z}$ (retraction)	ϵ	ϵ is the least positive element in Theorem A
$j_c = (X - c)$	J_a	$J(B)$ is the suspension of theorem 8.5

Kept unchanged: M for a model of arithmetic (as in theorem 16.6); X for an indeterminate, always positive infinite (unrelated to the variable of the type $\tau_*(X)$ in (37)); ct for the constant coefficient of $\mathbf{Oz}$ (section 1.4); q for a quotient in a division (unrelated to the sign sequences q_β). In $\mathcal{A}(k, \Gamma)$, Γ is an ordered abelian group of Hahn exponents, not the exponent class $\Gamma(G)$ of section 11, and k is a coefficient field. The shipped records of source 03 use its own letters: the compiler report data/03-standard-cut-compiler_report.json writes the bound Ω as H . Statements added at this write, with complete proofs, are marked [write], as statements added in the merge of source 02 are marked [merge].

A *discretely ordered ring* is a commutative unital ordered ring R in which 1 is the least positive element. Its order is total and compatible with addition and multiplication. Such a ring is a domain of characteristic zero, and its standard copy of $\mathbb{Z}$ is understood. Write $R_+ = \{a \in R : a \geq 0\}$. The word *standard* always refers to the external copy of $\mathbb{N}$ or $\mathbb{Z}$; it is not an additional predicate in the language. An element $h \in R_+$ is *positive infinite* if $h > n$ for every ordinary $n \in \mathbb{N}$. All displayed first-order formulas use the language $\{0, 1, +, \cdot, <\}$ on R_+ ; numerals and $\leq$ are abbreviations. A formula is Δ_0 if every quantifier is bounded by a term not containing its quantified variable. Such a bound need not make the range externally finite. PA^- is the arithmetic of nonnegative cones of discretely ordered commutative rings, IOpen its extension by induction for quantifier-free formulas, and $\text{I}\Delta_0$ its extension by induction for all bounded formulas, with parameters.

19.3 Ordered rings, division, and external standardness

Lemma 19.1 (Finite intervals). *For each ordinary $n \in \mathbb{N}$, the interval $[0, n]$ in R consists exactly of $0, 1, \dots, n$. Thus every nonstandard nonnegative element is positive infinite.*

Proof. Use external induction on n . If $0 \leq a \leq n + 1$, then either $a < 1$, in which case $a = 0$, or $1 \leq a$ and $0 \leq a - 1 \leq n$. The induction hypothesis applies to $a - 1$. If a nonnegative element is not above all standard integers, it lies in one such finite interval and is standard. $\square$

This is the finite induction already used for $\mathbb{Z}$ -groups in section 14.

Definition 19.2 (Standard division and integer characters). For an ordinary integer $n \geq 1$, the condition ED_n says that every $a \in R$ has a representation

$$a = nq + r, \quad q \in R, \quad r \in \{0, \dots, n - 1\}.$$

R has *standard division* if it satisfies ED_n for every ordinary $n \geq 1$. A *normalized integer character* is an additive-group homomorphism $\chi : R \rightarrow \mathbb{Z}$ such that $\chi(1) = 1$.

The quotient and remainder in ED_n are unique. Indeed, two such representations would give $n(q - q') = r' - r$ with $|r' - r| < n$; discreteness excludes $q - q' \neq 0$. For $a \geq 0$, the quotient is

nonnegative. If a is positive infinite, its quotient by an ordinary positive integer is positive infinite. These observations also justify the bounds on witnesses below. With standard division, $(R, +)$ is a $\mathbb{Z}$ -group with least positive element $e = 1$ in the sense of section 1.5: $a = nq + r$ shows that $R/\mathbb{Z}$ is divisible.

A normalized character is *not* assumed to preserve order. For example, evaluation at zero on $\mathbb{Z} + X\mathbb{Q}[X]$, ordered with X positive infinite, takes the positive element $X - 1$ to -1 . Confusing an integer character with an order-preserving standard-part map would invalidate the argument.

Lemma 19.3 (Residues detected by a character). *Suppose χ is a normalized integer character and ED_n holds. If $a = nq + r$ is its division representation, then*

$$\chi(a) = n\chi(q) + r, \quad n \mid a \text{ in } R \iff n \mid \chi(a) \text{ in } \mathbb{Z}.$$

Proof. Additivity and normalization give the displayed identity. Its residue modulo n is r . If $n \mid \chi(a)$, then $r = 0$ and $a = nq$. The reverse implication is immediate from additivity. $\square$

19.4 An elementary dyadic detector

Define two parameter-free bounded formulas:

$$\text{Pow}^{\text{dy}}(y) :\iff 0 < y \wedge (\forall u \leq y)(\forall v \leq y)(y \neq (2u + 3)v), \quad (38)$$

$$\text{Std}^{\text{dy}}(x) :\iff x = 0 \vee (\exists y \leq x)(x < 2y \wedge \text{Pow}^{\text{dy}}(y)). \quad (39)$$

The first formula says that y has no positive odd divisor at least 3. Source 03 calls it *divisor-theoretic dyadicity*; there is no exponentiation symbol in the formula. The second asks for a dyadic element between $x/2$ and x , with zero handled separately.

Theorem 19.4 (Exact dyadic and standard-cut detection). *Let R be a discretely ordered ring satisfying ED_2 and ED_3 , and let $\chi : R \rightarrow \mathbb{Z}$ be a normalized integer character. Then*

$$\begin{aligned} \{y \in R_+ : R_+ \models \text{Pow}^{\text{dy}}(y)\} &= \{2^k : k \in \mathbb{N}\}, \\ \{x \in R_+ : R_+ \models \text{Std}^{\text{dy}}(x)\} &= \mathbb{N}. \end{aligned}$$

The powers and the sets on the right are external standard ones. Neither χ nor an infinity parameter occurs in the formulas.

Proof. For a standard positive integer y , all the quantified variables in (38) are standard by lemma 19.1. Elementary integer factorization therefore gives $\text{Pow}^{\text{dy}}(y)$ exactly when y is an ordinary power of two. Zero fails the first conjunct.

Now let y be positive infinite and put $m = \chi(y) \in \mathbb{Z}$. If $m = 0$, division by 3 and lemma 19.3 give $y = 3v$ with $0 < v \leq y$. Taking $u = 0$ contradicts $\text{Pow}^{\text{dy}}(y)$.

Suppose instead that $m \neq 0$. Let k be the ordinary 2-adic order of $|m|$: thus $m = 2^k m_0$, where m_0 is an odd, possibly negative, integer. Repeated division by 2, exactly k times, gives

$$y = 2^k b, \quad \chi(b) = m_0.$$

At each step the character is even until the last, so the remainder is zero. The element b is positive infinite. Because $\chi(b - 3) = m_0 - 3$ is even, another application of division by 2 gives

$$b = 2u + 3, \quad u \in R_+.$$

Consequently $y = (2u + 3)v$ with $v = 2^k$. Both witnesses meet the formula's bounds: $v \geq 1$, $2u + 3 \geq 3$, and their product is y , so each factor is at most y , and $u \leq 2u + 3 \leq y$. Thus $\text{Pow}^{\text{dy}}(y)$ fails.

This proves the first equality. For each ordinary $x \geq 1$, the greatest ordinary power $2^k \leq x$ satisfies $x < 2^{k+1}$, so it witnesses $\text{Std}^{\text{dy}}(x)$. Conversely, a witness to $\text{Std}^{\text{dy}}(x)$ is a standard power of two, hence $x < 2y$ places x in a standard finite interval. Zero satisfies Std^{dy} by definition. $\square$

Remark 19.5 (What makes the proof small). The nonstandard case uses no prime distribution, four-square theorem, factorization of a nonstandard integer, or total exponentiation inside the structure. Only an ordinary integer m is factored. The value m may be zero or negative, and both possibilities are explicitly covered. The factorization certificate is finite even when the element being factored is surreal.

19.5 Dyadic bracketing is inductive without a character

Lemma 19.6 (Doubling and successor). *In every discretely ordered ring satisfying ED_2 ,*

$$\text{Pow}^{\text{dy}}(1), \quad (\forall y)(\text{Pow}^{\text{dy}}(y) \rightarrow \text{Pow}^{\text{dy}}(2y)), \quad \text{Std}^{\text{dy}}(0), \quad (\forall x)(\text{Std}^{\text{dy}}(x) \rightarrow \text{Std}^{\text{dy}}(x+1))$$

hold in the positive cone.

Proof. A positive product $(2u+3)v$ is at least 3, proving $\text{Pow}^{\text{dy}}(1)$. Suppose $\text{Pow}^{\text{dy}}(y)$, but $2y = (2u+3)v$ for nonnegative bounded witnesses. Reduction modulo 2 shows that v is even: the standard remainder map $R \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a ring homomorphism, and the odd factor has remainder 1. Write $v = 2w$. Cancellation gives $y = (2u+3)w$. Since $y > 0$, one has $w \geq 1$, and therefore $u \leq y$ and $w \leq y$. This contradicts $\text{Pow}^{\text{dy}}(y)$.

The assertion $\text{Std}^{\text{dy}}(0)$ is immediate, and $y = 1$ witnesses $\text{Std}^{\text{dy}}(1)$. For $x > 0$, let $y \leq x < 2y$ witness $\text{Std}^{\text{dy}}(x)$. Discreteness gives $x+1 \leq 2y$. If $x+1 < 2y$, reuse y . If $x+1 = 2y$, use $2y$ as the new witness, invoking closure of Pow^{dy} under doubling. Its positivity gives $x+1 = 2y < 4y$. In either case $\text{Std}^{\text{dy}}(x+1)$ follows. $\square$

Theorem 19.7 (A single induction sentence detects standardness). *Under the hypotheses of theorem 19.4, the following are equivalent:*

- (i) R is its standard copy of $\mathbb{Z}$;
- (ii) $R_+ \models \forall x \text{Std}^{\text{dy}}(x)$;
- (iii) R_+ satisfies the one sentence

$$(\text{Std}^{\text{dy}}(0) \wedge \forall x(\text{Std}^{\text{dy}}(x) \rightarrow \text{Std}^{\text{dy}}(x+1))) \longrightarrow \forall x \text{Std}^{\text{dy}}(x); \quad (40)$$

- (iv) R_+ is a model of $\text{I}\Delta_0$.

Proof. By theorem 19.4, (ii) says exactly that every nonnegative element is standard. Negating negative elements then gives (i). By lemma 19.6, the antecedent in (40) always holds, so (ii) and (iii) are equivalent. Since Std^{dy} is bounded, (iv) implies (iii), while (i) implies (iv) because ordinary arithmetic satisfies induction. $\square$

Write $\text{Ind}(\text{Std}^{\text{dy}})$ for the sentence (40). The formula Std^{dy} has bounded existential–universal form after expanding its abbreviations: one bounded existential witness followed by two bounded universal witnesses. No minimality claim about this quantifier pattern is made. The result identifies an explicit boundary, rather than only invoking a known strict inclusion between induction fragments. Any parameter-free formula that defines exactly $\mathbb{N}$ in every ring of this class, such as the four-square formula of [14, Theorem 2.9] (remark 20.5), gives the same equivalence within the class; what is particular to Std^{dy} is that its antecedent holds by lemma 19.6 in *every* ring with ED_2 , whether or not it has a character. Corollary 20.9 uses exactly that.

19.6 Why both small divisors occur

The assumptions ED_2 and ED_3 cannot simply be omitted from theorem 19.4 while leaving the detector unchanged. For a prime p , set

$$R_{(p)} = \mathbb{Z} + X\mathbb{Z}[1/p][X],$$

ordered by the sign of the leading coefficient with X positive infinite. These are discretely ordered rings with the character $f \mapsto f(0)$, and they satisfy ED_p .

Proposition 19.8 (Two sharpness examples). *In $R_{(2)}$, the positive infinite element X satisfies Pow^{dy} . In $R_{(3)}$, the positive infinite element $X + 1$ satisfies Pow^{dy} . Thus division by 2 alone, or by 3 alone, does not suffice.*

Proof. Suppose first that $X = (2u + 3)v$ in $R_{(2)}$ with $u, v \geq 0$. Both factors are nonzero, so polynomial degrees show that one is constant. If $2u + 3$ is constant, it is an ordinary odd integer $d \geq 3$. Then $v = X/d$, impossible because $1/d \notin \mathbb{Z}[1/2]$. If v is constant, it is a positive integer; comparing constant terms gives $0 = (2u(0) + 3)v$, impossible since the first factor is odd.

Next suppose $X + 1 = (2u + 3)v$ in $R_{(3)}$. If the odd factor is constant $d \geq 3$, the constant term of v would be $1/d$, not an integer. If v is constant, comparison of constant terms forces $v = 1$ and $2u(0) + 3 = 1$; comparison of the coefficient of X then forces a coefficient $1/2$ in u , which is not in $\mathbb{Z}[1/3]$. These contradictions do not even need the additional bounds $u, v \leq y$. $\square$

These examples concern the specific dyadic detector, not the existence of some other standard-cut definition. That distinction matters in view of the more general definability methods in the literature. [The write adds: both rings carry a unital ring retraction onto $\mathbb{Z}$, so the four-square formula of [14, Theorem 2.9] defines $\mathbb{N}$ in each of them (remark 20.5).]

20 Integer characters, bounded arithmetic and initial realizability

20.1 Classification of integer characters

Theorem 20.1 (Classification of integer characters). *Let R be a discretely ordered ring with standard division. Exactly one of the following holds:*

- (i) $\text{Hom}_{\mathbb{Z}}(R_{\text{add}}, \mathbb{Z}) = 0$;
- (ii) *there is a unique normalized integer character χ , it is a unital ring homomorphism, and*

$$\text{Hom}_{\mathbb{Z}}(R_{\text{add}}, \mathbb{Z}) = \{k\chi : k \in \mathbb{Z}\}.$$

In the second case, with $R^{\text{div}} = \ker \chi$, one has

$$R^{\text{div}} = \bigcap_{n \geq 1} nR, \quad R_{\text{add}} = \mathbb{Z} \oplus R^{\text{div}}, \quad R^{\text{div}} \text{ is a divisible additive group and an ideal.} \quad (41)$$

Proof. Let $f : R_{\text{add}} \rightarrow \mathbb{Z}$ be additive, and put $k = f(1)$. If $k = 0$, then $a = nq + r$ implies $f(a) = nf(q)$ for every standard n . An ordinary integer divisible by every positive integer is zero, so $f = 0$.

If $k \neq 0$, division by $|k|$ gives $f(a) = |k|f(q) + rk$, so every value of f is divisible by k in $\mathbb{Z}$. Hence $\chi(a) = f(a)/k$ is an integer-valued additive map and $\chi(1) = 1$. For any normalized character, lemma 19.3 shows that $\chi(a)$ is congruent modulo each standard n to the standard remainder of a . Two normalized characters must therefore agree at every a .

For multiplication, let a and b have remainders r and s modulo n . The remainder of ab is congruent to rs , so

$$\chi(ab) \equiv rs \equiv \chi(a)\chi(b) \pmod{n}.$$

As this holds for every standard n , the two ordinary integers are equal. Thus χ is multiplicative. Uniqueness and the normalization argument now give the asserted description of all f .

If $\chi(a) = 0$, division by n has remainder zero and gives $a = nq$. Moreover $0 = n\chi(q)$, so $q \in R^{\text{div}}$. This proves that R^{div} is divisible and contained in every nR . Conversely, membership in every nR makes $\chi(a)$ divisible by every n , hence zero. Finally, $a = \chi(a) + (a - \chi(a))$ is the unique additive decomposition into a standard integer and an element of R^{div} . Multiplicativity makes R^{div} an ideal. $\square$

In fact, multiplicativity of a normalized character already follows from ED₂: iterate division to obtain remainders modulo 2^j , and use that an ordinary integer divisible by every 2^j is zero (proposition 21.2). Standard division is retained in theorem 20.1 because it also normalizes an arbitrary nonzero character and gives the all-moduli formulation.

For the $\mathbb{Z}$ -group $(R, +)$, case (ii) is the split case of proposition 14.3: χ is the projection onto $\mathbb{Z}$ along the unique complement $R^{\text{div}} = G^{\text{div}}$, and (41) is lemma 14.2 together with condition (iii) there. What the ring structure adds is that this projection is automatically a ring homomorphism and R^{div} an ideal. [This comparison is the write's; source 03 does not cite this report.]

20.2 Profinite residues do not distinguish the rings by themselves

Standard division and discreteness identify R/nR with $\mathbb{Z}/n\mathbb{Z}$ for every ordinary $n \geq 1$, compatibly when one modulus divides another. Thus there is a canonical unital ring map

$$\text{res}_R : R \longrightarrow \widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z}.$$

Here the inverse system is indexed by divisibility. This is the ordinary-modulus completion; no topology on R has been assumed. On the additive group it is the coherent residue res_G of lemma 14.1 for $G = (R, +)$ and $e = 1$.

Proposition 20.2 (The diagonal-image criterion). *For a ring with standard division, a normalized integer character exists if and only if $\text{im } \text{res}_R$ is contained in the diagonal copy of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$. In that case $\text{im } \text{res}_R = \mathbb{Z}$ and $\text{res}_R = \iota \circ \chi$, where $\iota : \mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$.*

Proof. A character supplies exactly the standard remainders, proving one implication. Conversely, if $\text{res}_R(a) = \iota(m)$ for some integer m , that integer is unique. Define $\chi(a) = m$. Because res_R and ι preserve the ring operations and ι is injective, so does χ . The image contains every standard integer, proving equality rather than just inclusion. $\square$

For the additive group this is the equivalence (i) $\Leftrightarrow$ (ii) of proposition 14.3, in its ring version.

Every nonzero finite unital quotient of R is also an ordinary cyclic ring: its finite characteristic is some $n \geq 2$, and each element of R is congruent modulo nR to a standard integer. It follows that the quotient is generated by its unit and is isomorphic to $\mathbb{Z}/n\mathbb{Z}$, where n is its characteristic. Therefore neither these finite quotient rings nor the abstract inverse limit alone separate omnific arithmetic from nonstandard bounded arithmetic. The canonical image in the inverse limit is the extra information that matters, as for $\mathbb{Z}$ -groups after Theorem E.

20.3 No unit-preserving additive map from bounded arithmetic

For a model M of PA^- , its *difference ring* R_M is the group completion under addition, equipped with the usual induced multiplication and order. Its nonnegative cone is naturally M , and its additive group is the ordered additive group completion $G(M)$ of theorem 16.6.

Lemma 20.3 (Standard division in bounded arithmetic). *If $M \models \text{I}\Delta_0$, its difference ring has standard division.*

Proof. Fix a standard $n \geq 1$. In M , use bounded induction on a for

$$(\exists q \leq a)(\exists r < n) a = nq + r.$$

The base case is $q = r = 0$. For the successor, increment r if $r + 1 < n$; otherwise $r = n - 1$, so replace (q, r) by $(q + 1, 0)$. The witnesses remain bounded by $a + 1$. This proves division for nonnegative elements. If $b = nq + r \geq 0$, then $-b = n(-q) + 0$ when $r = 0$, and $-b = n(-q - 1) + (n - r)$ when $r > 0$. Thus division holds in R_M . $\square$

Theorem 20.4 (Vanishing of integer characters). *If M is a nonstandard model of $\text{I}\Delta_0$, then*

$$\text{Hom}_{\mathbb{Z}}((R_M)_{\text{add}}, \mathbb{Z}) = 0. \quad (42)$$

In particular, the inclusion of standard integers into $(R_M)_{\text{add}}$ does not split as an inclusion of abelian groups.

Proof. By lemma 20.3, standard division holds. A nonzero character would give a normalized character by theorem 20.1. The formula Std^{dy} would then define exactly the standard natural numbers by theorem 19.4. But lemma 19.6 makes that formula inductive, and $\text{I}\Delta_0$ contains its induction instance. It would follow that every element of M is standard, a contradiction. A splitting would itself be a normalized integer character. $\square$

Remark 20.5 (A second route through the four-square formula; [write]). A discretely ordered ring R with a unital ring homomorphism $\chi : R \rightarrow \mathbb{Z}$ is an ordered Dorroh ring in the sense of [14, Section 2.5], with witnessing ideal $\ker \chi$: the quotient is $\mathbb{Z}$, and a positive element of $\ker \chi$ is not a standard integer, so it is positive infinite by lemma 19.1, as their ordered condition requires. By proposition 21.2 a normalized character of a ring with ED_2 is such a homomorphism. Hence [14, Theorem 2.9] supplies, in every ring of theorem 19.4, a second parameter-free Δ_0 definition of $\mathbb{N}$ in R_+ , and the proof of theorem 20.4 goes through with it in place of Std^{dy} : a nonzero character is normalized and multiplicative by theorem 20.1, the four-square formula then defines $\mathbb{N}$ in M , $\mathbb{N}$ is closed under 0 and successor, and its Δ_0 induction instance makes M standard. The second route still needs theorem 20.1; it replaces only the detector. It does not give corollary 20.9, which uses the inductiveness of Std^{dy} in a ring *without* a character. Source 03 credits [14, Section 2.5] in general terms; this comparison with its Theorem 2.9 is the write's, from the v2 text of [14].

Corollary 20.6 (The additive-map obstruction). *Let $\mathcal{A}$ be a unital ring with a unital ring retraction $\chi_{\mathcal{A}} : \mathcal{A} \rightarrow \mathbb{Z}$, and let $M \models \text{I}\Delta_0$ be nonstandard. Every additive map $f : R_M \rightarrow \mathcal{A}$ satisfies*

$$\text{im } f \subseteq \ker \chi_{\mathcal{A}}.$$

There is no additive map $f : M \rightarrow \mathcal{A}$ preserving 0, addition, and 1. Consequently there is no unital ring homomorphism $R_M \rightarrow \mathcal{A}$ and no unital semiring homomorphism $M \rightarrow \mathcal{A}$.

Proof. The composite $\chi_{\mathcal{A}} f$ is an additive map to $\mathbb{Z}$, so it is zero by theorem 20.4. A unit-preserving additive map on M extends to R_M by $f(a - b) = f(a) - f(b)$; cancellation verifies that this is well-defined. Its composite with $\chi_{\mathcal{A}}$ would take 1 to 1, which is impossible. The multiplicative statements are special cases. $\square$

Corollary 20.7 (Even nonunital ring maps vanish). *If, additionally, $\mathcal{A}$ is a domain, then every ring homomorphism $R_M \rightarrow \mathcal{A}$, without a requirement to preserve 1, is zero.*

Proof. The image of 1 is an idempotent, hence 0 or 1 in the domain $\mathcal{A}$. The second option is excluded by corollary 20.6. In the first option, $f(a) = f(1a) = f(1)f(a) = 0$ for every a . $\square$

Remark 20.8 (Existence, not regularity). None of these maps is assumed continuous, Borel, definable, computable, injective, or order-preserving. The obstruction concerns the existence of a map, not its regularity. It is therefore complementary to Glazer's result, not a strengthening of his topological theorem for arbitrary models.

20.4 A concrete residue consequence without exponentiation

Corollary 20.9 (Non-diagonal residue profiles are forced). *In every nonstandard $M \models \mathsf{I}\Delta_0$, there is an $a \in M$ such that no ordinary integer has the same residues as a modulo all ordinary positive integers. Equivalently, $\text{im res}_{R_M} \not\subseteq \mathbb{Z}$. One can take any positive infinite y satisfying Pow^{dy} .*

Proof. The first conclusion follows from proposition 20.2 and theorem 20.4. For the explicit version, induction applied to lemma 19.6 yields $\forall x \text{Std}^{\text{dy}}(x)$ in M . Given a positive infinite h , choose a witness $y \leq h < 2y$. The witness y is positive infinite and satisfies Pow^{dy} . If its residues matched an integer $m = 0$, division by 3 would contradict Pow^{dy} . If $m \neq 0$, divide y by $2^{\nu_2(|m|)}$; the quotient has odd remainder modulo 2, and the same odd-factor construction as in theorem 19.4 gives a contradiction. This uses only the residue profile of this element, not a character defined on the entire ring. $\square$

This is an existence theorem for genuinely non-diagonal compatible residue profiles, not a claim that every profinite integer occurs. Nor does it assert that exponentiation is total in $\mathsf{I}\Delta_0$. In the language of section 16: the factorial type τ_* realized in a nonstandard Peano model by $F(h)$ (theorem 16.6) is replaced here by elements singled out in the *ring* language, by Pow^{dy} ; they witness that $\text{res}_{G(M)}(G(M)) \neq \mathbb{Z}$ without any recursion inside M .

20.5 Bounded arithmetic is not initially realizable

Source 03 never mentions initial realizability. Combined with source 02's criterion it gives the following, stated and proved at this write.

Definition 20.10 (The dyadic fragment; [write]). Let T^{dy} be the theory, in the language $\{0, 1, +, \cdot, <\}$, consisting of PA^- , the sentences

$$\text{ED}_n^+ : \quad \forall a \ (\exists q \leq a) \bigvee_{r=0}^{n-1} a = nq + \underline{r} \quad (n = 1, 2, \dots),$$

with $\underline{r}$ the numeral $1 + \dots + 1$, and the single sentence $\text{Ind}(\text{Std}^{\text{dy}})$ of (40).

T^{dy} is recursively axiomatized, and $\mathsf{I}\Delta_0 \vdash \mathsf{T}^{\text{dy}}$: each ED_n^+ is proved in the proof of lemma 20.3, and $\text{Ind}(\text{Std}^{\text{dy}})$ is an instance of bounded induction, since Std^{dy} is Δ_0 and has no parameters. Peano arithmetic therefore proves T^{dy} as well. For a model M of PA^- , the sentences ED_n^+ for all n give standard division in R_M , by the last step of the proof of lemma 20.3.

Corollary 20.11 (Nonstandard bounded arithmetic has no initial additive realization; [write]). *Let M be a nonstandard model of T^{dy} , for instance a nonstandard model of $\mathsf{I}\Delta_0$ or of Peano arithmetic, and let $G(M)$ be the ordered abelian group completion of its additive monoid. Then:*

- (i) $G(M)$ is a set-sized $\mathbb{Z}$ -group with least positive element 1, $\text{Hom}_{\mathbb{Z}}(G(M), \mathbb{Z}) = 0$, and $\text{res}_{G(M)}(G(M)) \neq \mathbb{Z}$;

- (ii) $G(M)$ is not initially realizable, and there is no additive homomorphism $f : G(M) \rightarrow \mathbf{Oz}$ with $f(1) = 1$, order-preserving or not.

Consequently, in theorem 16.6 the hypothesis “Peano arithmetic” can be replaced by ID_0 , and indeed by T^{dy} .

Proof. $G(M)$ is the additive group of the difference ring R_M , a set because M is. It is discretely ordered with least positive element 1, since R_M is a discretely ordered ring. By the remark after definition 20.10, R_M has standard division, so $G(M)/\mathbb{Z}1$ is divisible: $a = nq + r$ gives $a + \mathbb{Z}1 = n(q + \mathbb{Z}1)$. Thus $G(M)$ is a $\mathbb{Z}$ -group.

Suppose $f : G(M) \rightarrow \mathbb{Z}$ is a nonzero additive map. By theorem 20.1, R_M has a normalized integer character χ . R_M satisfies ED_2 and ED_3 , so theorem 19.4 shows that Std^{dy} defines exactly $\mathbb{N}$ in $M = (R_M)_+$. By lemma 19.6, which needs only ED_2 , the antecedent of $\text{Ind}(\text{Std}^{\text{dy}})$ holds in M , and $M \models \text{Ind}(\text{Std}^{\text{dy}})$ because $M \models \text{T}^{\text{dy}}$. Hence $M \models \forall x \text{Std}^{\text{dy}}(x)$, so every element of M is standard, contradicting the nonstandardness of M . Therefore $\text{Hom}_{\mathbb{Z}}(G(M), \mathbb{Z}) = 0$. This is the proof of theorem 20.4 with ID_0 replaced by the axioms of T^{dy} that it uses.

In particular $\mathbb{Z}1$ is not a direct summand of $G(M)$: a projection onto $\mathbb{Z}1 \cong \mathbb{Z}$ would be a nonzero additive map to $\mathbb{Z}$. By proposition 14.3, $\text{res}_{G(M)}(G(M)) \neq \mathbb{Z}$, which proves (i). By Theorem E, a set-sized $\mathbb{Z}$ -group whose bottom cyclic group is not a direct summand is not initially realizable. By corollary 17.2, it then has no additive homomorphism to $\mathbf{Oz}$ sending 1 to 1; alternatively, composing such a map with ct would be a nonzero additive map to $\mathbb{Z}$, which is the argument of corollary 21.4. This proves (ii). The final sentence follows from $\text{PA} \vdash \text{ID}_0 \vdash \text{T}^{\text{dy}}$. $\square$

At the lower end, open induction does not suffice, even with standard division.

Proposition 20.12 (Open induction is not enough; [write]). *The ring $\mathcal{A}_{\text{Sh}}$ of section 21.3 is countable and nonstandard, its nonnegative cone is a model of IOpen and of every ED_n^+ , and its additive group is an initially realizable $\mathbb{Z}$ -group:*

$$(\mathcal{A}_{\text{Sh}}, +) \cong V_{\text{Sh}} \overrightarrow{\times} \mathbb{Z}, \quad V_{\text{Sh}} = \left\{ \sum_{i=1}^d a_i X^{q_i} : a_i \in \mathbb{R}_{\text{alg}}, q_i \in \mathbb{Q}_{>0} \right\} \text{ divisible.}$$

Consequently $\text{IOpen} + \{\text{ED}_n^+ : n \geq 1\}$ does not prove $\text{Ind}(\text{Std}^{\text{dy}})$, and the conclusion of corollary 20.11 fails for nonstandard models of IOpen .

Proof. $\mathcal{A}_{\text{Sh}}$ is countable because $\mathbb{R}_{\text{alg}}$ and $\mathbb{Q}$ are, and it contains the positive infinite element X . Its cone satisfies IOpen by the imported integer-part theorem recalled in section 21.3, and it has standard division with the constant term as a ring retraction, as shown there. Every element is uniquely $m + v$ with $m \in \mathbb{Z}$ and $v \in V_{\text{Sh}}$; dividing the real algebraic coefficients by an ordinary integer shows that V_{Sh} is divisible. A nonzero v has a leading term aX^q with $q > 0$, so $m + v$ has the sign of a and $|v|$ exceeds every ordinary integer. The order is therefore lexicographic with V_{Sh} dominant, and $(\mathcal{A}_{\text{Sh}}, +)$ is a $\mathbb{Z}$ -group in which $\mathbb{Z}1$ is a direct summand. Theorem E makes it initially realizable. If $\text{IOpen} + \{\text{ED}_n^+\}$ proved $\text{Ind}(\text{Std}^{\text{dy}})$, the cone of $\mathcal{A}_{\text{Sh}}$ would be a nonstandard model of T^{dy} , and corollary 20.11 would forbid exactly this splitting. $\square$

Remark 20.13 (What this settles in question 28.7, and what it does not; [write]). Question 28.7 asks for the weakest arithmetic fragment whose nonstandard models have ordered additive group completions excluded by the factorial residue type, and for analogous computable residue types giving sharper bounds. For exclusion from Init by any means, corollary 20.11 and proposition 20.12 locate the threshold above open induction (even with standard division) and at or below $\text{T}^{\text{dy}} \subseteq \text{ID}_0$, well below the Peano arithmetic of theorem 16.6. They do not determine the weakest such fragment; whether T^{dy} is strictly weaker than ID_0 is not addressed (question 28.10). The route is not the factorial type: the factorial

function is not provably total in ID_0 , and whether every nonstandard model of ID_0 (or of T^{dy}) realizes τ_* in $G(M)$ is not addressed here. In place of τ_* , corollary 20.9 supplies the elements satisfying Pow^{dy} above the standard cut, whose residue profiles are non-diagonal. They form an analogous computable residue obstruction, but one stated in the ring language, not a type in the ordered-group language of (37). The question is re-scoped accordingly after its statement.

21 A uniqueness correction and concrete rings

21.1 A correction to a uniqueness lemma

There is a precise source-level correction relevant to the preceding character classification. In the terminology of [14, Section 2.5], a commutative ring is a *Dorroh ring* when some ideal has quotient isomorphic to $\mathbb{Z}$; that ideal is said to witness it. Lemma 2.7 of arXiv:2508.14758v2 asserts that the witnessing ideal is unique. The claim, as printed on pages 11–12 of that version, is false.

Proposition 21.1 (Distinct integer-quotient ideals). *For every $c \in \mathbb{Z}$, the ideal $\mathfrak{j}_c = (X - c)$ of $\mathbb{Z}[X]$ satisfies $\mathbb{Z}[X]/\mathfrak{j}_c \cong \mathbb{Z}$. These ideals are pairwise distinct. This remains a counterexample when $\mathbb{Z}[X]$ is ordered with X positive infinite and the extra ordered-Dorroh condition of [14] is imposed.*

Proof. Evaluation at c is a surjective unital ring homomorphism $\text{ev}_c : \mathbb{Z}[X] \rightarrow \mathbb{Z}$. Division by the monic polynomial $X - c$ shows that its kernel is $(X - c)$, proving the quotient statement. If $c \neq c'$, then $X - c \in \mathfrak{j}_c$ but $\text{ev}_{c'}(X - c) = c' - c \neq 0$, so $X - c \notin \mathfrak{j}_{c'}$.

In the leading-coefficient order with X positive infinite, every positive nonconstant polynomial exceeds every ordinary integer. No $\mathfrak{j}_c$ contains a nonzero constant, since evaluation fixes constants. Thus every positive element of each $\mathfrak{j}_c$ exceeds every integer, which is exactly the additional ordered condition in question. In particular, $\mathfrak{j}_0 = (X)$ and $\mathfrak{j}_1 = (X - 1)$ already suffice as a counterexample. $\square$

The error is a confusion between uniqueness of a decomposition *for a fixed kernel* and independence of the integer coordinate from the kernel. For the element X , the two legitimate decompositions are

$$X = X + 0 \quad \text{relative to } (X), \quad X = (X - 1) + 1 \quad \text{relative to } (X - 1).$$

Agreement of two retractions on the embedded integers does not make their values on X agree. The following replacement supplies a sufficient hypothesis directly relevant to omnific arithmetic.

Proposition 21.2 (A valid uniqueness replacement). *Let R be a discretely ordered ring satisfying ED_2 . There is at most one normalized additive map $R \rightarrow \mathbb{Z}$. Any such map is multiplicative. Consequently, there is at most one ideal $\mathfrak{a}$ with $R/\mathfrak{a} \cong \mathbb{Z}$.*

Proof. For each ordinary j , iterating division by 2 writes $a = 2^j q + r$ with $r \in \{0, \dots, 2^j - 1\}$. Every normalized additive map sends a to an integer congruent to r modulo 2^j . Therefore the difference of the values of two such maps is divisible by all powers of 2, and is zero. The same argument applied to $\chi(ab) - \chi(a)\chi(b)$ proves multiplicativity.

An isomorphism $R/\mathfrak{a} \cong \mathbb{Z}$ composed with the quotient map supplies a normalized ring map. It must preserve the identity: a surjective ring map takes the source identity to the target identity, even when this is not included in the definition of ring homomorphism. Thus any two such ideals are kernels of the same map and agree. $\square$

The polynomial ring in proposition 21.1 does not satisfy ED_2 : $X = 2q + r$ with $q \in \mathbb{Z}[X]$ and $r \in \{0, 1\}$ would require a coefficient $1/2$ in q . The added hypothesis therefore excludes the counterexample for a substantive reason.

This correction is local. The standard-cut argument in [14, Theorem 2.9] can be run using any *chosen* witnessing ideal and its retraction; uniqueness is unnecessary for that argument. No assertion is made that the counterexample invalidates the paper’s main completion theorems. A complete dependency audit of that paper is a separate task. The correction also does not establish that the error has not previously been noticed by its authors or by others.

For **Oz** the uniqueness is already in the collection: the constant term is the unique unital ring homomorphism $\mathbf{Oz} \rightarrow \mathbb{Z}$ (`odg:prop:canonicalct`), by the same 2-adic argument, and this is the Lean theorem `Surreal.Foundations.SignSequence.omnific_int_hom_eq_constant` in `Algebra/SurrealNumbers/Surreal/Foundations/OmnificConstantRigidity.lean`. [Proposition 21.2](#) extends it to normalized *additive* maps and to every discretely ordered ring with ED_2 ; that extension is not formalized. [The comparison is the write’s.]

21.2 A minimal computable example

Let

$$\mathcal{A}_0 = \mathbb{Z} + X\mathbb{Q}[X] = \left\{ a_0 + \sum_{i=1}^d a_i X^i : a_0 \in \mathbb{Z}, a_i \in \mathbb{Q} \right\}, \quad (43)$$

with the sign of a nonzero polynomial given by its leading coefficient. Thus X is positive infinite. Every positive nonconstant polynomial exceeds all ordinary integers, while positive constants are positive integers. The ring is discretely ordered, and

$$\chi(f) = f(0), \quad f = n \frac{f-r}{n} + r, \quad r \equiv f(0) \pmod{n}, \quad 0 \leq r < n$$

give a retraction and standard division. Rational coefficients have exact finite codes, so equality, order, addition, multiplication, and the nonnegative domain are computable. All the results above apply.

For clarity, $\mathcal{A}_0$ is *not* a model of open induction. Indeed, $x^2 < X$ defines the standard cut on its positive cone, with parameter X : it holds for standard integers, whereas the square of a positive nonconstant polynomial has degree at least two and exceeds X . Its induction instance fails. The ring is useful here because it isolates the character and computability phenomena with almost no background theory, not because it is a particularly strong arithmetic model. The same failed instance, in the real-coefficient ring $\mathbb{Z} + t\mathbb{R}[t]$ with t positive infinite, is proved independently by batch-86 manuscript 06, which is merged into the report `docs/foundations-and-computation/polish-models-of-omnific-arithmetic` (section 27.5).

Example 21.3 (Explicit certificates). For $y = X$, the forbidden factorization is $y = 3(X/3)$. For $y = X + 12$, one has

$$y = 4 \left(2 \frac{X}{8} + 3 \right).$$

For $y = X - 12$, whose character is negative, the certificate is

$$y = 4 \left(2 \left(\frac{X}{8} - 3 \right) + 3 \right).$$

In each case the displayed witnesses are nonnegative and bounded by y , although some of their integer constant terms are negative. This illustrates why positivity must be checked in the ordered ring, not after applying χ .

21.3 A classical open-induction example

A more structured example is the finite Puiseux-polynomial ring

$$\mathcal{A}_{\text{Sh}} = \left\{ a_0 + \sum_{i=1}^d a_i X^{q_i} : a_0 \in \mathbb{Z}, a_i \in \mathbb{R}_{\text{alg}}, q_i \in \mathbb{Q}_{>0} \right\},$$

where $\mathbb{R}_{\text{alg}}$ denotes the real algebraic numbers and equal exponents are combined. Order it with X positive infinite. This is the classical Shepherdson construction [17]; the integer-part characterization of open induction and the Puiseux description are recalled in [16].

Here is the structural reason. Put $t = X^{-1}$ and consider the real closed Puiseux field $K = \bigcup_{n \geq 1} \mathbb{R}_{\text{alg}}((t^{1/n}))$ with its natural order. Each series has only finitely many negative exponents. Remove its positive powers of t , retain its finite negative-exponent part, and take the ordinary integer floor of its algebraic constant. If the constant is an integer and the discarded infinitesimal tail is negative, subtract one. The resulting element of $\mathcal{A}_{\text{Sh}}$ is an integer part of the series. The classical integer-part theorem therefore gives $(\mathcal{A}_{\text{Sh}})_+ \models \text{IOpen}$. Its use here is an imported theorem, not a new proof of that characterization.

Constant extraction and division by ordinary integers work exactly as in $\mathcal{A}_0$. Thus $\mathcal{A}_{\text{Sh}}$ fails the single bounded induction instance (40) while satisfying every quantifier-free induction instance. Real algebraic numbers admit effective exact representations, so this ring also has a computable presentation. It consequently supplies an open-induction example for the complexity theorem below, and, by proposition 20.12, an initially realizable one.

21.4 Principal parts of Hahn series

Let k be an ordered field, Γ a nontrivial ordered abelian group, and use the Hahn convention in which t^γ is infinitesimal when $\gamma > 0$. In $k((t^\Gamma))$, define

$$\mathcal{A}(k, \Gamma) = \mathbb{Z} + \left\{ \sum_{\gamma < 0} a_\gamma t^\gamma : \text{supp}(a) \text{ is well ordered} \right\}.$$

Every displayed series is a Hahn series with set-sized support. The part supported on negative exponents is an ideal in this ring: sums of two negative exponents are negative. Constant extraction is a ring retraction onto $\mathbb{Z}$, and division of the coefficients by an ordinary integer, with the usual adjustment of the integer constant term, gives standard division. A positive nonconstant element has negative leading exponent and exceeds every ordinary integer. Thus the ring is discretely ordered and the results above apply.

No claim that this is an integer part of the whole Hahn field is needed. For example, arbitrary non-Archimedean coefficient fields require additional care with constants. The discrete ring, its character, and its division properties are all that the present argument uses.

21.5 Actual omnific integers

In Conway normal form, an omnific integer is a surreal number whose exponents are nonnegative and whose coefficient at exponent zero is an ordinary integer; its other coefficients are real (section 1.4). Schematically,

$$a = \sum_{i < \alpha} r_i \omega^{a_i}, \quad a_0 > a_1 > \cdots \geq 0,$$

where the final coefficient, when its exponent is zero, is integral. An absent zero exponent means zero constant term. The constant-term map $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ is additive and multiplicative: with no negative exponents present, exponent zero in a product arises only from multiplying two exponent-zero terms.

The positive finite omnific integers are precisely the ordinary positive integers, so the order is discrete. For an ordinary $n > 0$, choose the ordinary remainder r of $\text{ct}(a)$ modulo n ; then $(a - r)/n$ again has an integral constant term and is omnific. Hence $\mathbf{Oz}$ has standard division. These are the exact ingredients implemented in ProveIt's constant-term and ordinary-residue development [20] (section 27.3).

Corollary 21.4 (Omnific arithmetic obstruction). *The formulas Pow^{dy} and Std^{dy} define respectively the ordinary powers of two and the standard natural numbers in $\mathbf{Oz}_{\geq 0}$. If $M \models \text{Id}_0$ is nonstandard, there is no additive map $f : M \rightarrow \mathbf{Oz}$ satisfying $f(1) = 1$. In particular there is no unital semiring embedding of M into $\mathbf{Oz}_{\geq 0}$, and no unital ring embedding of its difference ring into $\mathbf{Oz}$.*

Proof. The detector applies by the preceding normal-form calculations. Any proposed additive map extends uniquely to R_M by $\tilde{f}(a - b) = f(a) - f(b)$. Composing with ct would be a normalized integer character, contrary to theorem 20.4. $\square$

The finite-polynomial map $\mathcal{A}_0 \hookrightarrow \mathbf{Oz}$ sending X to ω is an injective ordered ring homomorphism: distinct polynomial normal forms remain distinct surreal normal forms. Thus the target contains natural computable arithmetic-like rings even though it excludes the additive unit-preserving images of nonstandard bounded arithmetic. Embedding an ordered field into $\mathbf{No}$ is a different question: its integer part need not land in $\mathbf{Oz}$. For set-sized $\mathbb{Z}$ -groups, corollary 17.2 turns corollary 21.4 into the statement of corollary 20.11(ii) about initial realizability.

21.6 Set-sized workspaces and class language

All central theorems of source 03 are proved for *set-sized* rings. Their full omnific applications can be formulated in a class theory such as NBG, as in section 2, or as a scheme of statements about definable classes in ZFC. In the latter reading, a formula about $\mathbf{Oz}$ is an explicitly written first-order formula about set-coded surreal numbers; one does not postulate a set of all surreals or a truth predicate for all such formulas.

For the no-map result, the source M is a set. The image of a definable class map on this set is a set, and composing it with the definable constant-term operation gives an ordinary set function into $\mathbb{Z}$. The contradiction is therefore already set-theoretic. The detector's proof is pointwise and uses only finitely many divisions at each element. For the semantic and computability theorems that follow, the explicitly set-sized examples $\mathcal{A}_0$ and $\mathcal{A}_{\text{Sh}}$ suffice. No class satisfaction construction is hidden in their statements.

22 Compiling arithmetic into one infinite interval

Assume throughout this section the hypotheses of theorem 19.4, and fix a positive infinite $\Omega \in R$. Although the interval $[0, \Omega]$ is bounded inside the structure, it contains all ordinary natural numbers. The standard-cut predicate tells us exactly which elements to keep.

22.1 Relativization and correctness

For an arithmetic formula φ , define $\varphi^{[\Omega]}$ recursively. Leave atomic formulas unchanged, commute with Boolean connectives, and set

$$(\exists x \psi)^{[\Omega]} = \exists x \leq \Omega (\text{Std}^{\text{dy}}(x) \wedge \psi^{[\Omega]}), \quad (44)$$

$$(\forall x \psi)^{[\Omega]} = \forall x \leq \Omega (\text{Std}^{\text{dy}}(x) \rightarrow \psi^{[\Omega]}). \quad (45)$$

Bound variables in copies of Std^{dy} are renamed to prevent capture. For a sentence, the only possibly free variable in the output is Ω .

Lemma 22.1 (Truth-preserving relativization). *For every arithmetic formula $\varphi(\bar{x})$ and every standard tuple $\bar{a} \in \mathbb{N}$, one has*

$$\mathbb{N} \models \varphi(\bar{a}) \iff R_+ \models \varphi^{[\Omega]}(\bar{a}, \Omega).$$

The translation is effective and its output is bounded.

Proof. Induct on the construction of φ . Terms with standard inputs take the same standard values in $\mathbb{N}$ and R_+ , proving the atomic cases. The Boolean steps follow immediately. For an existential quantifier, the elements $x \leq \Omega$ satisfying Std^{dy} are exactly all of $\mathbb{N}$, by theorem 19.4; apply the induction hypothesis to each such value. For a universal quantifier the guard discards every nonstandard value, and the same induction hypothesis applies. Each new quantifier is bounded, and the fixed formula Std^{dy} itself uses only bounded quantifiers. The recursive procedure on finite syntax is an algorithm. $\square$

Before flattening atomic terms, the quantifier nesting depth of the output is at most $\text{qr}(\varphi) + 3$: Std^{dy} contributes at most three nested bounded quantifiers, while each original quantifier contributes one. This estimate is about depth, not a fixed bound on alternations over *all* input sentences.

22.2 Literally the same syntactic bound everywhere

In (44)–(45), the bounds internal to Std^{dy} are variables such as x and y . To make every written bound literally Ω , use

$$\text{Pow}_{\Omega}^{\text{dy}}(y) : \Longleftrightarrow 0 < y \wedge \forall u \leq \Omega \forall v \leq \Omega ((u \leq y \wedge v \leq y) \rightarrow y \neq (2u + 3)v), \quad (46)$$

$$\text{Std}_{\Omega}^{\text{dy}}(x) : \Longleftrightarrow x = 0 \vee \exists y \leq \Omega (y \leq x \wedge x < 2y \wedge \text{Pow}_{\Omega}^{\text{dy}}(y)). \quad (47)$$

For $0 \leq x \leq \Omega$, $\text{Std}_{\Omega}^{\text{dy}}(x)$ and $\text{Std}^{\text{dy}}(x)$ are equivalent: the guards reproduce the smaller original ranges exactly. Replacing Std^{dy} by $\text{Std}_{\Omega}^{\text{dy}}$ in the translation therefore preserves the proof of lemma 22.1.

22.3 Quadratic atomic normal form

High-degree input terms do not require high-degree output atoms. For each local term, introduce a fresh variable for every addition or multiplication gate. A gate is expressed by an equation such as

$$z = x + y \quad \text{or} \quad z = xy.$$

An input atomic comparison is replaced by a bounded existential block asserting all its gate equations and the final comparison. Every gate variable is bounded by Ω . For standard inputs, each intermediate gate value is standard and hence below Ω ; the equations also determine the gate values uniquely. This proves the local equivalence.

The existential blocks must remain *inside the atomic replacement*. For example, a negated atom becomes the negation of that whole replacement; one must not move its existential witnesses through negation without justification. Guards guarantee that the semantic comparison of source atoms is only required at standard source-variable assignments. The atoms already present in (46)–(47) have total polynomial degree at most two, so no further change is needed there.

Theorem 22.2 (Single-bound quadratic compiler). *There is a computable map $\varphi \mapsto C\varphi(\Omega)$ from arithmetic sentences to formulas with the following properties:*

- (i) *every quantifier is bounded by exactly the variable Ω ;*
- (ii) *every atomic equality or inequality has total polynomial degree at most two in its variables;*
- (iii) *the only possibly free variable is Ω ;*
- (iv) *for every ring under consideration and every positive infinite Ω ,*

$$R_+ \models C\varphi(\Omega) \iff \mathbb{N} \models \varphi.$$

Proof. Apply lemma 22.1, use the guarded equal-bound versions of the standard-cut formula, and flatten each source atom as described above. Fresh-variable generation makes every substitution capture-free. The local gate equivalences and structural induction prove correctness. Each operation on syntax is effective. Statements (i)–(iii) follow from the explicit construction. $\square$

Example 22.3 (Why the infinite bound is indispensable). The true sentence $\forall x \exists y (y = x + 1)$ translates to a formula in which both x and y are bounded by Ω but x is guarded by $\text{Std}_\Omega^{\text{dy}}(x)$. For every retained x , its successor is still standard and below Ω . The translation is therefore true. If Ω were an ordinary finite integer, the retained value $x = \Omega$ would have no allowed successor. A finite-range test cannot substitute for the infinite-bound proof.

This theorem does not provide a single formula $\text{Truth}(e, \Omega)$ that takes arbitrary syntax codes as inputs. Each sentence is compiled into a *different* formula of potentially greater length and quantifier depth. Consequently it is not a definable universal truth predicate, and there is no conflict with Tarski's undefinability theorem.

22.4 Exact truth complexity in a computable ring

Write TA for the set of codes of first-order sentences true in $(\mathbb{N}; 0, 1, +, \cdot, <)$. For sets of natural-number codes, $B \leq_m C$ means that a total computable function maps membership in B exactly to membership in C . Invalid syntax codes are assigned a fixed false sentence whenever necessary.

Let $T_{\text{one}}(\mathcal{A}_0, X)$ consist of the true formulas in the fragment of theorem 22.2, interpreted in $(\mathcal{A}_0)_+$ with $\Omega = X$. Its atoms have degree at most two, and every quantifier has the same bound X . Let $T_{\Delta_0}(\mathcal{A}_0, X)$ denote the entire bounded truth set in the language with the named element X .

Theorem 22.4 (True-arithmetic completeness). *For the computable presentation of $\mathcal{A}_0$ in (43),*

$$T_{\text{one}}(\mathcal{A}_0, X) \equiv_m T_{\Delta_0}(\mathcal{A}_0, X) \equiv_m \text{TA}.$$

More generally, the same conclusion holds for any computably presented nonstandard discretely ordered ring satisfying the detector hypotheses, with a specified computable name for a positive infinite element Ω . It holds, in particular, for $\mathcal{A}_{\text{Sh}}$.

Proof. The compiler supplies $\text{TA} \leq_m T_{\text{one}}$, and inclusion of the syntactic fragment gives $T_{\text{one}} \leq_m T_{\Delta_0}$.

For the other direction, a computable presentation has a recursive set of element codes, recursive operation graphs, and a recursive order relation. Each of these relations has an effective arithmetical definition: a finite computation trace of its deciding program can be expressed in ordinary arithmetic. Replace the atomic relations of a formula by these definitions and relativize its quantifiers to the presentation's domain. Terms can first be flattened to operation graphs. Structural induction shows that the resulting arithmetic sentence is true exactly when the original sentence is true in the presentation. This is a computable translation of all first-order truth, hence also of bounded truth, into TA . The specified name for Ω is a fixed ordinary code in this translation. No decision procedure for TA is used. $\square$

22.5 Even one outer unbounded quantifier suffices without parameters

The parameter Ω can be removed while keeping the matrix bounded. Let $\text{Inf}(h) := \neg \text{Std}^{\text{dy}}(h)$, on the positive cone, and set

$$E_\varphi := \exists h (\text{Inf}(h) \wedge C\varphi(h)), \quad (48)$$

$$U_\varphi := \forall h (\text{Inf}(h) \rightarrow C\varphi(h)). \quad (49)$$

The first sentence is syntactically Σ_1 and the second syntactically Π_1 in the usual unbounded-quantifier hierarchy over bounded matrices.

Corollary 22.5 (Parameter-free low-level truth). *For every computable ring in theorem 22.4, the truth sets of its parameter-free Σ_1 sentences and of its parameter-free Π_1 sentences are each many-one equivalent to TA .*

Proof. There is at least one positive infinite h , and Inf defines precisely those elements. By [theorem 22.2](#), $C\varphi(h)$ has the same truth value for every such h , namely the truth value of φ in $\mathbb{N}$. Hence each of (48) and (49) is true precisely when φ is true in $\mathbb{N}$. These give the lower reductions; the upper reductions are again the computable-presentation translation. $\square$

Syntactic boundedness in a nonstandard computable structure does not mean that bounded truth is recursive, or even arithmetical as a set of codes. The interval below an infinite element is externally infinite, and bounded quantifier alternations over it retain the complexity of arbitrary arithmetic sentences. The corollary is not a collapse of the ordinary arithmetical hierarchy in $\mathbb{N}$: it concerns truth in a different structure.

22.6 Why Diophantine retractions do not contradict this

For any unital ring retraction $\chi_{\mathcal{A}} : \mathcal{A} \rightarrow \mathbb{Z}$, a finite system of integer-coefficient polynomial equations has a solution in $\mathcal{A}$ if and only if it has a solution in $\mathbb{Z}$. Apply $\chi_{\mathcal{A}}$ to a solution for the forward direction, and use the constant inclusion for the reverse. This elementary observation underlies related claims in the repository; it is not new here. In this collection it is the existence transfer behind Hilbert’s tenth problem over $\mathbf{Oz}$ (odg:cor:H10).

The compiler uses inequalities, negations, and alternating bounded quantifiers. Retractions need not preserve order, as $X - 1$ illustrates, and do not transport these quantifiers. Therefore the equality-only existential observation places no upper bound on the bounded truth sets just studied. In particular, it must not be confused with the stronger notion of Diophantine correctness for formulas allowing inequalities.

23 Standard systems and computable scalar fields

23.1 The fourth source

Sections 23 to 26 are source 04, *Perfect Transcendence Gaps in Borel Arithmetic: Standard Systems, p -adic Shadows, Hahn Fields, and Omnific Integers* (26 pages, 3 October 2026). It does not mention initial subgroups or this report. It was placed here because its exact profinite image ([theorem 25.3](#)) answers [question 28.11](#), a question this report names, for nonstandard models of Peano arithmetic; the question asks about $\mathbf{l}\Delta_0$, and is re-scoped after its statement. Everything in these four sections is source 04’s, with the symbols renamed as in [section 23.2](#), except the corollary marked [write 04] and the bracketed comparisons, each of which says that it was added at this write.

The questions, as source 04 states them. The overlap is between two concrete themes. Glazer studies descriptive restrictions on models of arithmetic, including the failure of Borel models to have full standard system and automatic-continuity obstructions to topological arithmetic [21, 13]. ProveIt contains formal and exploratory work on arithmetic coding, computability, Conway normal forms, omnific integers and ordinary congruence completions [31]. Source 04 asks what a standard-system restriction says about the *coefficient fields* used in the latter constructions. Does failure to code every real merely omit some names, or does it leave an irreparable algebraic deficit? Can adjoining countably many external constants repair the deficit? Can adding arbitrary well-ordered Hahn supports repair it? Can finite congruence data detect what is missing? The first two questions are answered negatively, and the text explains exactly why the third can also have a negative answer.

A general theorem behind the arithmetic. Throughout [sections 23 to 26](#), “analytic” refers to a continuous image of a Polish space, or equivalently to a projection of a Borel set. It does not refer

to power-series analyticity. The work is in ordinary ZFC unless a different foundation is explicitly discussed.

Theorem 23.1 (Perfect gap after countably many parameters). *Let F be $\mathbb{R}$ or $\mathbb{Q}_p$, and let $k \subseteq F$ be an analytic subfield relatively algebraically closed in F . For any countable $Y \subset F$, put*

$$k_Y = \text{acl}_F(k(Y)) = \{x \in F : x \text{ is algebraic over } k(Y)\}.$$

Then k_Y is a proper analytic dense subfield of F , is null for additive Haar measure, and is meager. For each $n \geq 1$, the set of algebraically dependent n -tuples over k_Y is analytic, null, and meager in F^n . Every nonempty open $U \subset F$ therefore contains a Cantor set algebraically independent over k_Y , and

$$\text{trdeg}(F/k_Y) = \mathfrak{c}.$$

The word *countable* concerns an external set of parameters. There is no hypothesis that those parameters are computable or belong to a model. The theorem also permits k to be uncountable. Consequently a proof that simply enumerates all polynomials over k would be invalid. The aggregate dependence relations, rather than the individual polynomials, are the objects to which category and measure arguments are applied. The proof is assembled in [section 24.4](#).

The arithmetic and omnific conclusions. For a Turing ideal $\mathcal{S}$, the field $\mathcal{K}_F(\mathcal{S})$ is defined in [section 23.5](#). Write $\mathcal{K}_\infty(\mathcal{S}) = \mathcal{K}_\mathbb{R}(\mathcal{S})$ and $\mathcal{K}_p(\mathcal{S}) = \mathcal{K}_{\mathbb{Q}_p}(\mathcal{S})$.

Theorem 23.2 (Arithmetic-to-Hahn gap). *Let M be a nonstandard Borel model of PA and $\mathcal{S} = \text{SSy}(M)$. For each $F = \mathbb{R}, \mathbb{Q}_p$, the field $\mathcal{K}_F(\mathcal{S})$ satisfies the hypotheses, hence the conclusions, of [theorem 23.1](#). Moreover:*

- (1) *For the difference ring R_M , the residue homomorphism $\text{res}_p : R_M \rightarrow \mathbb{Z}_p$ has image $\mathcal{K}_p(\mathcal{S}) \cap \mathbb{Z}_p$. Its image is dense, null, and meager, and its fraction field is $\mathcal{K}_p(\mathcal{S})$.*
- (2) *If $Y \subset \mathbb{R}$ is countable and $k_{\mathbb{R},Y} = \text{acl}_\mathbb{R}(\mathcal{K}_\infty(\mathcal{S})(Y))$, there is a Cantor set $P \subset (1, 2)$ whose elements are algebraically independent even over the Hahn field $k_{\mathbb{R},Y}((t^\mathbb{Q}))$ inside $\mathbb{R}((t^\mathbb{Q}))$.*
- (3) *Under $t = \omega^{-1}$, the family $\{r\omega : r \in P\}$ consists of purely infinite omnific integers, is algebraically independent over $k_{\mathbb{R},Y}((t^\mathbb{Q}))$, and maps to zero modulo every positive ordinary integer.*

[Its parts are proved as [theorems 23.12](#), [25.1](#) and [26.3](#) and [corollaries 24.7](#), [25.2](#) and [26.2](#); this pointer is the write's.] In the last assertion the *coefficients* form a Cantor set in the real topology. The resulting omnific family is not asserted to be perfect in the surreal order topology; indeed, it is discrete in that topology. This distinction is proved in [section 26.3](#). A complementary reconstruction theorem identifies the full profinite image and recovers $\mathcal{S}$ as its Boolean algebra of idempotents ([theorem 25.3](#) and [corollary 25.4](#)).

Relation to previous work and novelty boundary, as [source 04](#) states it. Fatalini and Schindler explicitly credit independent observations of Ben de Bondt and Glazer that adding any new real to a suitable set-theoretic universe yields maximal transcendence degree over its old real field [[22](#), [Section 2](#)]. This is a close conceptual precursor, not a new result of [source 04](#). Their setting involves reals of nested universes; [source 04](#) uses external analytic fields attached to one arithmetic standard system and obtains perfect, measure, category, p -adic, and Hahn conclusions. Ye, Yu, and Zhao study analytic subfields of both $\mathbb{R}$ and $\mathbb{Q}_p$, including obstructions to maximal subfields omitting a specified point [[23](#), [Proposition 2.6](#)]. Their paper appeared online in 2026, so it is included in the novelty check rather than overlooked. Edgar and Miller's dimension theorem gives a further consequence for the real fields: their Hausdorff dimension is zero [[24](#)]. The existence of perfect independent sets once the relevant relations are meager is Mycielski's classical theorem [[25](#), [26](#)].

The results developed in source 04 are the countable-parameter derivation argument, the assembled perfect-gap theorem, the exact standard-system/residue identifications (including profinite idempotents), and their coefficient-preserving Hahn and congruence applications. Some individual ingredients are elementary or may already occur in the literature in another form. The search undertaken for source 04 does not establish first publication. The mathematical claims stand on their stated proofs and dependencies, not on a “breakthrough” label. In particular, neither the general Scott-set problem nor the reverse-mathematical questions in Glazer’s topological paper are claimed resolved.

Relation to this report and its neighbours. [This paragraph is the write’s; source 04 cites neither this report nor the other reports named here.] The difference ring R_M is that of section 20.3 (source 04 writes $D = M - M$), and $\text{res}_p, \text{res}_{R_M}$ are the coordinates and the whole of the canonical map res_R of section 20.2. Theorem 25.3 strengthens corollary 20.9, which says only that $\text{im } \text{res}_{R_M} \not\subseteq \mathbb{Z}$, to an exact description in the Peano case, and so answers question 28.11 for Peano arithmetic (section 25.4). Corollary 25.8, no unital ring map $R_M \rightarrow \mathbb{Z}$, is weaker than theorem 20.4 and corollaries 20.6 and 21.4, which exclude every additive map sending 1 to 1, already for Id_0 . The ring $\mathcal{A}(k, \mathbb{Q})$ of section 25.5 is source 03’s $\mathcal{A}(k, \Gamma)$ at $\Gamma = \mathbb{Q}$; proposition 25.6 and the $\mathcal{A}$ -half of theorem 25.7 are its set-sized versions of facts proved in Lean for the class-sized **Oz**, and the point of theorem 25.7 is the one made after proposition 20.2: the canonical image, not the completion, separates the rings. Theorem 26.1 is an immediate consequence of a lemma of the independent-surreal-copies report (`isc:cp:lem:coeffld`). These relations are recorded at the statements concerned and in section 27.5.

23.2 Conventions of the fourth source

Source 04 works in ZFC, with set-sized Hahn fields only (section 26.4). Several of its symbols are renamed here, because the letters are in use elsewhere in this report. The renaming changes no statement.

here	in source 04	reason for the change
$\mathcal{S}, \mathcal{T}$ (Turing ideals; $\mathcal{S} = \text{SSy}(M)$)	$\mathcal{S}, \mathcal{T}$	$\mathcal{S}_n$ are source 02’s factorial partial sums; $\mathcal{T}^{\text{dy}}$ is a theory
A, B, A_z (subsets of $\mathbb{N}$, oracles)	A, B, A_z	A is the initial group of Theorem A, B that of Theorem C
$\mathcal{K}_F(\mathcal{S}), \mathcal{K}_\infty(\mathcal{S}), \mathcal{K}_p(\mathcal{S})$	$K_F(\mathcal{S}), K_\infty(\mathcal{S}),$ $K_p(\mathcal{S})$	K_α is the dyadic-spine group, with $K_0 = \mathbb{Z}$ (Theorem B)
$k, k_Y, k_{F,Y}$ (fields)	$K, H_C, H_{F,C}, H$	$K_\alpha; H$ is the image group of Theorems A and B
Y (countable set of parameters)	C	$C \subseteq C'$ are convex subgroups (Theorem D); $\mathbb{C}$ is the complex field
$R_M, \text{res}_p, \text{res}_{R_M}$	$D = M - M, \rho_p, \widehat{\rho}$	$\mathbb{D}$ is $\mathbb{Z}[1/2]$; ρ is root deletion (lemma 8.2); R_M and res_R as in sections 20.2 and 20.3
$\mathbb{Z}_p(\mathcal{S}) = \mathcal{K}_p(\mathcal{S}) \cap \mathbb{Z}_p$	$A_p(\mathcal{S})$	A_p is the bottom coefficient group (proposition 3.2)
$\widehat{\mathbb{Z}}(\mathcal{S})$	$B(\mathcal{S})$	B as above
$r_{n!}(z)$ (residue modulo $n!$)	$r_n(z)$	$r_n(x)$ is the residue modulo n in lemma 14.1
$\mathbf{1}_A$ (idempotents)	e_A	e is the least positive element of a $\mathbb{Z}$ -group
$\tilde{d}_i, \tilde{r}_n$ (internally decoded digits and residues)	e_i, e_n	as for e_A
w (bit width of a digit block)	k	k is a field here
$\mathcal{A}(k, \mathbb{Q}) = \mathbb{Z} + \Pi_k, \Pi_k$	$R_L = \mathbb{Z} + I_L, I_L$	it is source 03’s $\mathcal{A}(k, \Gamma)$ at $\Gamma = \mathbb{Q}$ (section 21); Π_k is its purely infinite part, as Π in Oz $= \mathbb{Z} \oplus \Pi$
Γ (Hahn exponent group)	G	G is an ordered group throughout section 11; source 03 also writes Γ
$\partial, \widetilde{\partial}$ (derivations)	∂, D	D as above

here	in source 04	reason for the change
$\mathbf{b} = (b_j)_{j \in I}$ (transcendence basis)	$B = \{b_j : j \in J\}$	B as above; $J(B)$ is the suspension of theorem 8.5
Q (a polynomial in lemma 24.1 and in Dep_n)	P	P is also source 04's Cantor set, kept
$X, \mathcal{P}, \eta$ (product space, Cantor subset, homeomorphism)	X, Q, h	X is source 03's indeterminate; h a nonstandard element in theorem 16.6
$\mathcal{E}$ (finite set of exponent vectors)	E	E is an exponent interval in section 12

Kept unchanged: p , always an ordinary prime, as in section 19.6 (unrelated to the minimum exponent p of section 3); M for a model of arithmetic and N for a nonstandard element of it; F for $\mathbb{R}$ or $\mathbb{Q}_p$ (unrelated to $F(h)$ in theorem 16.6 and to the field F of section 17.2); P for an independent Cantor set; t for the Hahn variable, with t^γ infinitesimal for $\gamma > 0$ as in section 21; ct for constant extraction (section 1.4). Letters used only inside one proof (U, V for neighbourhoods, $m(X)$ for a minimal polynomial, a, c for elements of M , T_j for indeterminates) are local to that proof. In remark 24.3 source 04's t is written τ , since t is the Hahn variable, and where source 04 writes Y for a polynomial variable it is X_n or X_2 here, since Y is the parameter set. The shipped records of source 04 use its own letters and its own numbering. Its statements are numbered here as follows.

statement	source 04	here	statement	source 04	here
Theorem	1.1	23.1	Corollary	6.1	24.7
Theorem	1.2	23.2	Theorem	6.2	24.8
Lemmas	2.1–2.3	23.3–23.5	Theorem	7.1	25.1
Theorem	2.4	23.6 (imported fact)	Corollary	7.2	25.2
Corollary	2.5	23.7	Theorem	7.3	25.3
Definition	3.1	23.8	Corollary	7.4	25.4
Lemma	3.2	23.9	Proposition	8.1	25.6
Proposition	3.3	23.10	Theorem	8.2	25.7
Lemma	3.4	23.11	Corollary	8.3	25.8
Theorem	3.5	23.12	Theorem	9.1	26.1
Lemma	4.1	24.1	Corollary	9.2	26.2
Proposition	4.2	24.2	Theorem	10.1	26.3
Remark	4.3	24.3	Section	11	26.5
Lemma	5.1	24.4	Section	12	part of 27
Proposition	5.2	24.5	Questions	13.1–13.11	28.17–28.27
Corollary	5.3	24.6	Appendices	A, B	parts of 27

Source 04's equations were numbered within sections; here they continue the report's single sequence.

23.3 Borel models and external traces

A Borel model, in sections 23 to 26, has a standard Borel underlying set, with Borel addition and multiplication, and satisfies PA in the usual first-order sense. One may choose a compatible Polish presentation of the underlying standard Borel space. The topology is presentation data, not an order topology that is assumed to make arithmetic continuous.

A standard numeral is the interpretation of an ordinary $n \in \mathbb{N}$. An element is nonstandard when it exceeds every standard numeral. A model is nonstandard when it has such an element. The set of standard numerals is an external subset; it is not assumed internally definable.

For $a \in M$, define its standard binary trace

$$b_M(a) = \{n \in \mathbb{N} : M \models \text{Bit}(a, \bar{n}) = 1\}.$$

The standard system is

$$\mathcal{S} = \text{SSy}(M) = \{b_M(a) : a \in M\} \subseteq 2^{\mathbb{N}}.$$

Here bits are least-significant first, including zeros beyond the ordinary length for standard integers. This convention is useful for residues.

Lemma 23.3 (Coding definable traces). *In a nonstandard model of PA, every set $\{n \in \mathbb{N} : M \models \varphi(\bar{n}, \bar{a})\}$ belongs to $\text{SSy}(M)$. Consequently the binary-trace definition agrees with the definition using all parameter-definable subsets of the standard cut.*

Proof. For every arithmetic formula $\varphi(i, \bar{a})$, full induction proves finite binary comprehension:

$$\forall b \exists c \forall i < b [\text{Bit}(c, i) = 1 \longleftrightarrow \varphi(i, \bar{a})].$$

One proves this by induction on b , choosing whether to add 2^b at the successor step and maintaining $c < 2^b$. Classical excluded middle is used for the instance of φ ; full induction allows its complexity. Evaluate the theorem at any nonstandard $b \in M$. The resulting c has the required bits at every standard position. The converse is immediate because the bit relation is arithmetic. $\square$

Lemma 23.4 (Turing-ideal closure). *$\mathcal{S} = \text{SSy}(M)$ is a nonempty Turing ideal: it contains every computable set, is closed under finite Turing joins, and is downward closed under Turing reducibility.*

Proof. The even/odd interleaving of two traces is a definable trace, so finite joins are covered by lemma 23.3. Suppose an oracle program computes the characteristic function of B from $A = b_M(a)$, and is total on the standard inputs. In M , use the bit oracle of a and define $\varphi(n, a)$ to mean that the deterministic oracle computation halts with output one.

For a standard input, the actual computation is a standard finite run and uses only standard positions of A . That same run exists in M . Arithmetic proves determinism and the impossibility of two incompatible halting outputs. Thus a nonstandard purported run cannot change the answer already furnished by the standard halting run. The standard trace of φ is precisely B , and lemma 23.3 applies. Taking an oracle-free program proves that all computable sets occur. Nonemptiness is also witnessed by the empty set. $\square$

This argument does not need the stronger Scott-set path property. The Turing-ideal part is exactly what the subsequent computable-field construction consumes. Conversely, no claim is made that an arbitrary Turing ideal is the standard system of a model of arithmetic.

23.4 Analyticity and the external input from Glazer

Lemma 23.5 (The trace image is analytic). *If M is Borel, $b_M : M \rightarrow 2^{\mathbb{N}}$ is Borel, and $\text{SSy}(M)$ is analytic.*

Proof. For each fixed positive standard m , PA proves the existence and uniqueness of a remainder $r < m$ in division by m . For each standard $r < m$, the statement $a = mq + r$ for some $q \in M$ defines an analytic set of a , since arithmetic is Borel. Its complement is the finite union of the corresponding analytic sets for the other remainders. It is therefore Borel by the analytic/coanalytic separation theorem. The n th standard bit is determined by the remainder modulo 2^{n+1} , so each coordinate of b_M is Borel. Its image is analytic. $\square$

Source 04 uses the following external result in precisely the indicated strength.

Imported fact 23.6 (Glazer's standard-system theorem; input). *A Borel model of elementary function arithmetic does not have full standard system. In particular, for a nonstandard Borel model M of PA, one has $\text{SSy}(M) \neq 2^{\mathbb{N}}$.*

Glazer states the stronger Borel-quotient version in his February 2024 seminar slides [21, slide 4]. Source 04 neither reproves that result nor claims its weaker metatheory for the new arguments below. Its applications use ordinary Borel models of PA in ZFC. The distinction is substantive: the full induction used in lemma 23.3 should not silently be replaced by bounded induction. [Added at this write: source 04 cites the theorem from the seminar announcement and slides; this write consulted neither and did not look for a published proof. Every statement below about a Borel model depends on it, through the properness of $\text{SSy}(M)$. Statements that do not mention Borel models do not depend on it; among them are the exact images of theorems 25.1 and 25.3, the Boolean reconstruction of corollary 25.4, and lemmas 23.3, 23.4 and 23.11, proposition 23.10, corollaries 25.5 and 25.8, and theorem 26.1.]

Corollary 23.7 (A preliminary size gap). *If M is a nonstandard Borel model of PA, $\text{SSy}(M)$ is a dense null meager subset of Cantor space.*

Proof. Under symmetric difference it is a subgroup, by computability closure. It contains the finite sets, so is dense. It is analytic and proper. An analytic subgroup has the Baire property and is Haar measurable. A nonmeager or positive-measure subgroup is open by the category or measure difference-set theorem. An open subgroup is closed, and a dense closed subgroup is the whole group. This contradicts properness. $\square$

Nullness alone does not yet prove the desired algebraic independence. The missing work is to preserve properness after adjoining parameters and taking roots, and then to control dependence in every finite arity.

23.5 Fields computable from a Turing ideal

Use the usual rational presentations of $\mathbb{R}$ and $\mathbb{Q}_p$. A fast name for $x \in \mathbb{R}$ is a rational sequence (q_n) with $|q_n - x| \leq 2^{-n}$. For $x \in \mathbb{Q}_p$, use $|q_n - x|_p \leq p^{-n}$. A name is computable from $A \subseteq \mathbb{N}$ when an oracle program with oracle A outputs the rational q_n on input n .

Definition 23.8 (Computable scalar field). For a nonempty Turing ideal S and $F \in \{\mathbb{R}, \mathbb{Q}_p\}$, set

$$\mathcal{K}_F(S) = \{x \in F : \text{some fast name of } x \text{ is computable from some } A \in S\}.$$

No uniform choice of a name or an oracle for every element is part of the definition.

Rational field operations are computable on fast names. To invert a nonzero point, one searches for a certified positive lower bound on its absolute value; the search terminates because the point is nonzero. Finite joins combine the coefficient oracles. Hence $\mathcal{K}_F(S)$ is a subfield containing $\mathbb{Q}$, and it is dense in F .

Lemma 23.9 (Analyticity of the scalar field). *If $S \subseteq 2^{\mathbb{N}}$ is analytic, then $\mathcal{K}_F(S) \subseteq F$ is analytic.*

Proof. For a fixed oracle-program index e , the relation “the e th program with oracle A outputs a total rational sequence (q_n) , and $|q_n - x|_F \leq \epsilon_n$ for all n ” is Borel in (A, x) . Halting at each input is a countable Boolean combination of finite-oracle conditions, and each approximation inequality is closed in x once the finite computation is fixed. Intersect with $A \in S$, project to x , and take the countable union over e . Analytic sets are closed under these operations. $\square$

23.6 No information is lost by passing to a local field

For $A \in 2^{\mathbb{N}}$, consider the two encodings

$$c_{\infty}(A) = \sum_{n \geq 0} 2A(n)3^{-(n+1)}, \quad (50)$$

$$c_p(A) = \sum_{n \geq 0} A(n)p^n. \quad (51)$$

The first lies in the ordinary ternary Cantor set, and the second lies in $\mathbb{Z}_p$. Both have a fast name computable from A . Conversely, a fast name recovers the digits, uniformly on the indicated image. In the real case the two child intervals are separated by a nonempty middle gap. In the p -adic case a sufficiently accurate approximation determines the next residue, and hence the next digit. There is no binary-expansion ambiguity in either construction.

Proposition 23.10 (Faithful passage from ideals to fields). *For Turing ideals $\mathcal{S}, \mathcal{T}$ and $F \in \{\mathbb{R}, \mathbb{Q}_p\}$,*

$$\mathcal{S} \subseteq \mathcal{T} \iff \mathcal{K}_F(\mathcal{S}) \subseteq \mathcal{K}_F(\mathcal{T}).$$

In particular, $\mathcal{K}_F(\mathcal{S}) = F$ if and only if $\mathcal{S} = 2^{\mathbb{N}}$.

Proof. The forward implication is immediate. For the reverse, take $A \in \mathcal{S}$. Then $c_F(A) \in \mathcal{K}_F(\mathcal{T})$, so it has a fast name computable from some $B \in \mathcal{T}$. Digit recovery gives $A \leq_T B$, hence $A \in \mathcal{T}$. For the final assertion, every point of F has some fast name, and the graph of a name can itself be coded by a subset of $\mathbb{N}$. $\square$

23.7 Relative algebraic closure: the effective root argument

Lemma 23.11 (Individual roots are oracle-computable). *Let $f \in F[X]$ be nonzero, with its finitely many coefficients computable from one oracle A . Each root $\alpha \in F$ of f has a fast name computable from A . The assertion is nonuniform in the choice of the root and its finite isolating data.*

Proof. If α has multiplicity m , replace f by $f^{(m-1)}$. Its coefficients remain A -computable, and α is now a simple root. The integer m is a finite parameter; no algorithm to find all multiplicities uniformly is being asserted.

For $F = \mathbb{R}$, a rational root already has a computable name. For an irrational root, choose rational endpoints surrounding α in a small interval on which f' has constant nonzero sign and f has opposite signs at the endpoints. The endpoints are finite data. Bisection computes α : at any rational midpoint the value of f is nonzero, because α is the only root in the interval and is irrational. Its sign is therefore eventually certified by refining the coefficient approximations. The nested interval lengths provide a computable error bound.

For $F = \mathbb{Q}_p$, make a rational scaling of the variable and multiply the polynomial by a power of p so that the chosen root and all coefficients are integral. These are again finite choices; they can be undone at the end. Write the resulting polynomial as g . Choose a rational $a \in \mathbb{Z}_p$ sufficiently close to the simple root that

$$s = v_p(g'(a)) \geq 0, \quad \delta = v_p(g(a)) - 2s > 0.$$

A rational root is handled separately. For a nonrational root such an a exists by continuity and simplicity. Newton iteration

$$a_0 = a, \quad a_{j+1} = a_j - \frac{g(a_j)}{g'(a_j)}$$

is computable from A . Taylor expansion over $\mathbb{Z}_p$ shows inductively that the derivative valuation stays s , while

$$v_p(g(a_j)) \geq 2s + 2^j \delta, \quad v_p(a_j - \alpha) \geq s + 2^j \delta.$$

For the induction step, the linear term is canceled by Newton's correction; the remaining terms have at least twice the valuation of that correction. Its valuation exceeds s , so the derivative valuation does not change. The same estimates show that the limit is the unique root in the selected Hensel neighbourhood. The displayed error estimate turns the sequence into a fast name. All divisions are by nonzero quantities with the fixed certified valuation. No undecidable test of whether an intermediate residual is exactly zero is needed. $\square$

Theorem 23.12 (Relatively algebraically closed scalar fields). *For every Turing ideal S , $\mathcal{K}_F(S)$ is relatively algebraically closed in F . If S is proper and analytic, it is a proper analytic relatively algebraically closed dense subfield. Furthermore,*

$$\mathcal{K}_\infty(S) \preccurlyeq (\mathbb{R}, +, \cdot, <), \quad \mathcal{K}_p(S) \preccurlyeq (\mathbb{Q}_p, +, \cdot)$$

in their respective field languages.

Proof. If $\alpha \in F$ is algebraic over $\mathcal{K}_F(S)$, combine oracles for the finitely many coefficients of an annihilating polynomial. Their join belongs to S , and lemma 23.11 gives a name for α . Analyticity and properness follow from lemma 23.9 and proposition 23.10.

A relatively algebraically closed subfield of $\mathbb{R}$ is real closed: positive square roots and roots of odd-degree polynomials lie in $\mathbb{R}$ and are algebraic. Quantifier elimination for real closed fields then gives the elementary embedding.

For the p -adic assertion use Macintyre's quantifier elimination in the language with power predicates [27]. A unary formula over $\mathcal{K} = \mathcal{K}_p(S)$ is a Boolean combination of polynomial equalities and conditions that polynomial values are n th powers. If a witnessing point annuls a nonzero polynomial over $\mathcal{K}$, it belongs to $\mathcal{K}$ by relative algebraic closure. Otherwise all tested nonzero polynomial values stay nonzero locally, and their power classes are locally constant: $(\mathbb{Q}_p^\times)^n$ is open by Hensel's lemma. Thus the same formula holds on a neighbourhood containing a rational point. This proves the Tarski–Vaught condition. Power predicates agree on $\mathcal{K}$ and $\mathbb{Q}_p$ by relative algebraic closure, so the claim also holds in the pure ring language. $\square$

The elementary-subfield assertion is logically distinct from computability. It says that the small field satisfies the same first-order field theory with its own parameters. It does not make it topologically complete and does not provide a computable selector for solutions of arbitrary formulas.

24 The analytic-derivation obstruction and perfect gaps

24.1 No positive countable transcendence degree

The following lemma is the main mechanism for surviving external parameters. The zero-transcendence-degree case is intentionally excluded in its statement; relative algebraic closure handles that case in the application.

Lemma 24.1 (No positive countable transcendence degree). *Let $F = \mathbb{R}$ or $\mathbb{Q}_p$, and let $k \subseteq F$ be an analytic subfield. Then $\text{trdeg}(F/k)$ is not a positive finite or countably infinite cardinal.*

Proof. Suppose $\mathbf{b} = (b_j)_{j \in I}$ is a nonempty finite or countable transcendence basis, with $0 \in I \subseteq \mathbb{N}$. On the rational function field $k(\mathbf{b})$ define a k -derivation ∂ by

$$\partial b_0 = 1, \quad \partial b_j = 0 \quad (j \neq 0).$$

Algebraic independence makes formal differentiation well defined. Because the characteristic is zero, the algebraic extension $F/k(\mathbf{b})$ is separable, and ∂ has a unique extension $\tilde{\partial} : F \rightarrow F$. For completeness, if $m(X)$ is a separable minimal polynomial of α , the only possible extension is

$$\tilde{\partial}(\alpha) = -\frac{m^\partial(\alpha)}{m'(\alpha)},$$

where m^∂ differentiates the coefficients. This formula defines a compatible derivation on each simple finite extension; uniqueness makes the definitions agree on overlaps and hence on their union.

We now prove that $\tilde{\partial}$ is Borel. A pair $(x, y) \in F^2$ belongs to its graph if and only if there is a polynomial Q with coefficients in k , in a finite list of the indeterminates $(T_j)_{j \in I}$ and one more indeterminate X , such that

$$Q(\mathbf{b}, x) = 0, \tag{52}$$

$$Q_X(\mathbf{b}, x) \neq 0, \tag{53}$$

$$yQ_X(\mathbf{b}, x) + Q_{T_0}(\mathbf{b}, x) = 0. \tag{54}$$

Here $Q_{T_0} = 0$ when T_0 does not occur. Every x has such a witness: clear denominators in its separable minimal polynomial over $k(\mathbf{b})$. The clearing factor does not vanish at $\mathbf{b}$, so condition (53) survives. Differentiating (52) proves necessity of (54); it also proves that any witness yields the unique value $\tilde{\partial}(x)$.

There are countably many choices of finite variable lists and monomial patterns. For each pattern, the coefficient tuples form a finite power of the analytic set k , and the displayed constraints are Borel conditions given by continuous polynomial evaluation. Projection and a countable union show that the graph is analytic. No enumeration of the elements of k has been used.

A total single-valued function between Polish spaces with analytic graph is Borel: for any open target set V , both its inverse image and its complement are analytic, obtained by projecting the graph restricted to V and to its complement. They are therefore Borel. Since $\tilde{\partial}$ is additive, the automatic-continuity theorem for Borel homomorphisms of Polish groups makes it continuous. Finally $\tilde{\partial}$ vanishes on $\mathbb{Q}$, and $\mathbb{Q}$ is dense in both possible fields F . Hence $\tilde{\partial} = 0$, contradicting $\tilde{\partial}(b_0) = 1$. $\square$

Here is the small category argument underlying automatic continuity, to expose its exact role. Given a target neighbourhood U of zero, choose a symmetric neighbourhood V whose fourfold sum is contained in U . Cover the separable target by countably many translates of V . One of their inverse images is nonmeager and has the Baire property. The difference of that inverse image with itself contains a neighbourhood of zero by Pettis's theorem, and the image of this difference lies in $V - V \subseteq U$. This is continuity at zero. The Baire-property and analytic-graph facts used here are standard descriptive set theory [28].

Proposition 24.2 (Countable algebraic hulls remain proper). *Under the hypotheses of theorem 23.1, for every countable $Y \subset F$ the field $k_Y = \text{acl}_F(k(Y))$ is analytic, relatively algebraically closed in F , and proper.*

Proof. The field $k(Y)$ is the union of values of rational expressions with finitely many coefficients from k and finitely many parameters from Y , restricted to nonzero denominators. The family of expression patterns and parameter lists is countable. Each range is analytic. The set of roots in F of nonzero polynomials over this generated field is likewise analytic. It is a subfield, and transitivity of algebraic extensions proves relative algebraic closure.

If $k_Y = F$, then $\text{trdeg}(F/k) \leq \aleph_0$. It cannot be zero, because k is relatively algebraically closed in F and proper. It cannot be positive and countable by lemma 24.1. This contradiction proves properness. $\square$

Remark 24.3 (Why the analytic hypothesis matters). With unrestricted choice one can choose a transcendence basis T of $F/\mathbb{Q}$, remove one member τ , and form $k = \text{acl}_F(\mathbb{Q}(T \setminus \{\tau\}))$. This is a proper relatively algebraically closed subfield with $\text{trdeg}(F/k) = 1$ and $\text{acl}_F(k(\tau)) = F$. The lemma proves that this k cannot be analytic. Thus the hypothesis excludes an actual failure mode, not merely an inconvenient proof technique.

24.2 Nullness and meagerness of every hull

Lemma 24.4 (Proper analytic subfields are small). *Every proper analytic subfield of $\mathbb{R}$ or $\mathbb{Q}_p$ is dense, null, and meager.*

Proof. It contains $\mathbb{Q}$, hence is dense. It has the Baire property and is measurable for the completed additive Haar measure. A positive-measure set has a difference set containing a neighbourhood of zero; a nonmeager set with the Baire property has the same conclusion. For a subgroup the difference set is the subgroup itself. It would therefore be open, hence also closed, and density would force equality with the ambient field. This contradicts properness. $\square$

For $\mathbb{Q}_p$, density is important: a proper open additive subgroup such as $\mathbb{Z}_p$ certainly exists. It is the field's containment of the dense prime field, not an assertion that $\mathbb{Q}_p$ is connected, that rules out this possibility.

24.3 Aggregate polynomial dependence

For a subfield $k' \subseteq F$, define

$$\text{Dep}_n(k') = \{\bar{x} \in F^n : \exists Q \in k'[X_1, \dots, X_n] \setminus \{0\}, Q(\bar{x}) = 0\}.$$

Repeated coordinates are dependent, but the definition also makes sense for arbitrary ordered tuples. If k' is analytic, this set is analytic: use countably many finite monomial patterns and project over the nonzero coefficient tuples in finite powers of k' .

Proposition 24.5 (All dependence relations are null and meager). *Under the hypotheses of theorem 23.1, $\text{Dep}_n(k_Y)$ is null and meager in F^n for every $n \geq 1$.*

Proof. The case $n = 1$ is $\text{Dep}_1(k_Y) = k_Y$, because k_Y is relatively algebraically closed. It is null and meager by proposition 24.2 and lemma 24.4.

Suppose the claim holds for $n - 1$. Fix an $(n - 1)$ -tuple $\bar{a}$ algebraically independent over k_Y . In a nonzero polynomial $Q(\bar{X}, X_n)$ over k_Y , at least one coefficient as a polynomial in X_n is a nonzero polynomial in $\bar{X}$. Independence ensures it does not vanish at $\bar{a}$. Thus

$$\{x : (\bar{a}, x) \in \text{Dep}_n(k_Y)\} = \text{acl}_F(k_Y(\bar{a})) = \text{acl}_F(k(Y \cup \{a_1, \dots, a_{n-1}\})).$$

The last equality follows by algebraic transitivity in both directions. It is a proper null meager field by the preceding propositions. For dependent $\bar{a}$, the section is the whole of F , but those base points form a null meager set by induction.

The full relation is analytic, hence universally measurable and has the Baire property. Fubini's theorem, on finite-measure boxes and then on a countable cover of F^n , proves nullness. The Kuratowski–Ulam theorem proves meagerness. These arguments apply to completed measures; no Borelness stronger than analyticity is required. $\square$

24.4 Perfect sets and cardinality

Mycielski's theorem says that in a perfect completely metrizable space, a countable family of meager relations of finite arities has a perfect independent set; one can take a compact Cantor subset [25, 26]. Apply it in any nonempty open $U \subset F$ to the restrictions of the relations $\text{Dep}_n(k_Y)$. Open subspaces of a Polish space are Polish, and these spaces have no isolated points. The resulting Cantor set P has no polynomial dependence among any finite tuple of distinct members. Consequently $|P| = \mathfrak{c}$, so $\text{trdeg}(F/k_Y) \geq \mathfrak{c}$; the reverse inequality is $|F| = \mathfrak{c}$. This completes the proof of theorem 23.1.

There are two different “largeness” assertions here. The set of all independent n -tuples is conull and comeager, whereas the particular Cantor set furnished by the theorem may have measure zero. No positive-measure independent Cantor set is claimed. For any probability law on F^n absolutely continuous with respect to Haar measure, an n -tuple sampled from that law is independent over k_Y with probability one.

Corollary 24.6 (The real hull has dimension zero; inherited consequence). *For $F = \mathbb{R}$, every k_Y in theorem 23.1 has Hausdorff dimension zero.*

Proof. It is a proper analytic real closed subfield. Apply Edgar and Miller's theorem on such subfields [24]. $\square$

This is stronger than real nullness but is not a new dimension theorem. Source 04 does not transfer it without proof to p -adic Hausdorff dimension.

24.5 Arithmetic consequences

Lemmas 23.4 and 23.5, together with Glazer's theorem (imported fact 23.6), make $\mathcal{S} = \text{SSy}(M)$ a proper analytic Turing ideal for every nonstandard Borel $M \models \text{PA}$. Theorem 23.12 therefore supplies exactly the hypotheses of theorem 23.1 for every $\mathcal{K}_F(\mathcal{S})$.

Corollary 24.7 (Countable external advice does not close the gap). *For a nonstandard Borel $M \models \text{PA}$, a countable set $Y \subset F$, and $F = \mathbb{R}$ or $\mathbb{Q}_p$, let*

$$k_{F,Y} = \text{acl}_F(\mathcal{K}_F(\text{SSy}(M))(Y)).$$

Then every nonempty open set in F contains a Cantor set independent over $k_{F,Y}$, all finite dependence relations over $k_{F,Y}$ are null and meager, and $\text{trdeg}(F/k_{F,Y}) = \mathfrak{c}$.

Adjoining arbitrary countably many constants is stronger than adjoining only computable ones; the latter are already in the scalar field. It is also different from supplying a single oracle that encodes all those constants and then closing under *all* computations relative to that oracle. Such a Turing-ideal enlargement is a different construction, not covered by this field-generation assertion. The claim is about field generation and relative algebraic closure, as the formula states.

24.6 One Cantor parameter space for all local fields

There are countably many places under consideration. They can be treated simultaneously, without imposing an artificial identification of real and p -adic digits.

Theorem 24.8 (Simultaneous local perfect families). *Fix a nonstandard Borel $M \models \text{PA}$, and use the fields $k_{F,Y}$ of corollary 24.7. For each $v \in \{\infty\} \cup \{p : p \text{ prime}\}$, let $F_\infty = \mathbb{R}$, $F_p = \mathbb{Q}_p$, and let $Y_v \subset F_v$ be countable. There are continuous injections*

$$f_\infty : 2^{\mathbb{N}} \rightarrow (1, 2), \quad f_p : 2^{\mathbb{N}} \rightarrow \mathbb{Z}_p \quad (p \text{ prime})$$

such that, for every place v , the image of f_v is algebraically independent over k_{F_v, Y_v} . More generally one can prescribe a nonempty open real interval and a nonempty open p -adic ball at each place.

Proof. Let $U_\infty = (1, 2)$ and $U_p = \mathbb{Z}_p$, and form the countable product

$$\mathcal{X} = U_\infty \times \prod_p U_p.$$

It is a perfect Polish space. For each v, n , pull back the meager relation $\text{Dep}_n(k_{F_v, Y_v}) \cap U_v^n$ along the coordinate map $\mathcal{X}^n \rightarrow U_v^n$. This map is continuous, open, and surjective, so the pullback of a nowhere dense set is nowhere dense and the pullback of a meager set is meager. Mycielski's theorem supplies a Cantor subset $\mathcal{P} \subset \mathcal{X}$ independent for this countable family of relations.

Choose a homeomorphism $\eta : 2^\mathbb{N} \rightarrow \mathcal{P}$ and set $f_v = \pi_v \circ \eta$. These maps are continuous. They are injective because equality of two v -coordinates is the polynomial relation $X_1 - X_2 = 0$, which has been excluded for distinct members of $\mathcal{P}$. All the other independence requirements hold by construction. Replacing the factors by the prescribed open intervals or balls proves the last assertion. $\square$

The injections need not be computable, and the theorem does not say that one digit sequence, interpreted unchanged at every place, works. Those are stronger synchronization requirements. It does give a single compact parameter space with continuously varying, mutually independent families at all places.

25 The residue image of a nonstandard Peano model

25.1 The residue homomorphism

Let M be a nonstandard model of PA and R_M its difference ring, the ring of formal differences of elements of M (section 20.3). Two pairs represent the same difference when $(a, b) \sim (a', b')$ if $a + b' = a' + b$. Cancellativity and the ordinary arithmetic identities make this an integral domain. Only its algebraic structure is needed; no Polish quotient topology on R_M is assumed.

For every positive standard integer m , division with remainder gives

$$R_M / mR_M \simeq \mathbb{Z} / m\mathbb{Z}. \quad (55)$$

The isomorphism is the unique one carrying standard numerals to their ordinary residue classes. In particular, the compatible prime-power remainders give a unital ring homomorphism

$$\text{res}_p : R_M \longrightarrow \varprojlim_n \mathbb{Z} / p^n \mathbb{Z} = \mathbb{Z}_p. \quad (56)$$

It has kernel $\bigcap_{n \geq 1} p^n R_M$. This kernel is nonzero: for a nonstandard $N \in M$, the nonzero element p^N is divisible by every standard p^n . [Added at this write: (55) is the standard division of lemma 20.3, which needs only $\text{I}\Delta_0$, read through section 20.2; res_p is the p -adic coordinate of the map $\text{res}_{R_M} : R_M \rightarrow \widehat{\mathbb{Z}}$ there.]

25.2 Digits and the standard system

Fix a bit width $w \geq 1$ with $2^w \geq p$. For $z \in \mathbb{Z}_p$, write

$$z = \sum_{i \geq 0} d_i p^i, \quad 0 \leq d_i < p,$$

and encode each d_i into a block of w binary bits. Write $\text{Dig}_p(z) \in 2^{\mathbb{N}}$ for the resulting set. Fast p -adic names and these block-digit codes are mutually computable, uniformly for integral inputs. Thus

$$z \in \mathcal{K}_p(\mathcal{S}) \cap \mathbb{Z}_p \iff \text{Dig}_p(z) \in \mathcal{S}. \quad (57)$$

For the forward direction combine a name-computing oracle with Turing downward closure; for the reverse, the finite sums of the digit expansion are a fast name.

Theorem 25.1 (Exact residue-image theorem). *For any nonstandard $M \models \text{PA}$, whether Borel or not, and $\mathcal{S} = \text{SSy}(M)$, one has*

$$\text{res}_p(R_M) = \mathcal{K}_p(\mathcal{S}) \cap \mathbb{Z}_p =: \mathbb{Z}_p(\mathcal{S}), \quad \text{Frac}(\mathbb{Z}_p(\mathcal{S})) = \mathcal{K}_p(\mathcal{S}). \quad (58)$$

In fact every element of the image is the shadow of an element of M , not merely a formal difference.

Proof. For $a \in M$, each ordinary p -adic digit of $\text{res}_p(a)$ is specified by an arithmetic formula about division by standard powers of p . The corresponding binary block-digit set is a parameter-definable trace and hence belongs to $\mathcal{S}$ by lemma 23.3. Equation (57) gives $\text{res}_p(a) \in \mathbb{Z}_p(\mathcal{S})$. Since $\mathbb{Z}_p(\mathcal{S})$ is a subring, the same holds for all differences.

Conversely let $z \in \mathbb{Z}_p(\mathcal{S})$, and choose $c \in M$ whose binary trace is $\text{Dig}_p(z)$. Internally decode the i th block of w bits of c , and reduce that decoded number modulo p ; call the result $\tilde{d}_i$. This reduction ensures $0 \leq \tilde{d}_i < p$ at all internal indices, even where c has no externally prescribed meaning. At every standard i , one has $\tilde{d}_i = d_i$.

Choose any nonstandard $N \in M$. PA proves totality of the primitive recursive finite-sum operation, so it has the internal element

$$a = \sum_{i < N} \tilde{d}_i p^i.$$

For every standard n , reduction of this internal sum modulo p^n leaves exactly the first n terms, which agree with z . Hence $\text{res}_p(a) = z$. No interchange of an external infinite sum with a ring homomorphism has occurred: only finite arithmetic, internally evaluated at N , and standard residues have been used.

Finally every $x \in \mathcal{K}_p(\mathcal{S})$ becomes integral after multiplication by some standard power p^m . The scalar $p^m x$ still belongs to $\mathcal{K}_p(\mathcal{S})$, so $x = (p^m x)/p^m$ lies in $\text{Frac}(\mathbb{Z}_p(\mathcal{S}))$. The opposite inclusion holds because $\mathcal{K}_p(\mathcal{S})$ is a field. $\square$

Corollary 25.2 (Dense but incomplete residue information). *If M is nonstandard and Borel, $\mathbb{Z}_p(\mathcal{S})$ is a proper dense null meager subring of $\mathbb{Z}_p$. It is the valuation ring of $\mathcal{K}_p(\mathcal{S})$ for the restricted p -adic valuation. Its metric completion is $\mathbb{Z}_p$, while the metric completion of $\mathcal{K}_p(\mathcal{S})$ is $\mathbb{Q}_p$.*

Proof. Density follows from the standard integers. Properness follows from (58) and properness of $\mathcal{K}_p(\mathcal{S})$: if $\mathbb{Z}_p(\mathcal{S}) = \mathbb{Z}_p$, its fraction field would be $\mathbb{Q}_p$. Nullness and meagerness follow by intersecting the null meager field $\mathcal{K}_p(\mathcal{S})$ with the open set $\mathbb{Z}_p$. A valuation ring consists precisely of the nonnegative-valuation members of its field. The completion statements follow from density inside the indicated complete metric spaces. $\square$

There is a sharp distinction between having every *finite* residue and having every coherent infinite residue sequence. For every n , the map to $\mathbb{Z}/p^n\mathbb{Z}$ is onto; nevertheless, the set of coherent sequences actually realized by elements of R_M is only $\mathbb{Z}_p(\mathcal{S})$. Relative algebraic closure of its fraction field does not repair this topological omission.

25.3 The full profinite image and its idempotents

The same question has an exact answer simultaneously at all primes. Write $\widehat{\mathbb{Z}} = \varprojlim_m \mathbb{Z}/m\mathbb{Z} \simeq \prod_p \mathbb{Z}_p$, with its ordinary compact ring topology, and let $\text{res}_{R_M} : R_M \rightarrow \widehat{\mathbb{Z}}$ be the canonical residue map. A convenient effective representation of $z \in \widehat{\mathbb{Z}}$ is the sequence

$$r_{n!}(z) \in \{0, \dots, n! - 1\}, \quad r_{n!}(z) \equiv z \pmod{n!} \quad (n \geq 1).$$

These residues determine z because the factorial moduli are cofinal for divisibility. For a Turing ideal $\mathcal{S}$, put

$$\widehat{\mathbb{Z}}(\mathcal{S}) = \{z \in \widehat{\mathbb{Z}} : (r_{n!}(z))_{n \geq 1} \text{ is computable from some } A \in \mathcal{S}\}.$$

The arithmetic operations on these residue sequences are computable, so $\widehat{\mathbb{Z}}(\mathcal{S})$ is a subring. It contains the ordinary integers.

Theorem 25.3 (Exact profinite image). *For every nonstandard $M \models \text{PA}$ and $\mathcal{S} = \text{SSy}(M)$,*

$$\text{res}_{R_M}(R_M) = \text{res}_{R_M}(M) = \widehat{\mathbb{Z}}(\mathcal{S}).$$

Proof. Fix a primitive recursive injective pairing function $\langle n, j \rangle$ on natural numbers. Since $n! < 2^{n^2}$ for $n \geq 1$, code the factorial residue sequence by

$$A_z = \{\langle n, j \rangle : n \geq 1, j < n^2, \text{Bit}(r_{n!}(z), j) = 1\}.$$

This code and the residue sequence compute one another. Thus $z \in \widehat{\mathbb{Z}}(\mathcal{S})$ exactly when $A_z \in \mathcal{S}$. For $a \in M$, the code of $\text{res}_{R_M}(a)$ is a parameter-definable trace, so belongs to $\mathcal{S}$. Closure under differences gives $\text{res}_{R_M}(R_M) \subseteq \widehat{\mathbb{Z}}(\mathcal{S})$.

Conversely, let $z \in \widehat{\mathbb{Z}}(\mathcal{S})$ and choose $c \in M$ with binary trace A_z . For every internal $n \geq 1$, define in M

$$\tilde{r}_n = \left(\sum_{j < n^2} \text{Bit}(c, \langle n, j \rangle) 2^j \right) \bmod n!.$$

For standard n , this equals $r_{n!}(z)$. Hence the arithmetic property

$$\Phi(N) : \quad \forall 1 \leq i \leq j \leq N \quad \tilde{r}_j \equiv \tilde{r}_i \pmod{i!}$$

holds for every standard N . Overspill supplies a nonstandard N with $\Phi(N)$. In this elementary use of overspill, if there is a least failure of Φ , it is nonstandard and its predecessor is a nonstandard success; if there is no failure, any nonstandard N works. The least-number principle is available in PA with parameter c .

Set $a = \tilde{r}_N \in M$. For every standard n , the property $\Phi(N)$ gives $a \equiv r_{n!}(z) \pmod{n!}$. Since every ordinary modulus divides a factorial, $\text{res}_{R_M}(a) = z$. All coding and summation in this proof are internal finite arithmetic. Overspill is used here explicitly, rather than assuming that arbitrary nonstandard residue prefixes are already coherent. $\square$

Enumerate the ordinary primes as $p_0, p_1, \dots$. For $A \subseteq \mathbb{N}$, let $\mathbf{1}_A \in \prod_i \mathbb{Z}_{p_i}$ have i th coordinate 1 if $i \in A$ and 0 otherwise.

Corollary 25.4 (The standard system is the Boolean algebra of idempotents). *With the canonical prime coordinates understood,*

$$\{e \in \widehat{\mathbb{Z}}(\mathcal{S}) : e^2 = e\} = \{\mathbf{1}_A : A \in \mathcal{S}\}.$$

This identification is a Boolean-algebra isomorphism, where intersection is multiplication and complementation is $e \mapsto 1 - e$. If $\mathcal{S}$ is proper, then

$$\widehat{\mathbb{Z}}(\mathcal{S}) \subsetneq \prod_p \mathbb{Z}_p(\mathcal{S}).$$

If $\mathcal{S}$ is proper and analytic, $\widehat{\mathbb{Z}}(\mathcal{S})$ is a proper analytic dense null meager subring of $\widehat{\mathbb{Z}}$.

Proof. Every idempotent in the product has coordinates zero or one, since each $\mathbb{Z}_p$ is an integral domain. The factorial-residue code of $\mathbf{1}_A$ is computable from A : only finitely many membership bits are needed for each factorial modulus, and the finite Chinese remainder algorithm combines them. Conversely, the residue modulo p_i recovers $A(i)$. Consequently $\mathbf{1}_A \in \widehat{\mathbb{Z}}(\mathcal{S})$ if and only if $A \in \mathcal{S}$, proving the first assertion and its compatibility with Boolean operations.

Every coordinate projection of $\widehat{\mathbb{Z}}(\mathcal{S})$ is $\mathbb{Z}_p(\mathcal{S})$ by the two exact image theorems. Choose $A \notin \mathcal{S}$. Each coordinate of $\mathbf{1}_A$ is the ordinary integer zero or one, so $\mathbf{1}_A \in \prod_p \mathbb{Z}_p(\mathcal{S})$, but $\mathbf{1}_A \notin \widehat{\mathbb{Z}}(\mathcal{S})$. This proves strictness, not merely the possibility of strictness.

Analyticity follows by projecting the Borel oracle-computation relation over $\mathcal{S}$, as in lemma 23.9. The ordinary integers are dense in $\widehat{\mathbb{Z}}$ by finite Chinese remainders. A proper analytic dense subgroup of this compact group is null and meager, by the same difference-set argument as corollary 23.7. $\square$

The ordinary finite quotients of $\widehat{\mathbb{Z}}(\mathcal{S})$ itself are also unchanged. For $m \geq 1$, if $x \in \widehat{\mathbb{Z}}(\mathcal{S}) \cap m\widehat{\mathbb{Z}}$, the residue of x/m modulo $n!$ is computed by taking the residue of x modulo $mn!$ and dividing its integer representative by m . The representative is divisible by m , and factorial codes compute residues at every ordinary modulus. Thus $x/m \in \widehat{\mathbb{Z}}(\mathcal{S})$ and

$$m\widehat{\mathbb{Z}}(\mathcal{S}) = \widehat{\mathbb{Z}}(\mathcal{S}) \cap m\widehat{\mathbb{Z}}, \quad \widehat{\mathbb{Z}}(\mathcal{S})/m\widehat{\mathbb{Z}}(\mathcal{S}) \simeq \mathbb{Z}/m\mathbb{Z}.$$

Its ordinary profinite completion is therefore $\widehat{\mathbb{Z}}$. Thus the distinguished profinite image remembers the complete standard system through its idempotents, even though its finite quotients and its completion do not. The coordinatewise product loses the requirement of one shared oracle, and is strictly larger whenever the standard system is proper. This answers the full-profinite version of the residue-image question, rather than leaving simultaneous realization implicit.

25.4 Canonical residue images of Peano models

Question 28.11 asks which subrings of $\widehat{\mathbb{Z}}$ occur as im res_{R_M} for nonstandard $M \models \text{IA}_0$, and records that corollary 20.9 proves only that the image is not contained in $\mathbb{Z}$. For models of Peano arithmetic theorem 25.3 answers it: the image is $\widehat{\mathbb{Z}}(\text{SSy}(M))$, determined by the standard system alone. The image contains every profinite integer with a computable factorial residue sequence, because $\text{SSy}(M)$ contains every computable set (lemma 23.4); for instance the parameter ζ_* of (36), whose residues modulo $n!$ are the computable S_n , and which is the residue of the element $F(h)$ in the proof of theorem 16.6. Saturation and the choice of model enter only through $\text{SSy}(M)$. The following statement, added at this write, lists the images that occur among countable models. A *Scott set* is a nonempty Turing ideal $\mathcal{S} \subseteq 2^{\mathbb{N}}$ such that every infinite binary tree coded in $\mathcal{S}$ has an infinite path in $\mathcal{S}$.

Corollary 25.5 (Residue images of countable Peano models; [write 04]). *Distinct standard systems of nonstandard models of PA give distinct residue images. A subring $\Lambda \subseteq \widehat{\mathbb{Z}}$ is im res_{R_M} for some countable nonstandard $M \models \text{PA}$ if and only if $\Lambda = \widehat{\mathbb{Z}}(\mathcal{S})$ for a countable Scott set $\mathcal{S}$; and then $\mathcal{S}$ is the set of $A \subseteq \mathbb{N}$ with $\mathbf{1}_A \in \Lambda$.*

Proof. By theorem 25.3, $\text{im res}_{R_M} = \widehat{\mathbb{Z}}(\text{SSy}(M))$, and by the first assertion of corollary 25.4, which uses only that $\text{SSy}(M)$ is a Turing ideal, the idempotents of this ring are exactly the $\mathbf{1}_A$ with $A \in \text{SSy}(M)$. So the image determines $\text{SSy}(M)$, which proves the first and the last assertion. The standard system of every nonstandard model of PA is a Scott set, and by Scott's theorem [30] every countable Scott set is the standard system of a countable nonstandard model of PA; both are classical and imported here. A countable M has a countable standard system. This proves the equivalence. $\square$

The corollary is the write's combination of theorem 25.3 and corollary 25.4 with Scott's theorem; source 04 states neither it nor the Scott-set description, and says only that it does not claim to solve

Scott's general problem. For uncountable models the images are the rings $\widehat{\mathbb{Z}}(S)$ of the standard systems that occur; which uncountable Scott sets occur is Scott's problem, open in general and not addressed here. For models of $\mathsf{I}\Delta_0$ that are not models of PA the question stays open: the proofs of lemma 23.3 and theorems 25.1 and 25.3 use induction for arbitrary arithmetic formulas, overspill and the totality of the internal finite sums. See the note after question 28.11.

25.5 Equal completions, different arithmetic images

The comparison with omnific integers is clearest first in a set-sized ring where all constructions are elementary. Let $k \subseteq \mathbb{R}$ be a subfield and put

$$\Pi_k = \{f \in k((t^{\mathbb{Q}})) : \text{supp}(f) \subset \mathbb{Q}_{<0}\}, \quad (59)$$

$$\mathcal{A}(k, \mathbb{Q}) = \mathbb{Z} + \Pi_k. \quad (60)$$

The support of each Hahn series is well ordered in the increasing order on $\mathbb{Q}$. In the surreal interpretation $t = \omega^{-1}$, negative t -exponents represent purely infinite terms. There is no requirement that the negative support be finite. [Added at this write: this is source 03's ring $\mathcal{A}(k, \Gamma)$ of section 21 with $\Gamma = \mathbb{Q}$, where the ring retraction ct and standard division are already stated.]

Proposition 25.6 (The ordinary residue tower of $\mathcal{A}(k, \mathbb{Q})$). *$\mathcal{A}(k, \mathbb{Q})$ is a ring. Constant extraction $\text{ct} : \mathcal{A}(k, \mathbb{Q}) \rightarrow \mathbb{Z}$ is a surjective unital ring homomorphism with kernel Π_k . For every $n \geq 1$,*

$$\mathcal{A}(k, \mathbb{Q})/n\mathcal{A}(k, \mathbb{Q}) \simeq \mathbb{Z}/n\mathbb{Z},$$

compatibly with reduction maps. Every homomorphism from $\mathcal{A}(k, \mathbb{Q})$ to a finite ring annihilates Π_k and factors through ct .

Proof. Sums of negative exponents remain negative, so Π_k is closed under multiplication, and multiplication by an element of $\mathbb{Z} + \Pi_k$ preserves it. The constant term of a product in $\mathcal{A}(k, \mathbb{Q})$ is exactly the product of its integer constant terms: no pair of negative exponents can sum to zero. This proves the ring and homomorphism assertions.

For every nonzero ordinary integer n , division of all coefficients by n preserves Π_k , so $\Pi_k = n\Pi_k$. It follows that

$$n\mathcal{A}(k, \mathbb{Q}) = n\mathbb{Z} + \Pi_k = \text{ct}^{-1}(n\mathbb{Z}),$$

which proves the quotient formula. If a finite target ring has characteristic $m > 0$ and $f \in \Pi_k$, write $f = mg$ with $g \in \Pi_k$. Its image is zero. The factorization is then through the constant-term quotient. This last argument does not need the homomorphism to be surjective. $\square$

[Added at this write: for the class-sized **Oz** these statements are `odg:thm:finitequotients` and `proposition 17.1`, and they are proved in Lean: `omnificQuotientZModEquiv` (`OmnificResidues.lean`), and `omnific_finite_hom_kills_purelyInfinite` and `omnific_finite_hom_factors` (`OmnificFiniteQuotients.lean`), namespace `SurrealFoundations.SignSequence`. Under $t = \omega^{-1}$, $\mathcal{A}(\mathbb{R}, \mathbb{Q})$ is a set-sized subring of **Oz** with the same constant term (section 26.2); the proposition for $\mathcal{A}(k, \mathbb{Q})$ is not formalized.]

The ordinary p -adic completion means the inverse limit over the standard ideals $p^n R$. It follows that

$$\widehat{(R_M)}_p \simeq \mathbb{Z}_p \simeq \widehat{\mathcal{A}(k, \mathbb{Q})}_p. \quad (61)$$

These identifications are ring isomorphisms and homeomorphisms for the inverse-limit topologies. Nevertheless the canonical images are

$$\text{im}(R_M \rightarrow \widehat{(R_M)}_p) = \mathbb{Z}_p(S), \quad \text{im}(\mathcal{A}(k, \mathbb{Q}) \rightarrow \widehat{\mathcal{A}(k, \mathbb{Q})}_p) = \mathbb{Z}. \quad (62)$$

For example,

$$\frac{1}{p+1} \in \mathbb{Z}_p(S) \setminus \mathbb{Z},$$

because it is a rational p -adic integer and is computable. Therefore these two images are different even when both completions and all the finite stages are identified canonically.

Theorem 25.7 (Finite completion does not determine the canonical image). *A signed nonstandard PA ring R_M and the omnific-type ring $\mathcal{A}(k, \mathbb{Q})$ have identical ordinary finite residue systems and identical ordinary p -adic completions, but their canonical images in those completions are different. If R_M comes from a Borel model, its image has the proper analytic fraction field of theorem 23.12; the image of $\mathcal{A}(k, \mathbb{Q})$ is always the ordinary integers.*

Proof. Combine (55), proposition 25.6 and theorem 25.1. The transition maps are all the ordinary residue reductions, so the inverse limits agree as claimed. Equation (62) and the exhibited rational element separate the images. $\square$

[Added at this write: for **Oz** the second half is `odg:eq:profinite` and is proved in Lean: the canonical maps to the p -adic and to the profinite completion are constant extraction followed by the ordinary embedding of $\mathbb{Z}$ (`omnificPAdicCompletionEquiv_of` in `OmnificPAdicCompletion.lean`, `omnificProfiniteCompletionEquiv_of` in `OmnificProfiniteCompletion.lean`). The first half is theorem 25.1. The theorem is the exact Peano form of the remark after proposition 20.2: neither the finite quotients nor the abstract inverse limit separate omnific from nonstandard arithmetic, and the canonical image does.]

Corollary 25.8 (A useful nonidentification). *There is no unital ring homomorphism $R_M \rightarrow \mathbb{Z}$, and hence none $R_M \rightarrow \mathcal{A}(k, \mathbb{Q})$. This holds for every nonstandard PA model, without the Borel hypothesis.*

Proof. A unital homomorphism $h : R_M \rightarrow \mathbb{Z}$ preserves each standard remainder: from $d = p^n q + r$ it follows that $h(d) \equiv r \pmod{p^n}$. Therefore $\text{res}_p(d)$ would equal the ordinary integer $h(d)$ inside $\mathbb{Z}_p$, forcing $\mathbb{Z}_p(S) \subseteq \mathbb{Z}$, a contradiction. Composing a hypothetical map to $\mathcal{A}(k, \mathbb{Q})$ with `ct` gives the forbidden map to $\mathbb{Z}$. $\square$

[Added at this write: theorem 20.4 proves more, for every nonstandard model M of $\text{I}\Delta_0$, namely $\text{Hom}_{\mathbb{Z}}((R_M)_{\text{add}}, \mathbb{Z}) = 0$; so there is no additive map $R_M \rightarrow \mathbb{Z}$ sending 1 to 1, and by corollaries 20.6 and 21.4 none into $\mathcal{A}(k, \mathbb{Q})$ or **Oz**. The proof above is a second route to the ring-homomorphism case for Peano models, through the residue image: it is the criterion of proposition 20.2, applied with the explicit non-diagonal residue $1/(p+1)$.]

This corollary is included to prevent an incorrect interpretation of the bridge. Source 04 does *not* embed the original PA ring into the omnific integers. It transports algebraic information about the scalar field extracted from its standard system. The two routes must not be conflated.

26 Hahn coefficients and an invisible omnific family

26.1 Coefficientwise independence in Hahn fields

For a field k and an ordered abelian set-group Γ , the Hahn field $k((t^\Gamma))$ consists of formal sums

$$f = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma$$

with well-ordered support. Equality is coefficientwise. Addition is coefficientwise, and multiplication is the usual well-defined convolution. If $k \subseteq k'$, the series with coefficients in k form the subfield

$k((t^\Gamma))$ of $k'((t^\Gamma))$. Only standard Hahn arithmetic is used; no topology on the collection of all possible coefficient fields is needed.

Theorem 26.1 (Coefficient-preserving independence transfer). *Let $k \subseteq k'$ be fields and Γ an ordered abelian set-group. If $P \subseteq k'$ is algebraically independent over k , its constant images in $k'((t^\Gamma))$ are algebraically independent over $k((t^\Gamma))$. More generally, for any choices $u_r \in k((t^\Gamma))^\times$ indexed by $r \in P$, the family $\{u_r r : r \in P\}$ is algebraically independent over $k((t^\Gamma))$.*

Proof. Take distinct $r_1, \dots, r_m \in P$ and a polynomial

$$Q(X_1, \dots, X_m) = \sum_{v \in \mathcal{E}} a_v X^v, \quad a_v \in k((t^\Gamma)),$$

where $\mathcal{E} \subseteq \mathbb{N}^m$ is finite. Suppose $Q(\bar{r}) = 0$. Write $a_v = \sum_\gamma a_{v,\gamma} t^\gamma$. The coefficient of t^γ in the assumed equality is

$$\sum_{v \in \mathcal{E}} a_{v,\gamma} \bar{r}^v = 0. \quad (63)$$

This is a polynomial relation over k . Independence of the r_i forces $a_{v,\gamma} = 0$ for every v and every γ . Hence every $a_v = 0$ and $Q = 0$.

Only a finite outer sum was used in (63); this is not an illicit exchange of arbitrary infinite sums. For the final assertion, substitution $X_i \mapsto u_{r_i} X_i$ is an automorphism of the polynomial ring over $k((t^\Gamma))$, with inverse $X_i \mapsto u_{r_i}^{-1} X_i$. It preserves nonzero polynomials and reduces the assertion to the constant case. $\square$

[Added at this write: the first assertion is an immediate consequence of the coefficient-extension lemma of the independent-surreal-copies report, `isc:cp:lem:coeffld`: for $k \subseteq k'$, the fields k' and $k((t^\Gamma))$ are linearly disjoint over k inside $k'((t^\Gamma))$, in the same small- t convention. The distinct monomials in P are linearly independent over k , hence over $k((t^\Gamma))$, which is algebraic independence. The proof above is the same coefficient comparison.]

There is no bound on the cardinality or complexity of the supports beyond the set-sized Hahn definition. The theorem therefore identifies an exact limitation of support enrichment: introducing more monomials and longer well-ordered supports cannot create a polynomial relation between scalar constants whose coefficients are still restricted to k .

Corollary 26.2 (Persistent real coefficient gap). *For a nonstandard Borel $M \models \text{PA}$, countable $Y \subseteq \mathbb{R}$, and $k_{\mathbb{R},Y} = \text{acl}_{\mathbb{R}}(\mathcal{K}_\infty(\text{SSy}(M))(Y))$, choose a Cantor set $P \subset (1, 2)$ independent over $k_{\mathbb{R},Y}$. Then for every ordered abelian set-group Γ , the same P is independent over $k_{\mathbb{R},Y}((t^\Gamma))$ inside $\mathbb{R}((t^\Gamma))$.*

Proof. Use corollary 24.7 and theorem 26.1. $\square$

A corresponding complex-coefficient statement also holds. If P is real and independent over $k' \subseteq \mathbb{R}$, it is independent over $k'(i)$: split a putative relation into real and imaginary parts. Theorem 26.1 then gives independence over $k'(i)((t^\Gamma))$ in $\mathbb{C}((t^\Gamma))$. This is the relevant surcomplex analogue; it makes no assertion about independence over the whole surcomplex field.

26.2 A bounded-support route into the surreals

For rational exponents, Conway normal forms give the familiar embedding

$$\mathbb{R}((t^{\mathbb{Q}})) \longrightarrow \text{No}, \quad \sum_q a_q t^q \mapsto \sum_q a_q \omega^{-q}. \quad (64)$$

Increasing well-ordered t -support becomes decreasing well-ordered Conway support in the usual reverse-order convention. Normal-form addition and multiplication agree with Hahn arithmetic. The

identification between surreal normal forms and Hahn-style arithmetic is also part of the ProveIt development consulted by source 04 [31]; a primary modern treatment is [29].

Under (64), $\mathcal{A}(\mathbb{R}, \mathbb{Q}) = \mathbb{Z} + \Pi_{\mathbb{R}}$ is a set-sized subring of Conway's omnific integers: all nonconstant growth exponents are positive and the constant coefficient is an ordinary integer. In particular $r\omega$ is an omnific integer for every $r \in \mathbb{R}$.

Theorem 26.3 (An invisible independent family at one scale). *Let $M, S, Y, k_{\mathbb{R},Y}$ and P be as in corollary 26.2, with $\Gamma = \mathbb{Q}$. The family*

$$\mathcal{Z} = \{z_r = r\omega : r \in P\}$$

has cardinality $\mathfrak{c}$, is algebraically independent over the embedded field $k_{\mathbb{R},Y}((t^{\mathbb{Q}}))$, and has the following properties:

- (1) *Every z_r is purely infinite and lies in the set-sized ring $\mathcal{A}(\mathbb{R}, \mathbb{Q}) \subset \mathbf{Oz}$.*
- (2) *For every positive ordinary integer n , every z_r belongs to $n\mathcal{A}(\mathbb{R}, \mathbb{Q})$ and therefore maps to zero in $\mathbf{Oz}/n\mathbf{Oz}$.*
- (3) *Every homomorphism from $\mathcal{A}(\mathbb{R}, \mathbb{Q})$ to a finite ring kills the whole family. Nevertheless, for each finite distinct tuple $z_{r_1}, \dots, z_{r_m}$, evaluation gives an injective homomorphism*

$$k_{\mathbb{R},Y}((t^{\mathbb{Q}}))[X_1, \dots, X_m] \hookrightarrow \mathbb{R}((t^{\mathbb{Q}})).$$

Proof. Use theorem 26.1 with $u_r = t^{-1} = \omega$. Each element has negative t -support $\{-1\}$ and zero constant term. Since $z_r/n = (r/n)\omega \in \mathcal{A}(\mathbb{R}, \mathbb{Q})$, it is divisible by every ordinary n . The finite-image assertion is proposition 25.6. The evaluation map is injective exactly because the tuple is algebraically independent over the specified coefficient field. $\square$

The coefficient field is essential. Over the full real field, two nonzero members $r\omega$ and $s\omega$ satisfy $sX_1 - rX_2 = 0$. This does not contradict the theorem because those coefficients need not belong to $k_{\mathbb{R},Y}((t^{\mathbb{Q}}))$. The statement neither claims independence over $\mathbb{R}$ nor mistakes algebraic independence for independence of distinct valuation scales: all these elements use the *same* scale ω . [Added at this write: each z_r is, separately, transcendental over $\mathbb{R}$, as every infinite omnific integer is (Lean `omnific_transcendental_of_not_finite` in `OmnificPolynomialRoots.lean`); what fails over $\mathbb{R}$ is the joint independence. By `odg:thm:finitequotients` every element of Π , not only these, is zero in every $\mathbf{Oz}/n\mathbf{Oz}$.]

26.3 Which topology has the perfect set?

The Cantor structure lives on $P \subset \mathbb{R}$ and may be transported to $\mathcal{Z}$ as a coefficient-parameter topology. It is not the topology induced by the order on $\mathbf{No}$. If $r \neq s$ are real, then $|r - s|\omega$ exceeds every positive ordinary integer. Consequently

$$(r\omega - 1, r\omega + 1) \cap \mathcal{Z} = \{r\omega\}.$$

Thus $\mathcal{Z}$ is discrete in the surreal order topology. The same warning applies to the Hahn valuation topology: distinct nonzero constant differences have valuation zero, and the common monomial factor does not turn the real coefficient topology into that topology.

This is why “Cantor-parameterized omnific family” is the accurate phrase. The real coefficient set is perfect; the omnific image carries several inequivalent natural topologies, and perfectness is not invariant under changing them.

26.4 Set theory without class paradoxes

All fields, models, countable parameter sets, Cantor sets, and Hahn supports used in the main proofs are sets. For $\Gamma = \mathbb{Q}$ the domain of (64) is a set; its images are set-coded surreal sign sequences, and their birthdays have an ordinal bound by Replacement. One can therefore perform the entire application in a sufficiently large set-sized initial segment of the ambient class of surreal numbers, without treating the proper class **No** as an element of itself.

In a class theory, (64) is simply a class-defined map on a set domain. In a universe-polymorphic type theory, the same proof should first be stated for a set-sized Hahn field and an embedding into a larger universe. Neither a global well-order of all surreals nor quantification over a “set of all class embeddings” is needed. The general coefficient theorem is a scheme over set-sized Γ , not a Hahn construction whose exponent group is assumed to be a set while it is actually the whole proper class **No**. This is the same discipline as section 2 and section 21.6.

26.5 Scope, sharpness, and what is not constructed

Nonstandardness cannot simply be removed. For the standard model $M = \mathbb{N}$, binary traces of single integers are only finite sets. They are not a Turing ideal containing all computable sets. Its signed ring is $\mathbb{Z}$ and its residue image is $\mathbb{Z} \subset \mathbb{Z}_p$, not the valuation ring of all computable p -adic numbers. Thus the nonstandard hypothesis is already necessary for the exact coding bridge as stated.

Topological and algebraic closure point in opposite directions. For a Borel nonstandard model, $\mathcal{K}_p(S)$ is algebraically closed relative to $\mathbb{Q}_p$ but is not metrically complete. Its topological completion is all of $\mathbb{Q}_p$, whereas adjoining countably many constants and taking relative algebraic closure still leaves a perfect transcendence gap. The completion adds limits of arbitrary Cauchy sequences, not merely roots of polynomials or values of finitely many field operations. Similarly, allowing every Hahn support with coefficients in $k_{\mathbb{R},Y}$ is not the same operation as replacing $k_{\mathbb{R},Y}$ by its topological completion.

No named transcendental witness is supplied. Every computable real belongs to $\mathcal{K}_\infty(S)$, and every computable p -adic point belongs to $\mathcal{K}_p(S)$. Consequently no member of the independent sets can have a computable name. In particular the theorem does not produce a new independence result about a pair of familiar computable constants. It produces independence *over a field already containing all such constants*.

There is also no computable injection from all of Cantor space into one of the independent sets: evaluating it at the computable branch $0^\mathbb{N}$ would give a computable point of that set, a contradiction. A continuous injection exists by theorem 24.8, but a code for it carries additional noncomputable information. The proof is existential, not a finite approximation algorithm for the full perfect family.

No global theory of arbitrary presentations is assumed. The scalar field depends only on the standard system, not on a selected Polish topology on the presentation of M . The Borel hypothesis is used to obtain analyticity and properness, and not to select canonical oracles for elements. The argument also does not require continuous arithmetic operations, a Polish topology on the signed quotient ring, or a Borel global root-selection function. All such stronger assertions would need separate proofs.

27 Verification, provenance, and a formalization route

Each subsection below first gives source 01’s record, as delivered, and then those of sources 02, 03 and 04; “this manuscript” in source 01’s text means source 01.

27.1 The exact dependency chain

The proof separates into three kinds of input. Standard surreal normal forms identify the ambient additive spaces. The established Ehrlich–Kaplan criterion decides when a specified canonical Hahn image is initial. Everything connecting the original discrete group to the normalized image is then given by the explicit formulas and proofs in this manuscript.

The essential chain is

$$\begin{aligned} & \text{bottom coefficient structure} \longrightarrow \text{root relocation} \\ & \longrightarrow \text{strong Hahn transport} \longrightarrow \text{initiality of the image.} \end{aligned}$$

For the inverse classification, the additional chain is

$$\begin{aligned} & \text{missing ancestors of } p \longrightarrow \text{spine exponents} \\ & \longrightarrow \text{forced dyadic coefficients} \longrightarrow K_\alpha \subseteq H. \end{aligned}$$

The sign-sequence proof is independent of the arithmetic of exponents. In particular, no unproved assertion that Θ_α preserves addition of exponents occurs in the chain.

Source 02. The convex-factor theorem depends on normal forms and the initial Hahn criterion, then only on the proved projection and sign lemmas. The Presburger criterion adds the divisible initial-embedding theorem (imported fact 11.1). The explicit nonsplit examples add no further structural input. The elementary-equivalence consequences use the classical characterization of Presburger models; the ultrapower statement additionally uses Łoś’s theorem. The Peano statement uses the usual provably total factorial and summation recursions. The surcomplex module assertions are coordinatewise consequences of the real factor theorem. The merge statements proposition 13.5 and corollaries 13.6 and 17.3 add source 01’s sections 4 and 8 and, for the last, statements of the trigonometry report cited by label.

Source 03. The detector (theorem 19.4) uses only discreteness, ordinary division by 2 and 3, and a normalized additive map into the actual ordinary integers; its inductiveness (lemma 19.6) uses only division by 2. The character classification adds division by every ordinary integer; the no-map theorem adds bounded induction (lemma 20.3); the compiler adds an actually positive infinite bound; the complexity theorem adds an effective presentation and a specified code for that bound, but no effectiveness of the character. The open-induction example imports the classical integer-part theorem (section 21.3). The [write] statements of section 20.5 add source 02’s Theorem E, proposition 14.3, and corollary 17.2, and remark 20.5 adds [14, Theorem 2.9].

Source 03 lists hypotheses that must not be silently weakened. The detector needs a character with values in the *ordinary* integers: a map into a nonstandard copy of arithmetic does not support the external 2-adic case split used in the proof. Standard division means externally indexed division by every ordinary modulus; the detector itself needs only 2 and 3. A unit-preserving additive map is stronger than an arbitrary additive embedding, and no theorem here excludes every additive embedding without the unit condition. The character theorem uses standard division to normalize a nonzero character; the uniqueness replacement (proposition 21.2), in contrast, needs only division by 2. The truth compiler assumes an actually positive infinite bound. Its uniformity is over formulas, rings, and such bounds; no computable test for infinitude in arbitrary presentations is asserted. An effective presentation and a specified element code are needed for the upper many-one reduction.

Source 04. The perfect-gap theorem (theorem 23.1) depends on the analytic-derivation lemma (lemma 24.1: extension of a derivation through a separable algebraic extension, the analytic graph, and automatic continuity of Borel homomorphisms), on Fubini and Kuratowski–Ulam for the aggregate relations, and on Mycielski’s theorem, in the form of [26, Theorem 1]. The arithmetic applications add lemmas 23.3 to 23.5, which use full induction, Glazer’s theorem (imported fact 23.6), imported, and quantifier elimination for real closed fields and for $\mathbb{Q}_p$ [27], imported. The exact images (theorems 25.1 and 25.3) use only definable traces, internal finite sums and, for the profinite image, an explicit overspill; they need no Borel hypothesis. The Hahn and omnific statements use coefficient comparison and the rational Hahn slice of the surreal normal forms. The real dimension corollary imports Edgar and Miller’s theorem [24] and is not used elsewhere. Source 04’s dependency and novelty ledger, its Appendix B, reads as follows with this report’s numbers (its shipped 04-transcendence-gaps-theorem_ledger.md is a longer version with source 04’s numbers).

Claim	Principal dependency	Status
No full Borel standard system	Glazer’s seminar theorem	External input, not reproved (imported fact 23.6).
Scalar-field algebraic closure	Oracle root computation	Proved, using standard effective local methods (theorem 23.12).
No positive countable transcendence degree	Analytic graph of a derivation; automatic continuity	Full proof; priority not certified (lemma 24.1).
Perfect parameter-robust gap	Proper hulls, Fubini, Kuratowski–Ulam, Mycielski	Assembled theorem proved; classical inputs credited (theorem 23.1).
Real Hausdorff dimension zero	Edgar–Miller theorem	Inherited corollary, not a new dimension result (corollary 24.6).
Exact p -adic residue image	PA definable traces and internal recoding	Full proof, without a Borel hypothesis (theorem 25.1).
Full profinite image and idempotents	Factorial residue codes, overspill, and finite CRT	Exact shared-oracle image and Boolean reconstruction proved (theorem 25.3 and corollary 25.4).
Simultaneous places	Open coordinate projections and Mycielski	Proved; no uniform digit claim (theorem 24.8).
Hahn coefficient transfer	Coefficient equality and finite polynomial sums	Full elementary proof (theorem 26.1).
Invisible omnific family	Perfect real coefficients and Hahn transfer	Proved on an explicit set-sized slice (theorem 26.3).
Equal completions, different images	Exact residue image and constant-term quotient	Full comparison proof (theorem 25.7).
Lean/Rocq endpoints	Future formalization work	Not implemented or verified in source 04’s package.

The [write 04] corollary (corollary 25.5) adds the exact profinite image, the Boolean reconstruction and Scott’s theorem [30], imported.

27.2 Finite regression checks supplied with the article

The accompanying Python program `verify.py` (shipped in this report as `code/verify.py`) uses only the standard library, exact rational arithmetic, and a fixed random seed. Its recorded run performs **450,862 assertions**, all passing (shipped as `data/verification_report.json`). The tests cover finite sign words through length seven for all relevant finite cone minima; every prefix-closed subset of the binary tree through depth three, including the empty tree; inverse-spine tests on a further family of nonnegative initial trees; and 1,500 random pairs of finite rational normal forms.

The count of prefix-closed trees is independently specified by

$$T_{-1} = 1, \quad T_d = 1 + T_{d-1}^2,$$

so $T_0 = 2$, $T_1 = 5$, $T_2 = 26$, and $T_3 = 677$. The recurrence counts the empty tree, or a root with independently chosen initial left and right subtrees. This makes the finite-tree coverage exhaustive at the stated depth, rather than a random sample.

Tests check inverse identities, order preservation, the exact predecessor identity, the right-ancestor identity away from the minimum, loss of the minimum's right ancestors, initiality of transported trees, the exact inverse exponent criterion, and the source dyadic obligations recovered by adjoining the spine. The series tests check inverse transport, additivity, order, truncations, and integrality of the bottom coefficient. Separate negative tests reject the false global right-ancestor identity and unrestricted inversion.

The program does not represent arbitrary ordinals, arbitrary real coefficients, or infinite Hahn supports. It does not check the imported characterization theorem or establish absence of a transfinite proof gap. Its role is regression testing of the finite combinatorics and coefficient formulas. The written proofs, not these experiments, carry the general claims.

When this report was placed, the program was rerun on a copy of the directory with an explicit output file. It passed with the same 450,862 assertions, and its JSON output was identical in content to the shipped record. The program's default output file is `verification_report.json` in the current directory, and the delivered build script calls it with that name, so a rerun should always be made on a copy (the collection README gives the commands).

Source 02. Its program, shipped as `code/02-convex-factors-verify.py` (delivered as `code/verify.py`), uses only the Python standard library and exact integers and rational fractions. Its recorded run, shipped as `data/02-convex-factors-verification.json` (delivered as `data/verification.json`), returns PASS. The main coverage is:

Check family	Count
All nonempty convex intervals in the sign tree of lengths ≤ 5	2,016
All nonempty meet-closed subsets of the sign tree of lengths ≤ 2	61
Pairs checked for order, prefix reflection, and meet preservation	1,398,609
Nested convex interval pairs in the sign tree of lengths ≤ 4	46,376
Points checked for two-stage compression coherence	324,632
Factorial transition identities	100
Compatibility checks between factorial residues	5,050
Integer-parameter splitting identities	1,769
Finite projection composition identities	24,300

No row certifies a transfinite theorem. The program additionally checks initiality of every finite compressed image (145,740 prefix checks), the non-meet-closed collision example, the solvability of all finite retraction systems through sixty stages, the factorial bound used in lemma 15.3, and additivity of interval projections (2,970 checks); the full counts are in the JSON record. The successful finite retractions are a deliberate regression case: the code must not mistake finite solvability for a global splitting. The tests do not verify the Ehrlich–Kaplan theorem, arbitrary ordinals, class comprehension, completeness of Presburger arithmetic, recursive saturation, or the Peano-model argument. Those depend on the written proofs and the explicitly imported results, not on the test counts.

The program prints its JSON report and writes a file only when `-output` is given; the delivered README's rerun command names the delivered record as that output and would overwrite it, so it too should be rerun on a copy. At this write it was rerun on a copy with a scratch output file: it passed in about ten seconds, and the JSON content was identical to the shipped record.

Source 03. Its delivered files are shipped unchanged under these names:

shipped here	delivered as
code/03-standard-cut-verify.py	code/verify.py
code/03-standard-cut-compile_interval.py	code/compile_interval.py
code/03-standard-cut-build.sh	build.sh
data/03-standard-cut-verification.json	verification.json
data/03-standard-cut-compiler_report.json	compiler_report.json
data/03-standard-cut-BUILD_AUDIT.json	BUILD_AUDIT.json
03-standard-cut-SOURCE_AUDIT.md	SOURCE_AUDIT.md

The two Python programs use only the standard library. The first performs exact rational-polynomial checks of standard division and of the explicit odd-factor certificates in $\mathcal{A}_0$, including zero and negative constant terms; it also checks the finite integer dyadic predicate and the two evaluation kernels of proposition 21.1. The second implements a finite syntax tree and the single-bound compiler; its audit checks that every output quantifier has bound Ω (written H), that no free variable other than Ω remains for sentence inputs, and that every atom has total degree at most two. Local gate evaluations are checked on ordinary integer assignments. The recorded runs (seed 20261003) return PASS:

Check family	Count
Infinite-polynomial odd-factor certificates in $\mathcal{A}_0$	2,000
Division certificates	12,000
Character identities	4,000
Finite cases of the dyadic predicate Pow^{dy}	1,025
Finite cases of the standard-cut predicate Std^{dy}	1,025
Pairs of distinct evaluation kernels	272
Compiled sentence examples (with full outputs)	5
Local gate evaluations	768

Neither program evaluates all quantifiers in an infinite interval, checks all models, or constitutes a proof-assistant verification. The universal claims rest on the proofs; the code checks algebraic certificates and the concrete implementation of syntax transformations. The compiler takes formula trees through its constructors; it does not parse arbitrary strings or evaluate quantified formulas at infinite bounds. At this write both programs were rerun on a copy with scratch output files (Python 3.14.4): they passed in about two seconds and under one second, and both JSON outputs were identical in content to the shipped records. The programs' default output files are `verification.json` and `compiler_report.json` in the current directory, so a rerun should be made on a copy or with an explicit `-output`. The shipped `data/03-standard-cut-BUILD_AUDIT.json` records the delivered PDF (20 pages, no overfull boxes or unresolved references), which is not shipped.

Source 04. Its delivered files are shipped unchanged under these names:

shipped here	delivered as
code/04-transcendence-gaps-verify.py	verify.py
data/04-transcendence-gaps-verification.json	verification.json
04-transcendence-gaps-theorem_ledger.md	theorem_ledger.md
04-transcendence-gaps-source_audit.md	source_audit.md

The program uses only the Python standard library and exact arithmetic; there are no floating-point near-zero tests, and it does not use `assert`, so its checks also run under `python -O`. It checks finite base-digit recodings, factorial residues and idempotent Chinese remainders, arithmetic of finite rational Laurent tails and constant terms, the quotient-divisibility identities used for $\mathcal{A}(k, \mathbb{Q})$, and exact rational Newton iterates for a 7-adic root of $X^2 - 2$. Its recorded run returns PASS:

Check family	Count
Base- p digit words, $p = 2, 3, 5, 7$, lengths 0–5 (exhaustive)	23,941
Prefix-residue comparisons	139,168
Recodings with an uncontrolled tail	23,941
Exact finite Laurent-tail identities (1,000 seeded pairs, seed 20261003)	38,000
Exact Newton stages for a 7-adic root of $X^2 - 2$	7
Factorial-residue samples	720
Decoded factorial-residue checks	4,320
Finite coherence comparisons	15,120
Idempotents modulo 8!	16
Prime-coordinate recovery checks	64
Boolean-operation identities	272

These are regression checks of finite identities appearing in the proofs. They cannot certify algebraic independence over an uncountable field, existence of a perfect set, a Borel-model theorem, Glazer’s theorem, overspill or transfinite Hahn arithmetic; a finite check reporting “passed” must not be promoted into any of those assertions, and symbolic variables in a test are not numerical transcendental witnesses. The Newton check caps its modulus at 7^{12} by design, so its record shows the same checked modulus for iterations 4–6 while the residual valuations 16, 32, 64 are checked exactly. The program writes `verification.json` beside itself unless `-output` is given; shipped in `code/`, it would write `code/verification.json`, not the shipped record. At this write it was rerun on a copy with a scratch output file (Python 3.14.4): it passed in a few seconds, and its JSON content was identical to the shipped record. Source 04’s delivered checksum manifest `SHA256SUMS` (seven files) was verified at placement and dropped, as repository policy requires.

27.3 Relationship to the Surreal repository

The repository review examined the public README, the introductory coverage and dependency ledger, and targeted searches for overlap [6]. Those materials describe extensive sign-sequence, canonical cut, normal-form, Hahn-summation, and algebraic infrastructure. They also explicitly distinguish source manuscripts from checked Lean declarations. The present argument is designed to fit that distinction.

The review took place while the repository was changing. The initially inspected README and ledger had blob hashes

```
391afd7c64cebd48b47fbe43dae26c27e8ee8047
6a89497cb1de3e5c7a346201f7fc006d85082426.
```

A later metadata read reported commit

```
173eb522fdf5d4bf201c097f76b38273eb4f95e1.
```

The initially returned recursive-tree hash is not used as a commit identifier. This manuscript does not claim to have reviewed every repository article or every companion archive, and no repository result is silently promoted from “pending” to “proved.” The manuscript has no pinned commit. The commit 173eb52 above is an ancestor of the commit c6359e4 that placed the manuscript in the collection. The manuscript makes no claim about the content of any report, so no statement of it had to be corrected against the collection as it now stands.

A sound implementation would first formalize the following mathematical interfaces; the names are proposed work units, not declarations claimed to exist:

1. *Cone root relocation.* Define the three-way sign split, the inverse, the order equivalence, and (8) for ordinal-length sequences.

2. *Hahn reindexing with one coefficient multiplier.* Prove support transport, real-linearity, order preservation, truncation compatibility, and strong-sum interchange.
3. *Concrete initial-subgroup criterion.* Supply a checked version of [imported fact 2.2](#), including both its coefficient and right-ancestor conditions, before claiming the final theorem.
4. *Normalization and exact inverse.* Instantiate the criterion for the transformed groups; then derive the spine classification, quotient realization, and complex-coordinate extension.

The largest imported proof obligation is the third item. A checked conditional theorem using that criterion would be valuable, but would not by itself constitute a checked theorem about every actual initial subgroup of **No**. This package supplies no Lean file with that obligation hidden as an axiom or an unfinished proof.

Source 02. Its repository review was made at commit

9693b28c24e6fcb317ce47969a40185c5dfef402,

and included the main README, the formalization ledger, the article catalogue exposed there, and the guide of the omnific groups and lattices report; these distinguish mathematical drafts from Lean results. It did not constitute a claim-by-claim audit of the repository or a local Lean build. Source 02 read source 01 from the project library, including its quotient, suspension and relative splitting arguments, and recorded that its location in the inspected Git tree was not established, citing it as a preceding project draft without assigning it a repository path. That was correct at its pin: 9693b28 is an ancestor of the commit c6359e4 that placed source 01, and the preceding draft is now [sections 1 to 10](#) of this report (its “Section 8” is [section 8](#)). No theorem of source 02 is called Lean-verified. A sensible implementation order would formalize the finite and transfinite sign-tree compression first, followed by the support projection lemmas and the transfer of the initial Hahn criterion. The residue and factorial-presentation arguments can be formalized in ordinary abelian-group and ordered-group libraries independently of a full implementation of omnific integers. This is a dependency plan, not delivered formal code.

Source 03. Its repository review was made at the ProveIt commit

883e0b3b22263bd9d92fdbbf10376ca4b652b862,

an ancestor of its placement commit 0bd0e5527. It consulted the root and surreal-number reader’s guides and the Lean file `Algebra/SurrealNumbers/Surreal/Foundations/OmnificResidues.lean` (blob aacd3cac, unchanged at this write), naming the declarations `omnific_int_dvd_iff`, `omnific_prime_pow_dvd_iff`, `omnific_dvd_all_pos_int_iff` and `iInf_omnific_int_multiples` (namespace `Surreal.Foundations.SignSequence`); all four exist. They formalize, for the actual omnific ring, that divisibility by a nonzero ordinary integer is read off the constant term and that the intersection of the ideals $n\mathbf{Oz}$ is the purely infinite ideal Π : the $\mathbf{Oz}$ case of standard division and of (41), which is `odg:thm:finitequotients` and `odg:eq:divkernel`. The constant term is the ring homomorphism `omnificConstantCoeff` of `OmnificIntegers.lean`, and its uniqueness among unital ring homomorphisms to $\mathbb{Z}$ is `omnific_int_hom_eq_constant` ([section 21.1](#)). Nothing else of source 03 is formalized. It makes no claim about the content of any report, except that the Diophantine-retraction observation of [section 22.6](#) underlies related claims in the repository, which is true (`odg:cor:H10`). It was not a validation of every research claim in the repository.

Source 03 proposes a Lean plan; the names are proposed work units, not declarations claimed to exist. The elementary layer can be formalized independently of surreal numbers: a linear ordered commutative ring with least positive element 1, a predicate expressing ED_n , and an additive homomorphism $\chi : R \rightarrow \mathbb{Z}$ carrying 1 to 1; then finite-interval rigidity, residue detection and the finite odd-factor certificate, with the semantic predicates Pow^{dy} and Std^{dy} kept separate from their encoded first-order formulas, identifying their meanings by an explicit bridge theorem. The next layer proves successor closure and a semantic contradiction with the one bounded induction instance; a model-theoretic interface

then supplies standard division from ID_0 and forms the difference ring. Only after these reusable lemmas should the actual omnific constant-term map be supplied, where the declarations above already exist. For the compiler, one formalizes capture-avoiding fresh-variable generation, substitution, local gate uniqueness and structural truth preservation; a theorem merely stating correctness of an abstract translator is not enough, it should identify the executable translator and prove its bound and degree invariants. The package includes no Lean file and makes no claim that these theorems have been kernel-checked. In the repository's arithmetic syntax, a nonstandard model would be a `SetTheory.PA.PreModel` satisfying the relevant axioms, not a `SetTheory.PA.Model`, whose induction for every Lean predicate makes it standard (`Logic/Interpretability/PAHF/Lean/PAHF/PASyntax.lean`); the Presburger development `Logic/PresburgerArithmetic` decides additive arithmetic over the ordinary integers. Neither formalizes any statement of source 03. [This paragraph's last three sentences are the write's.]

Source 04. Its repository review was made at the ProveIt commit

7d9b8b957fa6a1cc28636ac8fce3cfaa6d230031,

a merge commit and an ancestor of its placement commit 12076b2e8, through a GitHub connector. It inspected `Algebra/SurrealNumbers/README.md` and `Logic/PeanoArithmetic/ListCoding/README.md`, both unchanged at this write, and the root README and directory metadata for navigation; it did not review all reports or formal source files, and no Lean or Rocq build or kernel audit was run for it. Its two claims about these files are true: the surreal guide documents normal forms, omnific constant extraction, finite quotients, the p -adic completion and its canonical image, and it separates formalized foundations from unrefereed reports; the list-coding guide distinguishes the standard-natural interpretation of its formulas from internally derived universal theorems. [Added at this write: the **Oz** facts named there are Lean theorems, not only documentation; the declarations corresponding to [proposition 25.6](#) and [theorems 25.7](#) and [26.3](#) are named at those statements. No statement of source 04, and not the [write 04] corollary, is formalized. As for source 03, a nonstandard model of arithmetic in the repository's syntax would be a `SetTheory.PA.PreModel`, not a `SetTheory.PA.Model`.]

Source 04's formalization plan, as it gives it. The list-coding distinction prevents a false formalization claim: a Lean proof about evaluating a formula in $\mathbb{N}$ does not, by itself, justify using its totality at a nonstandard $N \in M$. The exact-residue theorem requires a syntactic PA derivation, or a separate semantic theorem valid in every PA model, for the bounded digit-recoding and finite-sum operations. This additional obligation is recorded rather than silently imported from the standard-model coding documentation.

Layer	Mathematical interface	Verification obligation
Arithmetic traces	Parameter-definable traces and standard bits	PA finite comprehension; deterministic oracle runs; standard-system Turing closure.
Descriptive coding	Analytic $\mathcal{S}$ and analytic $\mathcal{K}_F(\mathcal{S})$	Borel remainder maps; represented-space computation; analytic projection lemmas.
Local root closure	Individual real and p -adic roots	Nonuniform isolating data; effective sign bisection; quantitative Newton estimates.
Derivations	Countable-basis differential extension	Extension through separable algebraic fields; exact analytic graph characterization.
Category and measure	Aggregate dependence relations	Completed Haar measures, Fubini, Kuratowski–Ulam, and Mycielski.
Residue bridge	$\text{res}_p(R_M) = \mathcal{K}_p(\mathcal{S}) \cap \mathbb{Z}_p$	Nonstandard digit recoding; inverse-limit compatibility; both inclusions of the image.
Hahn transfer	Coefficient extraction of finite relations	Theorem 26.1 ; scalar substitution by invertible monomials.

Layer	Mathematical interface	Verification obligation
Omnific application	Normal forms and constant extraction	Embedding the rational Hahn slice; comparison of topologies; finite quotient maps.

The most economical first formal target is [theorem 26.1](#). It has no descriptive set theory, no arithmetic-model coding, and no proper-class quantification. After that, [proposition 25.6](#) and the same-scale application provide a small end-to-end algebraic certificate. The analytic-derivation lemma is a second, independent high-value target. Its graph equivalence (52)–(54) should be formalized before any measurable-function wrapper is attached to it. No Lean file is included merely to give an impression of machine verification. Source 04’s proofs were developed and reviewed at the mathematical level, and the repository was not built for it.

27.4 Novelty and literature status

The new contributions claimed provisionally here are the explicit root-relocation map together with its exact predecessor identity, its use in omnific normalization, and the sharp dyadic-spine image classification. The extremal, quotient, and surcomplex statements are derived refinements of that construction. The known two-level example, the general initial-Hahn-subgroup characterization, and the classification of initial real coefficient groups are not claimed as new.

The literature check recovered the stated question in the original preprint, thesis, and workshop account [1, 3, 4], and examined adjacent sign-concatenation work [5]. Targeted searches did not locate a prior resolution or the specific root-relocation argument. That negative search result is limited evidence: it does not exclude an unindexed paper, unpublished proof, or a reformulation already known to a specialist. Neither mathematical correctness nor priority is certified by the absence of a search result.

Source 02. The convex-factor theorem and its surreal applications are proposed contributions with complete proofs; neither priority nor independent referee certification is claimed. The Ehrlich–Kaplan criterion, Ehrlich’s divisible embedding, Jeřábek’s residue theory and residue-injectivity rigidity mechanism [10], and $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ [11] are classical and credited; no new Ext computation is claimed, and the existence of rigid Presburger models is not claimed as new. Bagayoko and van der Hoeven’s surreal substructures [5] are a related viewpoint, not a dependency, and [12] is background on internal arithmetic only. A targeted source review did not find a prior formulation of the main convex-factor theorem; the search was not exhaustive.

Source 03. Its abstract claims “the explicit dyadic refinement and its assembled consequences, not a certified solution of a previously open major conjecture”. Standard-cut definability and the failure of induction beyond open induction in **Oz** are credited to Jeřábek [18] and to Enayat, Lelyk and Visser [14]; Phillips [15] is cited for historical context only. This report narrows the claim further. Theorem 2.9 of [14] already gives a parameter-free Δ_0 definition of $\mathbb{N}$ in every ordered Dorroh ring, which covers **Oz** (witnessing ideal Π) and every ring of [theorem 19.4](#) ([remark 20.5](#)); a failed bounded induction instance in **Oz** follows from it at once. Bounded definability of the standard cut in **Oz**, and with it a failed bounded induction instance there, are therefore not new. What remains proposed as new is: the dyadic formula Std^{dy} itself and its finite odd-factor certificate; its inductiveness under ED_2 alone, with the single-instance boundary and the sharpness examples; the character results ([theorems 20.1](#) and [20.4](#) and [proposition 20.2](#) and their corollaries, [corollaries 20.6](#), [20.7](#), [20.9](#) and [21.4](#)); the single-bound quadratic compiler with the complexity theorem and its corollary; and the correction of [14, Lemma 2.7] with its repair. [Remark 20.5](#) shows that the no-character theorem can also be reached through the four-square formula once the character classification is available. Source 03’s own literature search did not establish priority for any exact formulation, and the [write] statements of [section 20.5](#) add no claim of priority.

Source 04. Its claimed contributions are the countable-parameter derivation argument (lemma 24.1), the assembled perfect-gap theorem, the exact standard-system/residue identifications, including the profinite idempotents, and their coefficient-preserving Hahn and congruence applications. It credits Fatalini and Schindler’s attribution to Ben de Bondt and Glazer as a close conceptual precursor [22, Section 2], Ye, Yu and Zhao’s [23, Proposition 2.6] as nearby work on analytic subfields of $\mathbb{R}$ and $\mathbb{Q}_p$, Edgar and Miller for the dimension corollary, and Mycielski for perfect independent sets; it establishes no priority, for the assembled theorem or for its components. This report narrows the claim further. Corollary 25.8 is weaker than theorem 20.4 and corollaries 20.6 and 21.4 of source 03; theorem 26.1 follows at once from `isc:cp:lem:coeffld`; proposition 25.6 and the $\mathcal{A}$ -half of theorem 25.7 are set-sized versions of statements proved in Lean for **Oz** (`odg:thm:finitequotients`, `odg:eq:profinite`); and that the canonical image, not the completion, separates the rings is said after proposition 20.2. What remains proposed as new, within this collection, is lemma 24.1 with proposition 24.2 (whose relation to [23, Proposition 2.6] was not compared at this write), the perfect-gap and simultaneous-place theorems, the scalar fields $\mathcal{K}_F(\text{SSy}(M))$ with their faithfulness and elementarity, the exact residue images (theorems 25.1 and 25.3) and the idempotent reconstruction, the comparison of theorem 25.7 as a statement about Peano rings, and the same-scale omnific family. The [write 04] corollary adds no claim of priority; its combinatorial content is Scott’s theorem.

27.5 Provenance, placement review, and place in the collection

Four sources. This report is built from four manuscripts; sources 01 and 02 were merged in batch 29, source 03 was added in batch 86, and source 04 in batch 90.

Source 01 is *Discrete Initial Groups and Omnific Normalization* (24 pages, 23 September 2026), manuscript 09 of the collection’s batch 28. It was delivered with a README, a source audit, the finite checks `verify.py`, their recorded result and a build script. It was placed in commit `c6359e4` as this directory and written in `3d9dbe9`. Its LaTeX source is the basis of `article.tex`; its source audit is shipped unchanged as `source_audit.md`, the program and build script as `code/verify.py` and `code/build.sh`, and the record as `data/verification_report.json`. Its delivered PDF and README are not shipped. It has no pinned repository commit; its read identifiers are those recorded above. It contributes sections 1 to 10, Appendix A, Questions 28.1–28.4, non-claims (N1)–(N15) and most of the checklist of Appendix B.

Source 02 is *Convex Factors and Profinite Obstructions for Initial Surreal Groups* (23 pages, 23 September 2026), manuscript 09 of the collection’s batch 29. It was delivered with a README, a provenance record, the program `code/verify.py` and its record `data/verification.json`, and placed in commit `66d7e55` under the prefix `02-convex-factors-`: the provenance record is shipped unchanged as `02-convex-factors-PROVENANCE.md`, the program as `code/02-convex-factors-verify.py` and the record as `data/02-convex-factors-verification.json`. Its LaTeX source, PDF and README are not shipped; the provenance record names the delivered file names (`code/verify.py`, `data/verification.json`), which in this directory carry the prefix. Its repository pin is `9693b28`. It contributes section 1.5, sections 11 to 18 except the statements marked [merge], Appendix C, Questions 28.5–28.8, non-claims (N16)–(N29) and the corresponding rows of the checklist.

Source 03 is *Standard Integers in a Single Infinite Interval: Residue Characters, Bounded Induction, and Omnific Arithmetic* (20 pages, 3 October 2026), manuscript 10 of the collection’s batch 86, delivered as the archive `Glazer_ProveIt_Arithmetic_Research.zip` in the arrival commit `fb8414869` (inner directory `standard_integers_in_infinite_interval/`). It was delivered with a README, a source audit, two programs, their two records, a build audit, a build script and a checksum manifest. It was placed in commit `0bd0e5527` as an addition, with the prefix `03-standard-cut-`, continuing this directory’s local sequence after the unprefixed source 01 and `02-convex-factors-`; the seven staged files are listed, with their delivered names, in the source-03 paragraph of section 27 (finite checks). Its

LaTeX source, PDF and README are not shipped and survive in the arrival commit; its checksum manifest was verified (ten of ten files) and dropped, as repository policy requires. Its repository pin is 883e0b3b2. It contributes sections 19 to 22 except the statements marked [write] and the comparisons marked as the write's, the source-03 paragraphs of section 27, Questions 28.9–28.16, and non-claims (N30)–(N41).

Source 04 is *Perfect Transcendence Gaps in Borel Arithmetic: Standard Systems, p -adic Shadows, Hahn Fields, and Omnific Integers* (26 pages, 3 October 2026; “research manuscript prepared for Vladimir Reshetnikov”, marked AI-assisted and unrefereed), manuscript 06 of the collection's batch 90, delivered as the archive `Perfect_Transcendence_Gaps_Package.zip` in the arrival commit a162e4386 (inner directory `perfect_transcendence_gap/`). It was delivered with a README, a theorem ledger, a source audit, the program `verify.py`, its record `verification.json` and a checksum manifest. It was placed in commit 12076b2e8 as an addition, with the prefix `04-transcendence-gaps-`, continuing this directory's local sequence after `03-standard-cut-`; the four staged files are listed, with their delivered names, in the source-04 paragraph of section 27 (finite checks). Its LaTeX source, PDF and README are not shipped and survive in the arrival commit; its checksum manifest was verified (seven of seven files) and dropped. Its repository pin is 7d9b8b957. It contributes sections 23 to 26 except corollary 25.5 and the comparisons marked as the write's, the source-04 paragraphs of section 27, Questions 28.17–28.27, and non-claims (N42)–(N55).

How the merge was made (source 04). Source 04 was placed as an addition because its exact profinite image answers question 28.11, a question this report names, for nonstandard models of Peano arithmetic; the question asks about $\mathcal{I}\Delta_0$, so it is answered in part and re-scoped in a dated note after its statement. Source 04 does not cite this report, so its relation to the report is stated by the write: the paragraph “Relation to this report and its neighbours” of section 23.1, the bracketed comparisons after imported fact 23.6, (56), proposition 25.6, theorems 25.7, 26.1 and 26.3, and corollary 25.8 and in section 27, the subsection section 25.4 with the corollary corollary 25.5, marked [write 04], and (N56). The alternatives considered at placement were a part of the report docs/foundations-and-computation/polish-models-of-omnific-arithmetic, which studies Borel and Polish models of arithmetic motivated by Glazer but names no question that source 04 answers, and a new report; the perfect-gap material, the bulk of source 04, travels with it. Source 04's material is printed as it stands, in four new sections inserted after section 22; the former Sections 23–25 (verification, questions, conclusion) moved down by four, so the questions numbered 24.x at the write of source 03 are now 28.x, and no theorem-like number of Sections 1–22 or of the appendices changed. Its symbols are renamed as in section 23.2, which also gives the correspondence of its numbering, because its shipped ledger and audit use its own. Its introduction is sections 1.7 and 23.1; its dependency ledger, executable checks, formalization plan and novelty material are source-04 paragraphs of section 27; its eleven questions follow source 03's in section 28; its limitations are (N42)–(N55); its conclusion is a paragraph of section 29. The checklist of Appendix B gained six rows. No mathematical statement, proof or example of sources 01–03 changed, and no label was renamed or removed.

How the merge was made (source 03). Source 03 was placed as an addition because, with Theorem E, it answers part of question 28.7, a question this report names; it does not cite this report, so the combination is stated here, marked [write] (definition 20.10, corollary 20.11, and proposition 20.12, remark 20.13), and the question is re-scoped in a dated note after its statement. The alternative considered at placement was a part of the report docs/foundations-and-computation/polish-models-of-omnific-arithmetic, which collects the Glazer-motivated manuscripts of the same delivery; it names no question that source 03 answers. Source 03's material is printed as it stands, in four new sections inserted after section 18; the former Sections 19–21 (verification, questions, conclusion) moved down by four, and no theorem-like number of Sections 1–18 or of the appendices changed. Its symbols are renamed as in section 19.2, notably P, S to $\text{Pow}^{\text{dy}}, \text{Std}^{\text{dy}}$, which keeps

them apart from the Diophantine report’s existential Std (`odg:thm:standarddef`). Its introduction is sections 1.6 and 19.1; its verification, Lean-plan and novelty material is in the source-03 paragraphs of section 27, its questions follow source 02’s in section 28, its limitations are (N30)–(N41), and its conclusion is a paragraph of section 29. Added at this write, beyond the [write] statements: [remark 20.5](#), the comparisons with [proposition 14.3](#) and with the Lean theorem `omnific_int_hom_eq_constant`, the narrower novelty statement, the pointer to the batch-86 manuscript 06, and the repository paragraph’s last three sentences; each says so. The checklist of Appendix B gained six rows, five from source 03’s hypotheses section and one for the [write] corollary. Source 02’s text received three bracketed pointers (after [theorem 16.6](#), after [corollary 17.2](#) and in (N26)). No mathematical statement, proof or example of sources 01 or 02 changed, and no label was renamed or removed.

How the merge was made (source 02). Source 02 was placed as an addition because it answers two questions this report named, [questions 28.1](#) and [28.2](#). Its material is printed as it stands, with these choices. Results that both sources contain are printed once: the Ehrlich–Kaplan criterion ([imported fact 2.2](#)), the classification of initial real groups ([imported fact 2.3](#)), the caution against Hahn completion ([remarks 5.3](#) and [11.2](#)), the integer-bottom suspension ([theorem 8.5](#); source 02 re-proves it from the criterion, by the same argument), the description of **Oz** ((4)) and the rectangular convention ([definition 9.1](#)). Source 02’s foundations section is replaced by [section 2](#) and [section 11.1](#); its symbols that collide with source 01’s are renamed ([section 11.1](#)); its Theorems A–C are Theorems D–F. Its verification, repository, novelty, question and non-claim material is merged into the corresponding parts of [sections 27](#) and [28](#); its notation table is replaced by the renaming table of [section 11.1](#) and the definitions at their places, its dependency summary is in [section 27](#), and its independent-review checks are in Appendix B. Three statements are added in the merge, each with a complete proof and marked [merge]: [proposition 13.5](#) (source 02’s construction for $\mathbb{Z}\varepsilon$ is source 01’s $\rho \circ \Theta_\alpha$), [corollary 13.6](#) (limit stages of the iteration of [corollary 8.4](#)) and [corollary 17.3](#) (period groups of the trigonometry report); [remark 18.1](#) records how the two versions of Ehrlich and Kaplan’s optimality theorem differ. The Ehrlich–Kaplan numbering table of [section 1](#) gained the two rows for Section 8. No mathematical statement, proof or example of source 01 was changed. Where source 01 said that convex quotients and limit stages were not covered (after [corollaries 8.4](#) and [8.6](#), in [questions 28.1](#) and [28.2](#) and in (N10)), the text now says which source proves what; the title page, abstract, status box, conventions, this section and the conclusion mention the second source.

What the write changed (source 01). Every label now carries the prefix `isg:`. The journal numbering of the Ehrlich–Kaplan results ([section 1](#)) and the wording of the journal version of their criterion (after [imported fact 2.2](#)) were added, as were [section 1.4](#), this subsection, the shipped file names, a definition of `ct` (which the manuscript used without defining it), labels on formerly unlabelled statements, and the collected list of non-claims ([section 28.1](#)). Cross-references now name each kind of statement correctly; in the delivered source every reference to a lemma, proposition, corollary, remark or imported fact printed as “theorem”. No mathematical statement, proof or example was changed.

Placement review (source 01). When source 01 was placed, every step of the argument was checked by hand: the bottom-coefficient structure, both cases of the predecessor identity (8) and its failure at the old minimum, the ambient transport, every dyadic obligation in the proof of [Theorem A](#), both directions of [Theorem B](#) with its two counterexamples, the extremal groups, the size and depth corollaries, the splitting and suspension theorems, and every example of [section 10](#). No gap was found. The Ehrlich–Kaplan criterion was checked as a statement, against the arXiv text and the author-hosted text of the journal version: [imported fact 2.2](#) is a faithful reading of [1, Theorem 1 and its proof], equivalently of Theorem 5.1 of [2], and [imported fact 2.3](#) is [1, Lemma 3]. The criterion was not re-proved; the whole argument rests on it. This review is not a refereeing, and the solution proposed here has not been refereed. Its scope is source 01’s material; it does not cover [sections 11](#) to [18](#).

Placement review (source 02). When source 02 was placed, its proofs were checked by hand and no error was found: the exponent-cut lemmas and the canonical splitting, the compression theorem with its uniqueness and nested coherence, the transfer of every dyadic obligation in theorem 13.1 (coefficient groups are unchanged and every target right ancestor comes from a source one by (27)), the residue lemmas and proposition 14.3, the factorial family (its table, the transitions, $n!x_n = x_1 + S_n e$, $S_n \leq 2(n-1)!$ and the finite retractions), the extension classes and rigidity, the dense example (initiality of W : its exponents 0, 1, 2 have no right ancestors in the class, and $\mathbb{Q}$ and $\mathbb{Z}$ are initial in $\mathbb{R}$), the omitted type, the ultrapower and Peano arguments, the omnific residues and the rectangular transport. The compression was re-implemented independently and tested on all 496 nonempty convex intervals of the sign tree of lengths at most four (order, injectivity, initial image, prefix and right-ancestor reflection) and for nested coherence on the tree of lengths at most three. The elementary equivalence in theorem 16.2 rests on a standard two-sorted interpretation, checked as an argument. The Ehrlich–Kaplan criterion and Ehrlich’s embedding theorem were not re-proved. For the merge statements, (30) was confirmed by a finite computation for $\alpha \leq 3$ and words of length at most six, and the iteration in corollary 13.6 for three removal steps on those trees; that computation is not shipped. This review is not a refereeing, and it does not extend to source 01’s material beyond the consistency statements of sections 13.2 and 13.3.

Placement review (source 03). At this write, source 03’s proofs were checked by hand and no error was found: both cases of the odd-factor certificate (character zero, and character $2^k m_0$ with m_0 odd and possibly negative) with the witness bounds, the doubling and successor closure under ED_2 , the single-instance boundary, both sharpness rings, the character classification and the diagonal-image criterion, standard division in ID_0 , the vanishing theorem and its corollaries, the explicit non-diagonal profiles (including the passage from residues modulo 2^{k+j} to residues of $y/2^k$), the counterexample and the repair, the examples $\mathcal{A}_0$ (with the three certificates of example 21.3 recomputed), $\mathcal{A}(k, \Gamma)$ and **Oz**, the equal-bound guards, the compiler and the complexity reductions. The open-induction theorem for $\mathcal{A}_{\text{Sh}}$ and the effective arithmetical definition of a computable presentation were checked as imported statements, not re-proved. Lemma 2.7 and Theorem 2.9 of [14] were read at placement and again at this write in the v2 PDF (pages 11–12): Lemma 2.7 is printed as “The witnessing ideal of a Dorroh ring is unique”, its proof makes the step that proposition 21.1 refutes, and Theorem 2.9 states that the natural numbers are Δ_0 -definable in the nonnegative part of every ordered Dorroh ring. The [write] statements were proved at this write. This review is not a refereeing.

Placement review (source 04). At this write, source 04’s proofs were read in full and checked by hand, and no error was found: the trace and Turing-ideal lemmas, the analyticity of b_M and of the scalar fields, the size gap, faithfulness of the digit encodings, the real bisection and the p -adic Newton estimates of lemma 23.11, both elementarity arguments (the p -adic one through local constancy of power classes), the derivation lemma (the witness conditions (52)–(54), uniqueness of the value, analyticity of the graph without enumerating k , and the passage from analytic graph to Borel to continuous), properness of the hulls, the Fubini and Kuratowski–Ulam induction, the product argument for all places (injectivity from the relation $X_1 - X_2 = 0$), both inclusions of the p -adic and profinite images with the overspill step, the idempotents, the strict inclusion and the finite quotients of $\widehat{\mathbb{Z}}(S)$, the residue tower of $\mathcal{A}(k, \mathbb{Q})$, the comparison and the nonidentification, the Hahn transfer, the omnific family and its discreteness. Glazer’s theorem, Mycielski’s theorem in the form of [26, Theorem 1], Macintyre’s quantifier elimination, Edgar and Miller’s theorem and the classical descriptive set theory were taken as imported and not re-proved; Scott’s theorem, used by the [write 04] corollary, likewise. This review is not a refereeing.

Literature. The placement review of source 01 ran targeted web searches (“discrete initial subgroup”, “initial subgroup” together with “omnific integers”, and the title of [1]). They found only the Ehrlich–Kaplan paper, Kaplan’s thesis and unrelated pages, and no resolution of the question. The search was targeted, not exhaustive. The citations of Kaplan’s thesis (its Sections 6.2 and 7.1) and of the

Oberwolfach report (printed pages 3355–3356) are the manuscript’s; they were not re-checked at placement. For source 02 no separate literature search is recorded at placement; its own review was targeted and found no matching formulation of the convex-factor theorem. Its citations of Jeřábek, Strickland, Enayat–Hamkins–Wcisło and Bagayoko–van der Hoeven are the manuscript’s and were not re-checked. Its account of Ehrlich and Kaplan’s Section 8 was compared with the arXiv text and the authors’ text of the journal version at this write (remark 18.1). For source 03, the arXiv records of [14] (versions of 20 August and 30 September 2025) and of [16] were checked at this write, and the cited pages of [14] were read (section 27.5, placement review); the citations of Glazer [13] were compared with the arXiv v1 text, whose Theorem 2, Corollary 1 and Questions 1–2 say what is attributed to them here. Jeřábek’s MathOverflow answer [18], Phillips’s article [15] and Shepherdson’s paper [17] were not re-checked; source 03 itself consulted Phillips only through the publisher’s extract and Shepherdson only through [16]. For source 04, none of its citations was re-checked at this write: the seminar announcement and slides of [21], the versions of [22, 23, 24, 26, 29] and [25, 27, 28] are as source 04 records them in its shipped source audit, which also says that the original Mycielski PDF was not fetched. Its citation of Glazer’s *A Topological Tennenbaum Theorem* is [13], already checked for source 03.

Place in the collection. At the placement of source 01, no other report named Question 2 (Question 9.1) of Ehrlich and Kaplan, and none treated initial subgroups: a search of the reports and of the Lean sources for “initial subgroup”, “discrete initial” and “discretely ordered initial” found nothing outside this directory. The paper [1] is cited as background in two reports: the foundations report cites it among the initial-embedding results next to its sign-tree categoricity (`found:sub:signtree`), and the definable-surreals report for the simplicity theory of simplest separators (`dsn:sec:foundations`). The latter’s maximal initial elementary exponential core (`dsn:thm:core`) concerns initial *subfields* of definable surreals; it is related in spirit and neither overlaps nor conflicts with the present groups. The omnific integers used here are those of the Diophantine and set-sized-quotient reports (`odg:def:rings`, `osq:eq:Ozdef`); their discreteness is `odg:prop:units`, and the constant coefficient is the additive map of `osq:lem:ct`. The Diophantine report and the foundations report also cite a different Ehrlich–Kaplan paper, *Surreal ordered exponential fields*, for the initial integer part $\mathbf{Oz} \cap K$ of an initial subfield K (the status note after `odg:thm:floor`, and `found:sub:canonicalexpr`). That statement concerns subrings and integer parts. The groups $H \subseteq \mathbf{Oz}$ here are additive subgroups and $\mathcal{N}_{\alpha,n}$ is not multiplicative, so there is no conflict. No `isg:` statement has a Lean implementation.

Source 02 adds one neighbour. The trigonometry report studies the normalized period groups $\mathcal{I}(\Psi)$ of surcomplex phases, which are $\mathbb{Z}$ -groups for set-sized fields, with the same Jeřábek residue theory: its residue `resI` and maximal divisible subgroup $\mathcal{I}^{\text{div}}$ (`trigonometry:per:sec:defect`) are `resG` and G^{div} here, and its profinite defect is that residue modulo $\mathbb{Z}$. By corollary 17.3 a period group is initially realizable exactly when its profinite defect vanishes. There is no conflict: that report’s initiality statements concern the kernel-multiplier ring (`trigonometry:per:thm:robustness`), a subring, and the initial integer part $\mathbf{Oz} \cap F$, not the realizability of period groups. The divisible-kernel lemma is classical in both places. The set-sized quotient report (`osq:eq:Ozdef`) and the Diophantine report use the same Π and `ct` as proposition 17.1.

Related (batch 33). The exponential-relations report (`docs/foundations-and-computation/exponential-relations-over-omnific-integers`) reads the class-sized $\mathbf{Oz}$ formula by formula as a $\mathbb{Z}$ -group, and so as a model of Presburger arithmetic in the sense of section 1.5. Its division with remainder (`exr:prop:division`) is lemma 14.1 for $G = \mathbf{Oz}$, $e = 1$, re-derived independently, and its lexicographic splitting $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ is the split form of Theorem E with $G^{\text{div}} = \Pi$. It expands this group by the zero predicates of finite exponential sums with rational slopes and proves the result a definitional expansion of Presburger arithmetic, with $\mathbb{Z}$ an elementary substructure of $\mathbf{Oz}$ (`exr:thm:rational`). Nothing there concerns initiality, and no statement here concerns exponential predicates.

Source 03 (batch 86) adds three neighbours. The Diophantine report defines $\mathbb{Z}$ in $\mathbf{Oz}$ by an *existential* quartic formula (`odg:thm:standarddef`) and gives failed existential induction instances on

$\mathbf{Oz}_{\geq 0}$ (`odg:thm:induction`, `odg:rem:ringinduction`); Std^{dy} is bounded but not existential, so it bears on neither the Diophantine guard question (`odg:q:guard`) nor the Diophantine definitions there. The open-induction statements of the collection (the Shepherdson passage after `odg:thm:induction` for $\mathbf{Oz}$, and `dsn:cor:openinduction` for the definable integer part I_A of the definable-surreals report, whose constant-term retraction and standard division are `dsn:prop:ct`) are consistent with theorem 19.7: these rings satisfy open induction, have standard division and a ring retraction onto $\mathbb{Z}$, and therefore fail the bounded instance $\text{Ind}(\text{Std}^{\text{dy}})$. The uniqueness of the constant term (`odg:prop:canonicalct`) is the $\mathbf{Oz}$ case of proposition 21.2, and the existence transfer behind Hilbert’s tenth problem over $\mathbf{Oz}$ (`odg:cor:H10`) is the Diophantine retraction of section 22.6. Finally, the report `docs/foundations-and-computation/polish-models-of-omnific-arithmetic`, placed from the same delivery’s manuscripts and being written at the same time, studies Polish topologies on models of Presburger arithmetic and related obstructions, motivated by the same paper of Glazer; its first part proposes an affirmative construction for Glazer’s Question 2. Source 03 is not topological, and its obstruction precedes any topology (remark 20.8). Batch-86 manuscript 06, merged there, proves the failed open-induction instance $x^2 < t$ in $\mathbb{Z} + t\mathbb{R}[t]$, which is the instance given for $\mathcal{A}_0$ in section 21. No `isg:sc:` statement has a Lean implementation.

Source 04 (batch 90) adds these neighbours. The independent-surreal-copies report proves the coefficient-extension lemma `isc:cp:lem:coeffld` from which theorem 26.1 follows, and itself studies algebraic independence in Hahn fields; source 04’s coefficient fields $k_{\mathbb{R},Y}$ are a new instance, not a new method. The Diophantine report proves the finite quotients and the p -adic and profinite completions of $\mathbf{Oz}$ (`odg:thm:finitequotients`, `odg:eq:profinite`), of which proposition 25.6 and the $\mathcal{A}$ -half of theorem 25.7 are set-sized versions; its example of ω , $\sqrt{2}\omega$ and 0, identical in every finite quotient, is of the same kind as the family of theorem 26.3, whose members are moreover algebraically independent over a large coefficient field; ω and $\sqrt{2}\omega$ are not members, since 1 and $\sqrt{2}$ are computable. The report `docs/foundations-and-computation/polish-models-of-omnific-arithmetic` studies Borel and Polish models of arithmetic after Glazer’s topological paper; source 04 uses a different paper of Glazer’s, on standard systems, needs no topology on R_M (section 26.5), and answers none of that report’s questions. The report `docs/foundations-and-computation/cantor-families-of-surreal-subfields`, written from another manuscript of the same delivery, studies Cantor families of countable subfields of one countable surreal field; it shares words with source 04 (Cantor families, subfields), not theorems. No `isg:ptg:` statement has a Lean implementation.

28 Further questions and exact boundaries

The construction establishes an additive normal form while deliberately relinquishing several stronger kinds of compatibility. The following questions isolate possible further work without asserting that each is a previously published open problem.

Question 28.1 (Other convex quotients; answered by source 02). Which convex subgroups C of an initial group $A \subseteq \mathbf{No}$ admit an explicit initial realization of A/C by a normal-form transport?

Source 01 handles $C = \mathbb{Z}\varepsilon$. For a larger convex subgroup, one would need to identify the removed exponent region and determine whether its boundary can be relocated without creating unmet dyadic obligations. A numerical quotient of exponent chains alone would not settle the simplicity condition.

Status: *answered, by source 02.* All of them. Theorem D and theorem 13.1 realize A/C explicitly: the removed exponent region is the lower segment Γ_C (lemma 11.3), the quotient is represented by the upper projection P_{Γ_C} , a difference of leading truncations (lemma 11.4 and theorem 11.5), and compression of Γ^C creates no unmet dyadic obligation because it reflects right ancestors (27); it is not a numerical quotient of exponent chains (example 12.7). For $C = \mathbb{Z}\varepsilon$ the transport is source 01’s (proposition 13.5).

Question 28.2 (Limit iteration; answered by source 02). When an iterated removal of discrete bottom layers continues through infinitely many stages, which hypotheses give a coherent initial realization at a limit stage?

In source 01, finite iteration follows from [corollary 8.4](#); limit iteration requires a separate control of supports and of the changing sign-tree presentations. No commutation with such an unproved transfinite limit is used in source 01.

Status: answered, by source 02 with a corollary of the merge. No hypothesis is needed. The union of the removed layers is a convex subgroup, whose quotient is realized directly from its exponent cut ([corollary 13.2](#)), coherently with every earlier stage ([corollary 13.3](#)); [corollary 13.6](#) states this for the iteration of [corollary 8.4](#). The required control is supplied by compression, which is defined position by position and never as a limit of successive deletions ([remark 12.5](#)).

Question 28.3 (Multiplicative compatibility on restricted classes). For which special initial groups does some normalization to **O** also preserve products wherever those products are part of a specified algebraic structure?

The counterexample with $1 \cdot 1$ rules out a general claim for our normalization. Any useful positive result must impose additional conditions on exponent addition and coefficient rescaling, not merely on the order of supports.

Question 28.4 (Formal certification and uniqueness principles). Can the root-relocation proof and the concrete initiality criterion be fully verified in the repository's foundational model, and what natural extra conditions single out this normalization among the possible ordered-group isomorphisms to initial omnific groups?

Our formulas are canonical relative to the given sign model, normal forms, and least positive scale. They are not asserted to be invariant under arbitrary abstract group isomorphisms. The uniqueness in [Theorem B](#) concerns the inverse under the fixed map, not uniqueness of all possible initial embeddings.

The next four questions are source 02's. They mark extensions of the proved statements rather than prerequisites left unproved in their proofs, and they are not asserted to be previously published open problems (the second is related to the published [Question 1 / 8.1](#) of Ehrlich and Kaplan, [section 18.2](#)).

Question 28.5 (Intrinsic characterization). Which initially realizable ordered groups are characterized by intrinsic conditions on all convex factors and the splitting of convex extensions, without first choosing a Hahn presentation?

Question 28.6 (Intermediate theories). Can the nonsplit convex-extension obstruction be arranged inside a set-sized model of every proper intermediate theory appearing in the Ehrlich–Kaplan optimality question?

Question 28.7 (Arithmetic fragments). What is the weakest arithmetic fragment whose nonstandard models have ordered additive group completions excluded by the factorial residue type? Which analogous computable residue types yield sharper bounds?

Note (3 October 2026, at the write of source 03): partly answered, re-scoped. For exclusion from Init by any argument, the threshold is now located: the additive completion of every nonstandard model of $\text{I}\Delta_0$, indeed of the theory T^{dy} whose only induction axiom is the one bounded sentence $\text{Ind}(\text{Std}^{\text{dy}})$, is not initially realizable ([corollary 20.11](#)), while a countable nonstandard model of open induction with standard division has an initially realizable additive group ([proposition 20.12](#)). So the threshold lies strictly above IOpen and at or below $\text{T}^{\text{dy}} \subseteq \text{I}\Delta_0$. Still open: the exact weakest fragment (whether T^{dy} is strictly weaker than $\text{I}\Delta_0$ is [question 28.10](#)); and the question as literally posed, for exclusion by the factorial type τ_* , since whether nonstandard models of $\text{I}\Delta_0$ realize τ_* is not addressed. For the second sentence, [corollary 20.9](#) supplies an analogous computable residue obstruction, the elements satisfying Pow^{dy} above the standard cut, but it is stated in the ring language, not as a type of the ordered-group language ([remark 20.13](#)).

Question 28.8 (Compression and multiplication). Under what additional compatibility conditions does exponent-tree compression preserve a specified multiplication or a specified action of a coefficient domain? The unrestricted additive result provides no such compatibility automatically.

This is the counterpart for compression of [question 28.3](#), which asks the same for the normalization $\mathcal{N}_{\alpha,n}$.

The next eight questions are source 03's. They are proposed directions arising from its proofs, not all asserted to be previously unposed or established open problems; a focused literature check should precede any priority claim.

Question 28.9 (Minimal detector complexity). Under standard division and an integer retraction, can one choose a uniform standard-cut formula with fewer bounded variables or fewer alternations than Std^{dy} ? Which additional algebraic assumptions permit bounded-existential definitions? Distinguish the exact displayed detector from the broader definability methods already available.

The four-square formula of [14, Theorem 2.9] is one such broader method ([remark 20.5](#)); the existential definitions of the Diophantine report (`odg:thm:standarddef`) are unbounded.

Question 28.10 (The induction threshold for additive rigidity). The one instance $\text{Ind}(\text{Std}^{\text{dy}})$, together with standard division, already excludes every nonzero integer character in a nonstandard ring. Determine which familiar weak induction theories prove enough of these assumptions, and compare their strengths over a precisely fixed base theory.

In the terms of [definition 20.10](#): $\text{ID}_0 \vdash \text{T}^{\text{dy}}$, and $\text{IOpen} + \{\text{ED}_n^+\} \not\vdash \text{T}^{\text{dy}}$ ([proposition 20.12](#)); this question bears directly on [question 28.7](#).

Question 28.11 (Canonical residue images). Which subrings of $\widehat{\mathbb{Z}}$ occur as im res_{R_M} for nonstandard $M \models \text{ID}_0$? [Corollary 20.9](#) proves only that this image contains more than the diagonal integers. Investigate how induction, saturation, and the chosen model constrain the extra residue profiles.

Note (4 October 2026, at the write of source 04): answered for Peano arithmetic, re-scoped for ID_0 . For every nonstandard model M of Peano arithmetic, $\text{im res}_{R_M} = \widehat{\mathbb{Z}}(\text{SSy}(M))$, the ring of profinite integers whose factorial residue sequences are computable from one member of the standard system ([theorem 25.3](#)); its idempotents are exactly the $\mathbf{1}_A$ with $A \in \text{SSy}(M)$ ([corollary 25.4](#)), so the image determines the standard system and depends on nothing else, and it contains every profinite integer with a computable factorial residue sequence, ζ_* among them. With Scott's theorem, the images of the countable nonstandard models of PA are exactly the $\widehat{\mathbb{Z}}(\mathcal{S})$ for countable Scott sets $\mathcal{S}$ ([corollary 25.5](#), [write 04]). Still open: the question as posed, for models of ID_0 that are not models of PA, where the proofs of source 04 do not apply (they use induction for arbitrary arithmetic formulas, overspill and internal finite sums; [question 28.17](#) asks for the weakest theory that suffices), and the uncountable Peano case beyond the Scott-set description, which is Scott's problem. For ID_0 , [corollary 20.9](#) remains the only information here. [Question 28.26](#) asks a related question about completions with a distinguished image.

Question 28.12 (Fixed syntactic resources). Classify truth complexity when the single-bound quadratic fragment has a fixed number of variables, a fixed alternation bound, or a fixed gate-depth bound. [Theorem 22.4](#) allows these quantities to grow with the input sentence and does not answer the fixed-resource versions.

Question 28.13 (Weaker division hypotheses). Division by 2 suffices for uniqueness of a normalized character, whereas normalization of arbitrary characters used all standard moduli. Determine exact character groups when division is available only for selected prime powers, and identify the weakest assumptions supporting the no-additive-map conclusion.

Question 28.14 (Integer parts inside surreal fields). When a nonstandard arithmetic field embeds in **No**, characterize the integer parts that can be its arithmetic image. Such an image cannot lie in **Oz** under the hypotheses of [corollary 21.4](#); which structural features distinguish the admissible integer parts from the canonical omnific one?

Question 28.15 (Regularity of residue maps). In presentations of weaker arithmetic on Polish spaces, how do continuity or Borel measurability of residue maps and integer characters constrain multiplication? A useful comparison with Glazer’s work must state the induction and regularity hypotheses separately, rather than infer one from the other.

Question 28.16 (A focused proof-assistant milestone). Formalize the source-level counterexample, the ED_2 uniqueness repair, and the dyadic certificate before attempting the full interpretation compiler. These are short, reusable targets with explicit assumptions and little dependence on heavy surreal or model-theoretic libraries.

For **Oz**, the repair’s ring-homomorphism case is already a Lean theorem (`omnific_int_hom_eq_constant`, section 21.1). Source 03 names as immediate priorities the last question and a targeted comparison of the exact additive-character statement with older work on nonstandard arithmetic. A separate dependency audit should identify every use of the incorrect unrestricted uniqueness lemma in [14] and replace it either by a chosen witnessing ideal or by a valid additional hypothesis.

The next eleven questions are source 04’s. Except where a cited literature problem is explicitly named, they are not asserted to be previously published open problems. They are ordered by the clarity of the next proof obligation rather than by an unsupported prediction of difficulty.

Question 28.17 (Weaker arithmetic). What is the weakest arithmetical theory for which its nonstandard Borel models have the complete exact-residue and local-field package proved in sections 23 to 26?

The specific tasks are to separate definable-trace coding, downward Turing closure, and nonstandard finite recoding. Glazer’s no-full-system input already reaches EFA, but the Peano proof uses stronger induction. A successful weakening must replace those uses, not simply change “PA” to “EFA” in the statement. [Added at this write: for the residue images, this bears directly on question 28.11, which asks about $|\Delta_0|$.]

Question 28.18 (Effective strength of the perfect-gap construction). Given codes for the analytic field and its relative-closure property, what oracle strength is sufficient to produce a code for an independent Cantor set? Which reverse-mathematical subsystem proves the countable-parameter version?

The derivation proof and the Mycielski extraction have different logical costs. The inability to give a computable parametrization without extra information is already established in section 26.5; it does not settle the optimal relative construction.

Question 28.19 (Descriptive complexity of the scalar image). Which Borel ranks or analytic non-Borel complexities occur for $SSy(M)$, $\mathcal{K}_\infty(SSy(M))$, and $\mathcal{K}_p(SSy(M))$ as M varies through Borel nonstandard PA models?

Proposition 23.10 recovers ideal inclusion from field inclusion, but does not identify an optimal descriptive complexity. The digit embeddings give natural lower-bound reductions that a classification should exploit.

Question 28.20 (A shared digit family at all places). Can one choose a Cantor family of binary sets A such that the fixed encodings $c_\infty(A)$ and $c_p(A)$ are simultaneously algebraically independent over all the corresponding model fields?

Theorem 24.8 allows different continuous coordinate maps. It does not answer this version. Pullback of a meager relation by an arbitrary continuous embedding need not be meager, so the earlier product argument cannot simply be reused on the diagonal digit image.

Question 28.21 (p -adic dimension). What are the possible Hausdorff and packing dimensions of proper analytic relatively algebraically closed subfields of $\mathbb{Q}_p$, especially those arising from standard systems?

The proof establishes Haar-nullness and meagerness. It does not show p -adic dimension zero. A comparison with the real ring-dimension phenomena studied by Edgar–Miller and Ye–Yu–Zhao is a concrete starting point, not a license to transfer their real conclusion.

Question 28.22 (Realization of local shadows). Which proper analytic subfields $k \preccurlyeq \mathbb{Q}_p$ are of the form $\mathcal{K}_p(\text{SSy}(M))$ for a Borel nonstandard PA model? Which compatible collections of real and p -adic fields arise from one standard system?

Being elementary, analytic, and relatively algebraically closed is necessary, but the oracle closure encoded by [proposition 23.10](#) imposes additional structure. The realization question should keep the underlying Scott-set requirements separate from local field theory.

Question 28.23 (Intrinsic reconstruction of computation). Can Turing reducibility on S be reconstructed from the ring $\widehat{\mathbb{Z}}(S)$ with its named prime-coordinate residue maps, without supplying an external oracle-coding convention?

[Theorem 25.3](#) and [corollary 25.4](#) answer the exact-image and Boolean-reconstruction questions. They do not automatically recover the additional computability relation on the idempotents. A useful next result would specify the allowed language and distinguish definability, interpretability, and external effective reconstruction.

Question 28.24 (Regularity below analyticity). Which regularity assumptions on k suffice for the full perfect-gap theorem when analyticity is unavailable?

A useful formulation should state regularity for the aggregate polynomial relations and the graph of the induced derivation, not merely measurability of k itself. The transcendence-degree-one counterexample of [remark 24.3](#) shows why unrestricted subfields cannot satisfy the theorem.

Question 28.25 (Beyond coefficient-preserving Hahn extensions). Which transformations that mix coefficients and exponents preserve a perfect scalar independence gap? What is the corresponding statement for exponential or differential Hahn fields?

The coefficient proof answers the coefficient-preserving case exactly. An exponential operation can convert coefficient information into an exponent or vice versa, so a proposed extension must specify which operations and which normal forms are available.

Question 28.26 (Completions with a distinguished image). What additional invariant, beyond the abstract profinite completion, classifies the pairs $((R_M)_p, \text{im } R_M)$ for Borel PA models, and distinguishes them from omnific-type completion pairs?

[Theorem 25.7](#) proves that the distinguished image is indispensable. Through the exact-image theorem it carries a standard-system field, so computability and model-theoretic invariants have a direct entry point. [Added at this write: by [theorem 25.1](#) the distinguished image is $\mathbb{Z}_p(\text{SSy}(M))$, so the pair determines the field $\mathcal{K}_p(\text{SSy}(M))$, and by [proposition 23.10](#) the standard system, as a subset of $\mathbb{Z}_p$; the profinite version is [corollary 25.5](#).]

Question 28.27 (A machine-checked bridge). Can the Hahn independence theorem and the exact p -adic trace-image theorem be verified in Lean with an explicit dependency ledger, without adding the analytic corollaries as axioms?

The first target is coefficient algebra; the second requires correct nonstandard arithmetic semantics. A useful initial contribution would prove the finite recoding operation in every PA model, explicitly closing the gap between standard-model formula evaluation and internal PA derivability identified in the source-04 paragraph of [section 27](#). [Added at this write: for **Oz** the finite quotients and completions are already Lean theorems ([section 25.5](#)); the repository has no Lean development of nonstandard models of PA ([section 27.3](#)).]

28.1 Non-claims, collected

The limitations stated at their places in the text, in the shipped source audit and in the shipped provenance record of source 02 are collected here, so that none is lost. Each stays in force where it was stated.

Items (N1)–(N15) are source 01’s; in them “the manuscript” means source 01. Items (N16)–(N29) are source 02’s. Items (N30)–(N41) are source 03’s, with its shipped source audit, except (N41), which limits the [write] statements. Items (N42)–(N56) are source 04’s, with its shipped source audit and theorem ledger, except (N56), which limits the [write 04] statement.

- (N1) *Status*. The manuscript is unrefereed. It is a proposed solution with full proposed proofs, not an independently certified resolution; neither correctness nor novelty is certified, and no priority is claimed. The placement review (section 27.5) is not a refereeing.
- (N2) *Literature*. The literature and repository searches were targeted, not exhaustive. A negative search result is limited evidence and does not exclude an unindexed paper, an unpublished proof or a reformulation known to specialists. That a question was posed in 2015 and 2018 does not establish that it was still unresolved on the review date.
- (N3) *Imported criterion*. Imported fact 2.2 is imported, not re-proved, by the manuscript or at placement. No proof here replaces it by the weaker assertion that initiality of the exponent class alone suffices.
- (N4) *Not claimed as new*. The two-level example and its map, the initial-Hahn-subgroup characterization, the classification of initial real groups (imported fact 2.3) and Conway normal forms are imported and cited. The extremal, quotient and surcomplex statements are derived refinements of the construction.
- (N5) *Not a simplicity isomorphism*. $N_{\alpha,n}$ is an ordered-group isomorphism onto an initial image. It need not preserve the simplicity relation (remark 6.3); it is not multiplicative, does not preserve norms, and does not fix ordinary constants: $N_{\alpha,n}(1) \neq 1$ in general (section 10).
- (N6) *No classification*. No classification of all ordered abelian groups admitting initial embeddings is claimed, and the initial-monoid problem (Question 3 of [1], Question 9.2 of the journal version) is not addressed.
- (N7) *Exponent maps*. Θ_α is not a homomorphism of exponent groups, and (15) is not preservation of a multiplicative valuation (remark 5.4). Proposition 4.8 bounds exponent lengths, not the birthday of a transformed normal form.
- (N8) *No internal summation*. The strong-sum statements concern the ambient Hahn spaces; they give no internal summation on a subgroup A (remark 5.3), and lemma 2.4 does not license arbitrary infinite subseries.
- (N9) *Invariance and uniqueness*. The ordinal α is not an invariant of the abstract ordered group. The uniqueness in Theorem B is relative to the fixed map, not uniqueness of all initial embeddings, and the formulas are not claimed invariant under abstract isomorphisms.
- (N10) *Quotients*. It is not claimed that every convex quotient of an initial group has an initial realization. Iteration is covered only at finite stages (corollary 8.4), with no claim at a limit stage, and corollary 8.6 is a relative criterion that does not decide realizability of every quotient. (This is a statement about source 01. Source 02 proves both missing statements: Theorem D and corollaries 13.2 and 13.6. The relative criterion remains relative; Theorem E decides the case of $\mathbb{Z}$ -groups.)
- (N11) *Surcomplex scope*. Theorem 9.2 is Gaussian-linear and coordinatewise initial. It gives no field isomorphism, no canonical sign-tree or simplicity relation on $\mathbf{No}[i]$, no preservation of the modulus, and no compatibility with multiplication by general surcomplex scalars. Corollary 9.3 transports the right-hand side as well and does not apply to equations with ordinary constants held fixed, to nonlinear polynomials, to arbitrary surreal scalar matrices, or to norms and inner products.

- (N12) *Finite checks.* The finite checks do not verify transfinite concatenation or limit cases, arbitrary set-supported Hahn sums, the imported criterion, or the absence of a transfinite proof gap; the written proofs carry the general claims. Compiling and rendering the document establish its integrity, not its correctness.
- (N13) *No Lean.* No new Lean formalization and no Lean build are claimed. The proposed formalization units of [section 27](#) are not existing declarations, and a checked conditional theorem using the imported criterion would not by itself be a checked theorem about every initial subgroup of **No**.
- (N14) *Repository review.* The manuscript’s repository review was targeted: it did not review every report or archive, it treated truncated metadata as neither a complete inventory nor a proof of absence, it took no repository description as a certificate, and it promoted no repository result from pending to proved. The argument relies on no repository manuscript, and no repository Lean build was run for it.
- (N15) *Sources used only for context.* Bagayoko and van der Hoeven’s sign-sequence conventions are context, not a source of the normalization or of its range theorem. The questions of this section are not asserted to be previously published open problems, and $d(H) = \mathbf{On}$ is a formal marker, not a surreal number or a set ordinal.
- (N16) *Status of source 02.* The convex-factor theorem and its applications are proposed contributions with complete proofs. Neither priority nor independent referee certification is claimed, no Lean formalization or Lean build is claimed, and the placement review ([section 27.5](#)) is not a refereeing.
- (N17) *Credited, not new.* The Ehrlich–Kaplan criterion, Ehrlich’s divisible embedding ([imported fact 11.1](#)), Jeřábek’s residue theory and rigidity mechanism, and $\text{Ext}(\mathbb{Q}, \mathbb{Z}) \cong \widehat{\mathbb{Z}}/\mathbb{Z}$ are classical and credited; no new computation of Ext is claimed, and the existence of rigid Presburger models is not claimed as a new phenomenon. The results of source 01 that source 02 uses (the suspension, the relative splitting criterion) are not claimed again as new.
- (N18) *No classification.* There is no classification of initially realizable groups. Convex splitting is necessary, and its sufficiency is not proved ([section 18](#)).
- (N19) *Optimality question.* Source 02 does not resolve the Ehrlich–Kaplan set-model optimality question (Question 1 of [1], Question 8.1 of the journal version) and gives no construction uniform over the intermediate theories ([section 18.2](#) and [remark 18.1](#)). It answers none of Ehrlich and Kaplan’s questions.
- (N20) *Realization-dependent splitting.* The splitting of [theorem 11.5](#) depends on the chosen realization and is not invariant under abstract automorphisms; only the Presburger split case has the uniqueness of [proposition 14.3](#).
- (N21) *Realizability, not literal initiality.* The results concern initial *realizability*, not literal initiality of C or of a complement in its original position.
- (N22) *No Hahn product, no completion.* An initial group is not identified with the full Hahn product of its factors, and no closure under ambient strong sums is asserted ([remarks 11.2](#) and [13.4](#)).
- (N23) *Additive only.* The results are additive and module-theoretic. There is no multiplicative, field, ring, differential or exponential equivalence; the compression need not respect addition of exponents.
- (N24) *Surcomplex scope of source 02.* The rectangular notion is coordinate-dependent; no canonical sign tree on $\mathbf{No}[i]$ is postulated.
- (N25) *Finite checks of source 02.* The finite regression does not verify the Ehrlich–Kaplan theorem, arbitrary ordinals, class comprehension, completeness of Presburger arithmetic, recursive satu-

- ration or the Peano-model argument; the successful finite retractions are not a global splitting (remark 15.4). The delivered build and rendering checks concern presentation, not correctness.
- (N26) *Arithmetic fragment.* The fragment of arithmetic used for theorem 16.6 is not optimized. (This is a statement about source 02. With source 03, corollary 20.11 reaches $\text{I}\Delta_0$ by a different argument; the fragment for the factorial type itself is still not optimized.)
- (N27) *Repository review of source 02.* Its review, at pin 9693b28, was targeted: it was not a claim-by-claim audit of the repository, and no local Lean build was run.
- (N28) *Predecessor citation.* Source 02 cited source 01 without inventing a repository path for it; the current location (section 27) was supplied at this write.
- (N29) *Finite solvability.* Solvability of every finite retraction system is not a global splitting (remark 15.4).
- (N30) *Status of source 03.* It is an AI-assisted, unrefereed research draft. Its proofs are human-readable arguments, not a Lean or Rocq verification; no new Lean or Rocq proof is included or claimed, and the hand check of section 27.5 is not a refereeing.
- (N31) *Known phenomena credited.* Standard-cut definability and the failure of induction beyond open induction in $\mathbf{Oz}$ were already known (Jeřábek 2011, Enayat–Lętyk–Visser) and are credited; bounded definability of $\mathbb{N}$ in $\mathbf{Oz}_{\geq 0}$ is [14, Theorem 2.9]. The claimed contribution is the explicit dyadic refinement and its assembled consequences, not a certified solution of a previously open major conjecture. Global priority is not established for the exact formulations.
- (N32) *The correction is local.* The counterexample to [14, Lemma 2.7] does not claim that the paper’s main theorems fail; Theorem 2.9 there needs only a chosen witnessing ideal. No full dependency audit of [14] was performed, and it is not claimed that the error was previously unnoticed by its authors or by others.
- (N33) *Characters are not order-preserving.* A normalized integer character is not assumed to preserve order, and confusing it with a standard-part map would invalidate the arguments.
- (N34) *Unit condition.* The no-map theorems exclude unit-preserving additive maps; no theorem here excludes additive embeddings without the unit condition, which may land in the kernel of the target’s character.
- (N35) *No regularity claimed.* The obstruction concerns the existence of maps, not their continuity, measurability, definability or computability; it is complementary to Glazer’s topological theorem, not a strengthening of it, and answers neither of Glazer’s questions.
- (N36) *Sharpness is about this detector.* The examples of proposition 19.8 concern the dyadic formula only, not the existence of other definitions of the standard cut. No minimality of the quantifier pattern of Std^{dy} is claimed.
- (N37) *Not a truth predicate.* The compiler produces a different formula for each input sentence; it is not a single uniform truth predicate taking syntax codes, and there is no conflict with Tarski’s theorem. It needs an actually positive infinite bound; no computable test for infinitude is asserted; effective presentations are needed only for the upper reductions.
- (N38) *Imported results.* The open-induction property of $\mathcal{A}_{\text{Sh}}$ is the classical integer-part theorem, consulted through [16]; Phillips’s article is used only for the historical context of its publisher’s extract, and no uninspected theorem of it is used.
- (N39) *Finite checks of source 03.* They are finite exact checks and syntactic audits, not machine-checked proofs of the universal theorems, and do not evaluate quantifiers at infinite bounds.

- (N40) *Repository review of source 03.* It consulted the reader’s guides and `Omni ficResidues.lean` at its pin and is not a validation of every research claim in the repository. Elliot Glazer is not an author of source 03, and no review or endorsement by him is implied.
- (N41) *Scope of the [write] statements.* They do not determine the weakest fragment asked for in question 28.7, do not show that T^{dy} is strictly weaker than $\text{I}\Delta_0$, and do not address whether nonstandard models of $\text{I}\Delta_0$ realize the factorial type τ_* .
- (N42) *Status of source 04.* It is an AI-assisted, unrefereed research manuscript. Its proofs are ordinary mathematics; no Lean or Rocq verification of its new deductions is included or claimed, and the repository was not rebuilt for it. The hand check of section 27.5 is not a refereeing.
- (N43) *External inputs.* Glazer’s standard-system theorem (imported fact 23.6), cited from seminar slides, Mycielski’s theorem, classical descriptive set theory, quantifier elimination for real closed fields and for $\mathbb{Q}_p$, and Edgar and Miller’s dimension theorem are inputs, not new proofs. The EFA and Borel-quotient strength of Glazer’s theorem is not inherited by the new deductions, which use full Peano arithmetic.
- (N44) *Priority.* Historical priority is not established for the assembled theorem or for any component. Fatalini and Schindler’s attribution to de Bondt and Glazer is a close precursor, credited; lemma 24.1 is near [23, Proposition 2.6], and the relation was not compared at this write. Some ingredients may occur in the literature in another form; no “breakthrough” label is claimed.
- (N45) *Not Scott’s problem, not Glazer’s questions.* Neither the general Scott-set problem nor the reverse-mathematical questions of Glazer’s topological paper is claimed resolved, and no arbitrary Turing ideal is claimed to be a standard system.
- (N46) *Topology of the omnific family.* The family of theorem 26.3 is Cantor-parameterized by its real coefficients but discrete in the surreal order topology and in the valuation topology (section 26.3).
- (N47) *Restricted coefficient field.* Its independence is over $k_{\mathbb{R},Y}((t^{\mathbb{Q}}))$, not over $\mathbb{R}$ or **No**, and it is not independence of distinct valuation scales.
- (N48) *Set-sized only.* Only set-sized Hahn fields are used; **No** is never treated as a set, and no global well-order of the surreals is needed (section 26.4).
- (N49) *No effective witnesses.* The root lemma is nonuniform: no uniform all-roots selector and no algorithm for multiplicities. No member of an independent set is computable, no computable parametrization or injection of the Cantor sets exists, the continuous injections of theorem 24.8 need not be computable, and no single digit sequence is claimed to work at every place.
- (N50) *Measure and dimension.* No positive-measure independent Cantor set is claimed. Real Hausdorff dimension zero is inherited from Edgar and Miller, not a new theorem, and no p -adic dimension statement is made.
- (N51) *No embedding into **Oz**.* Source 04 does not embed the Peano ring into **Oz**; corollary 25.8 is a safeguard. What is transported is the scalar field of the standard system.
- (N52) *Scope of the countable advice.* Corollary 24.7 concerns field generation and relative algebraic closure, not the enlargement of the Turing ideal by an oracle. The elementary subfields are not complete and come with no computable selector. No Polish topology on R_M , continuity of arithmetic, or Borel global root selection is assumed or claimed.
- (N53) *Questions.* The eleven questions are source 04’s own, not established open problems.
- (N54) *Finite checks of source 04.* They check finite identities only, and certify no independence over an uncountable field, perfect set, Borel-model theorem, Glazer’s theorem, overspill, standard system or transfinite Hahn arithmetic.

- (N55) *Repository review of source 04.* It read two repository guides at its pin through a connector, reviewed no report or formal file in full, and ran no build; it did not know this report or the independent-surreal-copies report, and its weaker or derivable statements are credited at their places. No third-party papers are redistributed with it. Elliot Glazer is not an author of source 04, and no review or endorsement by him is implied.
- (N56) *Scope of the [write 04] statement.* Corollary 25.5 covers countable models of Peano arithmetic and rests on Scott’s theorem, imported. It does not address uncountable models beyond the Scott-set description, or models of Id_0 .

The statements marked [merge] add no claim of priority; they are proved from the two sources and, for corollary 17.3, from statements of the trigonometry report cited by label, whose own non-claims remain in force there. The same holds for the statements marked [write], proved from sources 02 and 03 and, for remark 20.5, from [14, Theorem 2.9]. The statement marked [write 04] adds no claim of priority either; it is proved from source 04 and Scott’s theorem [30].

29 Conclusion

The obstruction in the Ehrlich–Kaplan question is not the numerical size of the least positive element. Scalar multiplication removes that size immediately. The obstruction is the interaction of the exponent tree with coefficient groups forced by right ancestors.

The explicit map Θ_α moves the minimum exponent to zero and preserves all right-ancestor relations away from it. The coefficient map then normalizes the discrete bottom group without disturbing the higher groups. This gives the proposed affirmative proof. The relations lost at the minimum are exactly those recorded by the dyadic spine, which yields a necessary-and-sufficient image criterion rather than only a sufficient construction.

The resulting theory is additive but strong: it transports arbitrary set-supported normal forms without changing their support order types, commutes with strong sums, identifies extremal initial groups at each least positive scale, and produces initial models of the bottom cyclic quotients and rectangular surcomplex modules. The remaining obligations are independent mathematical review, a more exhaustive priority check, and a complete formal verification of the identified dependency chain.

Source 02’s compression construction makes convex-factor closure a concrete normal-form theorem: retain an exponent interval, delete precisely the ancestor positions outside it, and invoke the initial Hahn criterion. It is coherent for nested intervals and works at arbitrary transfinite cuts; for the bottom cyclic layer it is source 01’s $\rho \circ \Theta_\alpha$. At the same time, normal-form projection forces all convex extensions inside an initial realization to split. In additive arithmetic this yields a sharp boundary: a Presburger group has an initial surreal realization exactly when each element’s coherent standard residues are ordinary integers. The explicit factorial family places continuum many countable rigid models on the other side of that boundary. Dense examples and omitted-type arguments show that the distinction survives beyond discreteness and cannot be captured by first-order theory alone. None of this is a certified resolution of the full classification problem or of Ehrlich and Kaplan’s optimality question.

Source 03 shows that ordinary residue behaviour is not merely a benign shadow of omnific arithmetic. An integer-valued character forces every nonstandard element to have an explicit forbidden odd factor for the dyadic predicate, turning a small bounded formula into an exact standard-cut detector. Successor closure then converts that detector into an induction obstruction and, after normalization of characters, a prohibition on unit-preserving additive maps from nonstandard bounded arithmetic. The same detector has the opposite effect on interpretation power: inside any positive infinite interval it lets bounded formulas recover all standard arithmetic truth, so a computable ring can have true-arithmetic-complete bounded truth with a single common bound and quadratic atoms. These are semantic and computability statements, not claims of low-cost evaluation. Its polynomial counterexample separates having a

retraction onto $\mathbb{Z}$ from having a unique one, and division by 2 repairs the uniqueness. Read through source 02's criterion, the additive group of a nonstandard model of bounded arithmetic has no initial surreal realization, while open induction leaves room for one; the exact threshold between them is open.

Source 04 reads the negative standard-system theorem positively, as a source of large independent families. A Borel arithmetic model's accessible local scalars form dense elementary subfields, but analytic regularity prevents a finite or countably infinite transcendence basis from closing the ambient local field. This yields conull and comeager independent tuples and perfect independent sets, even after arbitrary countable algebraic supplementation. The p -adic image of the Peano ring is not merely bounded by that scalar field: it is exactly its valuation ring. The full profinite image additionally recovers the standard system as its Boolean algebra of idempotents and is strictly smaller than the product of its local images when that system is proper; for Peano arithmetic this answers the residue-image question of source 03, which stays open for $\text{I}\Delta_0$. On the surreal side, coefficientwise Hahn arithmetic transports the deficit without weakening it, and the resulting same-scale omnific elements separate algebraic information from ordinary finite congruence information. Identical completions can conceal different canonical images, and Cantor coefficient structure can coexist with a discrete surreal order topology.

A A compact proof for independent checking

This appendix compresses the main existence argument without changing its hypotheses. The full case analysis and support arguments are in the body of the paper.

Let A be a discrete initial subgroup. Its least positive member has a single normal-form term: otherwise a nonzero tail in A would be a smaller positive element after taking absolute value. Write that term as $c\omega^p$. The exponent class has minimum p , which, being the minimum of an initial tree, is all-minus, say $p = -\alpha$. The bottom real coefficient group is discrete and initial, hence $c\mathbb{Z}$ with $c = 2^{-n}$.

On $[p, \infty)$ define

$$p \mapsto 0, \quad p \frown \langle + \rangle \frown u \mapsto \langle + \rangle \frown p \frown u, \quad x \mapsto \langle + \rangle \frown x \quad (p \not\leq x).$$

The inverse removes the initial plus and reinserts a plus immediately after the block p whenever the remaining word starts with p . The two maps are increasing inverse bijections. For $x > p$,

$$\text{Pred}(\Theta_\alpha x) = \Theta_\alpha [\text{Pred}(x) \cup \{p\}], \quad \text{R}_s(\Theta_\alpha x) = \Theta_\alpha [\text{R}_s(x)].$$

Therefore an initial exponent class with minimum p maps to an initial one with minimum zero, and every new right-ancestor obligation comes from an old one above p .

Transport each nonbottom term $r\omega^\gamma$ to $r\omega^{\Theta_\alpha \gamma}$, and transport the bottom term $r\omega^p$ to r/c . This is a real-linear order-isomorphism on the ambient Hahn spaces. Its image of A is truncation closed and cross-sectional, has coefficient group $\mathbb{Z}$ at zero, and has unchanged initial real coefficient groups at positive exponents. The right-ancestor identity preserves all required dyadic containments. By the concrete Ehrlich–Kaplan criterion, the image is initial. It lies in **Oz** because all exponents are nonnegative and the constant coefficient is integral.

For the inverse classification, only the ancestors of the old minimum were discarded. They are $-\beta$ for $\beta < \alpha$, and their images are $q_\beta = \langle + \rangle \frown \langle - \rangle^\beta$. Initial inversion requires those exponents, while the right-ancestor condition at p requires all their dyadic coefficients. Together these requirements are exactly $K_\alpha \subseteq H$.

B Proof-review checklist

An independent review can focus on the following specific potential failure points rather than checking experimental output as though it were a proof.

Issue	Location and required check
Concrete versus existential initiality	Imported fact 2.2: the imported sufficiency statement must apply to the specified canonical image, not only to an unspecified isomorphic copy.
Minimum exponent	Proposition 3.2: truncation and subtraction isolate a nonzero tail; its absolute value is smaller than the positive leading monomial.
Limit sign lengths	Lemma 4.2 and theorem 4.4: concatenated tails use ordinal interval order types. No subtraction of ordinal lengths or removal of a last limit sign is permitted.
Exceptional old minimum	Theorem 4.4 and remark 4.5: the right-ancestor equality excludes $x = p$. Its failure is used, not ignored, in the inverse classification.
Coefficient group scaling	Lemma 6.1: only the bottom coefficient is rescaled; all right-ancestor obligations in the target are at unchanged groups.
Infinite subgroup membership	Section 5: transport the actual subgroup. Never replace it with all normal forms allowed by its coefficient groups.
Sharp inverse criterion	Theorem B: inverse exponent initiality is not enough; the missing minimum's ancestors must carry all dyadic coefficients.
Quotient order	Lemma 8.1 and proposition 8.3: higher nonzero terms dominate the cyclic bottom group, and positive-root deletion supplies an initial image.
Class-sized scope	All supports and summation indices are sets; correspondences of class-sized groups are expressed with class parameters.
Surcomplex scope	Theorem 9.2: the conclusion is Gaussian-linear and coordinatewise initial, not a field isomorphism or a new canonical complex simplicity relation.
<i>Source 02 (convex factors).</i>	
Actual normal-form image	Imported fact 2.2: its sufficiency part must apply to the actual transported image $\mathcal{F}_E(G)$, not just to an abstract copy.
Restriction stays in G	Lemma 11.4: P_E is the difference of two leading truncations, so no infinite subseries is used.
Meet retention	Theorem 12.4: compression must retain the sign at the meet of an incomparable pair; convexity puts the meet in E (lemma 12.2 and example 12.8).
Every target right ancestor	Theorems 12.4 and 13.1: every target right ancestor must be accounted for, not merely every source ancestor; this is the equality (27).
Residue sign	Proposition 15.1: the residue of x_1 is $-\zeta$, not ζ ; factorial moduli are cofinal for divisibility.
Divisible complement	Proposition 14.3: a complement to $\mathbb{Z}e$ is divisible because it is isomorphic to the quotient, not because every torsion-free subgroup is divisible.
Noninteger parameter	Lemma 15.3: ζ_* is shown to be neither a nonnegative nor a negative ordinary integer.
Internal sums	Theorem 16.6: the sum in a nonstandard Peano model is an internally coded finite sum, not an external infinite sum in G .

Issue	Location and required check
Merge statements	Proposition 13.5 and corollaries 13.6 and 17.3: the identification with $\rho \circ \Theta_\alpha$, the one-point layers of the iteration, and the order of $I/\mathbb{Z} \cong \Pi_F$. <i>Source 03 (standard cut) and the write.</i>
Ordinary-integer character	Theorem 19.4: χ takes values in the ordinary $\mathbb{Z}$; the 2-adic case split is external, and $m = \chi(y)$ may be zero or negative.
Positivity of witnesses	Theorem 19.4 and example 21.3: $u, v \geq 0$ and the bounds are checked in the ordered ring, not after applying χ , which need not preserve order.
Unit condition	Corollaries 20.6 and 21.4: only unit-preserving additive maps are excluded.
Infinite bound	Theorem 22.2 and example 22.3: the bound Ω must be positive infinite; existential gate blocks stay inside each atomic replacement.
Version of the corrected lemma	Proposition 21.1: Lemma 2.7 of arXiv:2508.14758v2; the repair needs ED_2 , which $\mathbb{Z}[X]$ lacks.
The [write] corollary	Corollary 20.11: $G(M)$ is a set-sized $\mathbb{Z}$ -group, so Theorem E applies; the argument uses no factorial type.
<i>Source 04 (perfect transcendence gaps).</i>	
Uncountable fields, countable parameters	Propositions 24.2 and 24.5: category and measure are applied to the aggregate relations Dep_n , never to an enumeration of polynomials over an uncountable k .
Analytic graph	Lemma 24.1: the graph of $\tilde{\partial}$ is analytic; Borelness comes from totality and single-valuedness, continuity from additivity.
Internal sums and overspill	Theorems 25.1 and 25.3: the sums are internally coded finite sums at a nonstandard N ; the coherent prefix comes from an explicit overspill, not an external infinite sum.
One shared oracle	Corollary 25.4: $\widehat{\mathbb{Z}}(\mathcal{S})$ requires one oracle for all moduli and is strictly smaller than $\prod_p \mathbb{Z}_p(\mathcal{S})$.
Coefficient field and topology	Theorem 26.3: independence over $k_{\mathbb{R},Y}((t^{\mathbb{Q}}))$, not over $\mathbb{R}$; the family is discrete in the surreal order topology.
Strength of the imported theorem	Imported fact 23.6: used only for nonstandard Borel models of PA; the trace lemmas use full induction.

C The core convex-factor proof in one place

This appendix is source 02's. For independent checking, the logical chain can be compressed without changing its hypotheses. Let $G \subseteq \mathbf{No}$ be initial and let $C \subseteq C'$ be convex. The active exponents form an initial class Γ . Convexity says that membership of a nonzero element in C or C' is determined exactly by whether its leading monomial belongs to that subgroup. Thus each has a lower exponent segment, Γ_C and $\Gamma_{C'}$.

The interval $E = \Gamma_{C'} \setminus \Gamma_C$ represents C'/C : coefficient restriction to E stays in G , because it is a difference of two leading truncations. The kernel on C' is C , and the leading sign gives the quotient order.

Any two incomparable elements of E have their common prefix between them numerically. It is in Γ by initiality and in E by convexity. Thus E is meet closed. In each $\gamma \in E$, retain the signs at precisely those positions β for which $\gamma \upharpoonright \beta \in E$. Index them by their ordinal order type. A source prefix in E becomes a target prefix; incomparable points keep opposite signs at their retained meet. Every target

prefix comes from one of the retained source prefixes. Hence the map is an order-isomorphism onto an initial exponent class and preserves and reflects right ancestry.

Reindex the normal forms supported in E by this map. The coefficient groups do not change, truncation closure and cross-sectionality persist, and every target dyadic right-ancestor requirement is one of the original requirements. The Ehrlich–Kaplan criterion makes this actual image an initial subgroup of **No**. This proves the convex-factor assertion. Upper-exponent restriction simultaneously supplies the split section for each convex subgroup sequence.

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